

there is question as follows, i want the solutions as interviewer expected in the interview of 2-2.5 year experience candidate and the solutions first is basic approach then other approach then next another better approach as like that possible approaches which patterns used and why that patterns used and at last best optimize approach as per the interviewer.

Q1. Two Sum

Interview question

Given an integer array and target, return indices of two numbers whose sum equals the target.

Short interview answer

Use a `HashMap<Integer, Integer>` storing value -> index. For every element, check whether

target - current already exists.

Detailed explanation

Brute force checks every pair:

text

$O(n^2)$

Hashing reduces the lookup to approximately $O(1)$ per element.

Example:

text

nums = [2,7,11,15]

target = 9

2 → store

7 → 9 - 7 = 2 → found

Java

java

```
public int[] twoSum(int[] nums, int target) {  
    Map<Integer, Integer> map = new HashMap<>();  
  
    for (int i = 0; i < nums.length; i++) {  
        int needed = target - nums[i];  
  
        if (map.containsKey(needed)) {  
            return new int[]{map.get(needed), i};  
        }  
    }  
}
```

```
map.put(nums[i], i);  
}
```

```
return new int[0];  
}
```

Complexity

text

Time: $O(n)$

Space: $O(n)$

Real backend example

Finding two transaction amounts that combine to a reconciliation target.

Follow-up

What if the interviewer asks for the values instead of indexes?

Return:

java

```
return new int[]{needed, nums[i]};
```

Tricky counter-question

Why not use HashSet?

Because we need the index, not only existence.



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Absolutely. For a **2–2.5 year Java/SDE candidate**, the interviewer generally does not want only the final `HashMap` solution. A strong answer shows that you can:

1. Clarify the problem and assumptions.
2. Start with the brute-force/basic approach.
3. Improve it step by step.
4. Explain the pattern behind the optimization.
5. Discuss edge cases and trade-offs.
6. Arrive at the best practical solution.
7. Explain why you chose that solution.
8. Handle follow-up variations.

Here is how I would present **Q1 – Two Sum** in an actual interview.

Q1. Two Sum

Problem

Given an integer array `nums` and an integer `target`, return the **indices of two numbers** such that:

text

```
nums[i] + nums[j] = target
```

Assume exactly one solution exists and the same element cannot be used twice.

Example

text

Input:

```
nums = [2, 7, 11, 15]
```

```
target = 9
```

Output:

```
[0, 1]
```

Because:

text

```
nums[0] + nums[1]
= 2 + 7
= 9
```

1. How I would start speaking in the interview

A good opening would be:

"We need to find two elements whose sum equals the target and return their indices.

The straightforward approach is to check every possible pair, which takes $O(n^2)$.

We can then optimize this using additional space. Since for every number `x` we need to know whether `target - x` has already appeared, a HashMap is a natural fit because it gives average $O(1)$ lookup. That brings the solution to $O(n)$ time and $O(n)$ space."

This immediately tells the interviewer that you understand both the **problem** and the **optimization path**.

2. Approach 1 — Brute Force

Idea

Check every possible pair.

For each `i`, check every `j > i`.

text

```
nums[i] + nums[j] == target
```

If yes, return their indices.

Java

java

```
public int[] twoSum(int[] nums, int target) {  
    for (int i = 0; i < nums.length; i++) {  
        for (int j = i + 1; j < nums.length; j++) {  
            if (nums[i] + nums[j] == target) {  
                return new int[]{i, j};  
            }  
        }  
    }  
    return new int[0];  
}
```

Dry run

text

```
nums = [2, 7, 11, 15]  
target = 9
```

First iteration:

text

```
2 + 7 = 9
```

Return:

text

```
[0, 1]
```

Complexity

There can be approximately:

```
text
```

```
n(n - 1) / 2
```

pairs.

Therefore:

```
text
```

```
Time: O(n2)
```

```
Space: O(1)
```

Interview evaluation

This is a perfectly valid first solution.

But for a 2–2.5 year candidate, **stopping here is usually not enough** when the interviewer expects optimization.

So I would immediately say:

"This works, but it is $O(n^2)$. We can do better using additional space."

That naturally leads to the next approach.

3. Approach 2 – Sorting + Two Pointers

There is another important approach that is worth discussing because it demonstrates knowledge of the **Two Pointer pattern**.

Idea

If we sort the array, we can use two pointers:

```
text
```

```
left -> beginning
```

```
right -> end
```

Then:

```
text
```

```
sum < target
    move left++

sum > target
    move right--

sum == target
    found
```

For example:

```
text
```

```
nums = [2, 7, 11, 15]
target = 9
```

Start:

```
text
```

```
2 + 15 = 17
```

Too large:

```
text
```

```
right--
```

Now:

```
text
```

```
2 + 11 = 13
```

Still large:

```
text
```

```
right--
```

Now:

```
text
```

```
2 + 7 = 9
```

Found.

The problem

The original question asks for **indices**.

If we directly sort the array:

```
text
```

```
[3, 2, 4]
```

becomes:

```
text
```

```
[2, 3, 4]
```

The original indices have changed.

So we have to preserve the indices.

Solution with a pair of value/index objects

For example:

```
java
```

```
public int[] twoSum(int[] nums, int target) {

    int n = nums.length;

    int[][] arr = new int[n][2];

    for (int i = 0; i < n; i++) {
        arr[i][0] = nums[i]; // value
        arr[i][1] = i;       // original index
    }

    Arrays.sort(arr, Comparator.comparingInt(a -> a[0]));

    int left = 0;
    int right = n - 1;

    while (left < right) {

        int sum = arr[left][0] + arr[right][0];
```

```

        if (sum == target) {
            return new int[]{
                arr[left][1],
                arr[right][1]
            };
        }

        if (sum < target) {
            left++;
        } else {
            right--;
        }
    }

    return new int[0];
}

```

Complexity

Sorting:

text

$O(n \log n)$

Two-pointer scan:

text

$O(n)$

Total:

text

Time: $O(n \log n)$

Space: $O(n)$

depending on how the data is represented/sorted.

4. Why use Two Pointers?

This is an important interview discussion.

Pattern

Two Pointers

Why does it work?

Because after sorting, the array has an ordering property.

Suppose:

```
text
```

```
a[left] + a[right] < target
```

Since `left` is already the smallest available value, keeping `left` where it is while decreasing `right` would make the sum even smaller.

Therefore, to increase the sum, we move:

```
text
```

```
left++
```

Similarly, if:

```
text
```

```
a[left] + a[right] > target
```

we decrease:

```
text
```

```
right--
```

The sorting gives us the monotonic property needed for this optimization.

5. Approach 3 – HashSet

A candidate may say:

```
"I can use a HashSet."
```

This is partially correct.

We can determine whether the complement exists:

```
text
```

```
needed = target - nums[i]
```

and store values in a `HashSet` .

But there is an issue.

The question asks for:

```
text
```

```
indices
```

A `HashSet<Integer>` only stores:

```
text
```

```
value
```

not:

```
text
```

```
value -> index
```

So a `HashSet` is not sufficient by itself.

Example

```
text
```

```
nums = [2, 7, 11, 15]
```

The set could contain:

```
text
```

```
{2, 7, 11}
```

But if we find `2`, we don't know its index unless we maintain another structure.

We could maintain:

```
text
```

```
HashSet + additional index tracking
```

but then the `HashMap` becomes the cleaner representation.

6. Approach 4 – HashMap

This is the standard optimal solution for the normal Two Sum problem.

Key observation

For each:

text

```
nums[i] = x
```

we need:

text

```
target - x
```

So instead of searching the entire remaining array, we store previously seen numbers.

The mapping is:

text

```
value -> index
```

Then checking whether the complement exists becomes average $O(1)$.

7. HashMap solution

java

```
public int[] twoSum(int[] nums, int target) {  
  
    Map<Integer, Integer> map = new HashMap<>();  
  
    for (int i = 0; i < nums.length; i++) {  
  
        int needed = target - nums[i];  
  
        if (map.containsKey(needed)) {  
            return new int[]{  
                map.get(needed),  
                i  
            };  
        }  
  
        map.put(nums[i], i);  
    }  
  
    return new int[0];  
}
```

```
}
```

8. Dry Run — How HashMap works

Consider:

```
text
```

```
nums = [2, 7, 11, 15]  
target = 9
```

Initially:

```
text
```

```
map = {}
```

i = 0

```
text
```

```
nums[i] = 2  
needed = 9 - 2 = 7
```

Check:

```
text
```

```
7 exists?  
No
```

Store:

```
text
```

```
2 -> 0
```

Map:

```
text
```

```
{2=0}
```

i = 1

```
text
```

```
nums[i] = 7  
needed = 9 - 7 = 2
```

Check:

```
text
```

```
2 exists?  
Yes
```

Map tells us:

```
text
```

```
2 -> 0
```

Current index:

```
text
```

```
1
```

Return:

```
text
```

```
[0, 1]
```

9. Why HashMap is better here

Brute force does this:

```
text
```

```
current element  
  ↓  
search all remaining elements  
  ↓  
O(n)
```

for every element.

Therefore:

```
text
```

```
0(n²)
```

HashMap changes the search strategy.

Instead of:

```
text
```

```
"Search the array for the complement."
```

we do:

```
text
```

```
"Have I already seen the complement?"
```

HashMap provides approximately:

```
text
```

```
0(1)
```

average lookup.

Therefore:

```
text
```

```
n elements × 0(1) lookup  
=  
0(n)
```

10. Best Approach

For the **standard Two Sum problem where we need original indices**, the best practical approach is:

HashMap

```
text
```

```
Time: 0(n) average  
Space: 0(n)
```

This is better than:

```
text
```

```
Brute Force    ->  $O(n^2)$ 
Sorting        ->  $O(n \log n)$ 
HashMap        ->  $O(n)$ 
```

So in the interview, I would conclude:

"Among the approaches, I would choose the HashMap solution because the problem requires original indices, and we can solve it in one pass with average $O(1)$ lookup per element. The overall complexity is $O(n)$ time and $O(n)$ extra space."

11. Why is it called a One-Pass solution?

This is a good interviewer follow-up.

We don't need:

```
text
```

```
first pass -> store everything
second pass -> search
```

We can simultaneously:

```
text
```

```
check complement
  ↓
store current value
```

in the same loop.

That's why it is commonly described as a **one-pass HashMap solution**.

12. Important subtlety — Why check before put?

Consider:

```
text
```

```
nums = [3, 3]
target = 6
```

At index 0 :

```
text
```

```
needed = 3
```

Not found.

Then:

```
text  
  
map.put(3, 0)
```

At index 1 :

```
text  
  
needed = 3
```

Found:

```
text  
  
map.get(3) = 0
```

Return:

```
text  
  
[0, 1]
```

So:

```
java  
  
if (map.containsKey(needed)) {  
    ...  
}  
  
map.put(nums[i], i);
```

is a clean ordering.

It ensures that we only match against a **previous element**, not the current element itself.

13. What if we put first?

Suppose we wrote:

```
java
```



```
map.put(nums[i], i);

if (map.containsKey(target - nums[i])) {
    ...
}
```

For:

text

```
nums = [3]
target = 6
```

we would insert:

text

```
3 -> 0
```

and then ask whether 3 exists.

It does — but it is the **same element**.

That violates:

text

```
i != j
```

So checking first is safer and clearer.

14. Duplicate values — Important interview edge case

Consider:

text

```
nums = [3, 3]
target = 6
```

Our approach handles it correctly.

text

```
i = 0
3 -> needed 3
not found
store 3 -> 0
```

```
i = 1
3 -> needed 3
found at 0

return [0, 1]
```

This is an important edge case the interviewer may intentionally test.

15. Negative numbers

HashMap solution also works with negative numbers.

Example:

```
text

nums = [-3, 4, 3, 90]
target = 0
```

At:

```
text

4
```

needed:

```
text

-4
```

At:

```
text

3
```

needed:

```
text

-3
```

and `-3` is already stored.

Return:

```
text  
[0, 2]
```

So the algorithm is not limited to positive integers.

16. Follow-up: What if interviewer asks for values instead of indices?

Then the HashMap is still useful, but we don't actually need the index.

We can use a `HashSet`.

```
java  
  
public int[] twoSumValues(int[] nums, int target) {  
  
    Set<Integer> set = new HashSet<>();  
  
    for (int num : nums) {  
  
        int needed = target - num;  
  
        if (set.contains(needed)) {  
            return new int[]{needed, num};  
        }  
  
        set.add(num);  
    }  
  
    return new int[0];  
}
```

Complexity:

```
text  
  
Time: O(n) average  
Space: O(n)
```

This is exactly why the distinction between **HashSet** and **HashMap** matters.

HashSet

Use when you primarily need:

```
text  
  
"Does this value exist?"
```

HashMap

Use when you need:

```
text
```

```
"Does this value exist, and what information is associated with it?"
```

Here:

```
text
```

```
value -> index
```

17. Follow-up: What if multiple answers exist?

Example:

```
text
```

```
nums = [1, 2, 3, 4]  
target = 5
```

Possible answers:

```
text
```

```
1 + 4  
2 + 3
```

If the requirement is:

Return any valid pair.

The existing algorithm can return the first valid pair it encounters.

If the interviewer asks:

Return **all pairs**.

Then the problem changes substantially.

We cannot simply return immediately.

For example, we'd need to collect results, carefully handle duplicates, and define whether duplicate index pairs or duplicate value pairs should be returned.

This becomes a different interview problem.

18. Follow-up: What if the array is already sorted?

This is an important optimization question.

Suppose:

text

```
nums = [1, 2, 4, 7, 11]
target = 9
```

Since it is already sorted, use **Two Pointers**.

java

```
public int[] twoSumSorted(int[] nums, int target) {

    int left = 0;
    int right = nums.length - 1;

    while (left < right) {

        int sum = nums[left] + nums[right];

        if (sum == target) {
            return new int[]{left, right};
        }

        if (sum < target) {
            left++;
        } else {
            right--;
        }
    }

    return new int[0];
}
```

Complexity:

text

```
Time: O(n)
Space: O(1)
```

Very important interview distinction

For an **already sorted array**:

text

Two Pointer

$O(n)$ time

$O(1)$ space

can be better than HashMap because it uses constant extra memory.

For an **unsorted array where original indices are required**:

text

HashMap

$O(n)$ average time

$O(n)$ space

is generally the preferred solution.

19. Interview trap — Can we sort the original array?

Not directly when the original indices are required.

Example:

text

nums = [3, 2, 4]

target = 6

Original answer:

text

[1, 2]

because:

text

$2 + 4 = 6$

After sorting:

text

[2, 3, 4]

the positions become:

```
text
```

```
0, 1, 2
```

So we would lose the original index information.

That is why the sorting approach needs to preserve:

```
text
```

```
(value, originalIndex)
```

pairs.

20. Integer overflow — advanced follow-up

A stronger interviewer may ask:

What if the integers are very large?

In Java, `int + int` can overflow.

Example conceptually:

```
java
```

```
int sum = nums[i] + nums[j];
```

could overflow the 32-bit integer range.

For defensive handling, we can use `long` for arithmetic:

```
java
```

```
long needed = (long) target - nums[i];
```

However, note that if the map stores `Integer` keys, the key type should remain compatible with the input domain.

For normal LeetCode/interview constraints where values and target fit into `int` and the intended arithmetic doesn't overflow, the regular implementation is sufficient.

This is more of a **robustness discussion** than something you need to complicate in the standard solution.

21. Pattern Identification

This problem is a classic example of:

Pattern 1 — Hashing / Frequency Lookup

Core idea:

text

Need some complementary value



target - current



look it up in HashMap

Pattern 2 — Complement Search

Whenever you see something like:

text

Find x and y such that

$x + y = \text{target}$

rewrite it as:

text

$y = \text{target} - x$

Then the problem becomes:

text

Have I seen y before?

This transformation is extremely useful.

Pattern 3 — Two Pointers

Useful when:

text

array is sorted

or when sorting is acceptable.

22. Why HashMap and not HashSet?

This is one of the most common interview questions.

HashSet

Stores:

```
text
```

```
value
```

Example:

```
text
```

```
{2, 7, 11}
```

It answers:

```
text
```

```
Does 7 exist?
```

But we need:

```
text
```

```
Where does 7 occur?
```

So we need:

```
text
```

```
value -> index
```

which is exactly a `HashMap` .

```
java
```

```
Map<Integer, Integer> map = new HashMap<>();
```

23. HashMap internal reasoning

A 2–2.5 year Java interviewer may ask:

Why is HashMap lookup approximately $O(1)$?

At a high level:

```
text
  |
  v
key
  |
  v
hashCode()
  |
  v
hash
  |
  v
bucket
  |
  v
find key in bucket
```

For a well-distributed hash function, lookup is approximately:

```
text
  |
  v
0(1)
```

average.

In modern Java, if collisions become sufficiently high, bucket structures can transition from linked-node chains to balanced tree structures, improving worst-case behavior within a bucket.

For this problem, however, the standard interview complexity remains:

```
text
  |
  v
Average time = O(n)
```

for the complete algorithm.

24. Does HashMap really guarantee O(1)?

A precise interview answer is:

"HashMap provides average or expected O(1) lookup, not an unconditional mathematical guarantee. Its performance depends on hash distribution and collisions."

That is a better answer than simply saying:

"HashMap is always O(1)."

25. Which solution should YOU give first?

For an actual interview, I recommend this sequence:

Step 1 — State brute force

```
text
```

```
O(n²) time  
O(1) space
```

Then say:

"We can optimize the repeated search using additional memory."

Step 2 — Mention sorting + two pointers

```
text
```

```
O(n log n)
```

but note the original-index problem.

Step 3 — Identify complement lookup

```
text
```

```
needed = target - nums[i]
```

Step 4 — Choose HashMap

```
text
```

```
value -> index
```

Step 5 — Code it

Step 6 — Dry-run it

Step 7 — Discuss complexity and edge cases

This demonstrates much stronger problem-solving ability than immediately writing a memorized HashMap solution.

26. Interviewer Expected Answer — Final Version

If the interviewer says:

"Give me your optimal approach."

A strong 2–2.5 YOE answer would be:

"The brute-force solution checks every pair, giving $O(n^2)$ time and $O(1)$ space.

Another approach is sorting and using two pointers, which gives $O(n \log n)$, but since the problem asks for original indices, we'd need to preserve them while sorting.

The better approach is to use a HashMap storing `value -> index`. For every element `nums[i]`, I calculate `needed = target - nums[i]`. If `needed` is already in the map, I have found the required pair and can return its stored index together with the current index. Otherwise, I store the current value and index in the map.

This gives $O(n)$ average time and $O(n)$ space, and it is the best practical solution for the standard unsorted Two Sum problem."

That is the kind of response I would expect from a **2–2.5 year Java developer**.

27. Final Code I would write in the interview

```
java

public int[] twoSum(int[] nums, int target) {

    Map<Integer, Integer> map = new HashMap<>();

    for (int i = 0; i < nums.length; i++) {

        int needed = target - nums[i];

        if (map.containsKey(needed)) {
            return new int[]{
                map.get(needed),
                i
            };
        }

        map.put(nums[i], i);
    }

    return new int[0];
}
```

Complexity

text

Time: $O(n)$ average

Space: $O(n)$

Pattern

text

Hashing
+
Complement Search

Why this pattern?

Because for every current value x :

text

$x + y = \text{target}$

becomes:

text

$y = \text{target} - x$

and we need fast existence/index lookup for y .

Best approach

text

HashMap<Value, Index>

for an unsorted array when original indices are required.

28. What the interviewer is really testing

Two Sum looks easy, but for a 2–2.5 YOE candidate the interviewer can use it to test several things simultaneously:

text

Can you identify brute force?

↓

Can you improve complexity?

↓

Do you know hashing?

↓

Can you identify complement patterns?

↓

Do you understand HashMap vs HashSet?



Can you preserve indices?



Can you handle duplicates?



Can you explain time/space complexity?



Can you adapt when the array is sorted?

So don't treat Two Sum as just a memorized HashMap problem.

The **real pattern to remember** is:

text

PAIR / COMPLEMENT PROBLEM



target - current



Have I seen the complement?



Hashing

And when the array is sorted:

text

SORTED ARRAY + PAIR SUM



TWO POINTERS

That distinction is very useful because the interviewer may deliberately change one condition and expect you to recognize that the optimal pattern has changed.

2026-09-02 20:12:29 You 

ok as aboved deep and real interview, continue for next question -

Q2. Contains Duplicate

Question

Determine whether an array contains duplicates.

Best answer

Use a HashSet.

java

```
public boolean containsDuplicate(int[] nums) {
```

```
Set<Integer> seen = new HashSet<>();
for (int n : nums) {
    if (!seen.add(n)) {
        return true;
    }
}
return false;
}
```

Complexity

text

Time: $O(n)$

Space: $O(n)$

Follow-up

Can you solve without extra space?

Yes, sort first:

text

sort $\rightarrow O(n \log n)$

scan $\rightarrow O(n)$

Total:

text

$O(n \log n)$

Tricky question

Why does `Set.add()` help?

`add()` returns false if the element already exists.

Production example

Detecting duplicate request IDs before processing a batch.

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Q2. Contains Duplicate

Problem

Given an integer array `nums`, determine whether the array contains **at least one duplicate value**.

Example 1

text

Input:

`nums = [1, 2, 3, 1]`

```
Output:
true
```

Because 1 appears more than once.

Example 2

```
text
```

```
Input:
nums = [1, 2, 3, 4]
```

```
Output:
false
```

Every element appears only once.

1. How I would approach this in a real interview

I would not immediately jump to `HashSet` .

I would first say:

"The requirement is only to determine whether any value occurs more than once. A straightforward solution is to compare every pair, but that takes $O(n^2)$. Since we only need to know whether a value has appeared before, we can maintain a set of previously seen values. A `HashSet` gives average $O(1)$ lookup and insertion, so we can solve this in $O(n)$ time with $O(n)$ extra space."

That opening demonstrates the important part: **you identified the operation the problem actually needs — existence checking.**

2. Approach 1 — Brute Force

The most basic solution is to compare every element with every other element.

Idea

For each `i` , compare:

```
text
```

```
nums[i]
```

with every element after it:

text

nums[j]

If:

text

nums[i] == nums[j]

we have found a duplicate.

Java

java

```
public boolean containsDuplicate(int[] nums) {  
    for (int i = 0; i < nums.length; i++) {  
        for (int j = i + 1; j < nums.length; j++) {  
            if (nums[i] == nums[j]) {  
                return true;  
            }  
        }  
    }  
    return false;  
}
```

3. Dry Run — Brute Force

Consider:

text

nums = [1, 2, 3, 1]

Start:

text

i = 0

Compare:

```
text
```

```
1 vs 2
```

```
1 vs 3
```

```
1 vs 1
```

The last comparison matches.

Therefore:

```
text
```

```
return true
```

4. Complexity

There can be approximately:

```
text
```

```
n(n - 1) / 2
```

comparisons.

Therefore:

```
text
```

```
Time:  $O(n^2)$ 
```

```
Space:  $O(1)$ 
```

This is correct but inefficient for large arrays.

For example, with:

```
text
```

```
n = 100,000
```

quadratic comparisons become impractical.

So I would tell the interviewer:

"The brute-force approach uses constant extra space, but its $O(n^2)$ time is unnecessary because we only care whether we've seen a value before."

That naturally leads to hashing.

5. Approach 2 – Sorting

A second improvement is to sort the array.

Key observation

After sorting, duplicate elements become adjacent.

Example:

text

Original:

[4, 1, 3, 2, 1]

After sorting:

[1, 1, 2, 3, 4]

Now we only need to scan adjacent elements.

text

```
nums[i] == nums[i - 1]
```

means duplicate found.

Java

java

```
public boolean containsDuplicate(int[] nums) {  
  
    Arrays.sort(nums);  
  
    for (int i = 1; i < nums.length; i++) {  
  
        if (nums[i] == nums[i - 1]) {  
            return true;  
        }  
    }  
  
    return false;  
}
```

6. Dry Run – Sorting

```
text
nums = [4, 1, 3, 2, 1]
```

After sorting:

```
text

[1, 1, 2, 3, 4]
```

Scan:

```
text

1 == 1
```

Duplicate found.

Return:

```
text

true
```

7. Complexity of Sorting Approach

Sorting costs:

```
text

 $O(n \log n)$ 
```

Scanning costs:

```
text

 $O(n)$ 
```

Therefore:

```
text

 $O(n \log n) + O(n)$ 
```

which simplifies to:

```
text
```

Time: $O(n \log n)$

Extra space depends on the sorting implementation and language/runtime details.

For an interview, the important high-level point is:

text

Time: $O(n \log n)$

8. Important Interview Issue — Sorting Mutates the Input

There is a subtle issue here.

Suppose:

text

```
nums = [4, 1, 3, 2, 1]
```

After:

java

```
Arrays.sort(nums);
```

the caller's array has been changed to:

text

```
[1, 1, 2, 3, 4]
```

If the original order matters elsewhere in the application, this may be undesirable.

A candidate with 2–2.5 years of experience should be able to recognize this trade-off.

You could make a copy first:

java

```
public boolean containsDuplicate(int[] nums) {  
  
    int[] copy = Arrays.copyOf(nums, nums.length);  
  
    Arrays.sort(copy);  
  
    for (int i = 1; i < copy.length; i++) {
```

```

        if (copy[i] == copy[i - 1]) {
            return true;
        }
    }

    return false;
}

```

But now you are using:

text

$O(n)$

extra memory for the copy.

9. Approach 3 – HashSet

Now we identify what the problem really asks.

We don't need:

text

value -> index

as in Two Sum.

We only need:

text

"Have I already seen this value?"

That is exactly what a `Set` represents.

So:

text

`HashSet<Integer>`

is the natural data structure.

10. HashSet Solution

java

```
public boolean containsDuplicate(int[] nums) {

    Set<Integer> seen = new HashSet<>();

    for (int n : nums) {

        if (!seen.add(n)) {
            return true;
        }

    }

    return false;
}
```

This is the standard interview solution.

11. Why does `Set.add()` solve the problem?

This is one of the most important follow-up questions.

Normally, many candidates write:

```
java

if (seen.contains(n)) {
    return true;
}

seen.add(n);
```

That works.

But we can combine both operations:

```
java

if (!seen.add(n)) {
    return true;
}
```

Why?

Because `Set.add()` has a useful contract:

When element is new

```
java
```

```
seen.add(n)
```

returns:

```
text
```

```
true
```

and inserts it.

When element already exists

```
java
```

```
seen.add(n)
```

returns:

```
text
```

```
false
```

and the set remains unchanged.

Therefore:

```
java
```

```
if (!seen.add(n))
```

means:

```
text
```

```
"If adding this value failed because it already existed,  
then we have found a duplicate."
```

This is concise and idiomatic Java.

12. Dry Run – HashSet

Consider:

```
text
```

```
nums = [4, 1, 3, 2, 1]
```


Initially:

```
text
```

```
seen = {}
```

n = 4

```
java
```

```
seen.add(4)
```

returns:

```
text
```

```
true
```

Set:

```
text
```

```
{4}
```

n = 1

```
java
```

```
seen.add(1)
```

returns:

```
text
```

```
true
```

Set:

```
text
```

```
{4, 1}
```

n = 3

```
text
true
```

Set:

```
text

{4, 1, 3}
```

n = 2

```
text

true
```

Set:

```
text

{4, 1, 3, 2}
```

n = 1

Now:

```
java

seen.add(1)
```

returns:

```
text

false
```

because 1 is already present.

Therefore:

```
java

!false
```

is:

```
text
```

```
true
```

and we return:

```
text
```

```
true
```

13. Why HashSet is better than HashMap here

This is a very useful connection with Q1.

Two Sum

We needed:

```
text
```

```
value -> index
```

so:

```
java
```

```
HashMap<Integer, Integer>
```

was appropriate.

Contains Duplicate

We only need:

```
text
```

```
value
```

so:

```
java
```

```
HashSet<Integer>
```

is sufficient.

The general principle is:

```
text
```

Need existence only



HashSet

versus:

text

Need associated information



HashMap

This is exactly the kind of data-structure selection reasoning interviewers like to hear.

14. Complexity of HashSet

We process each element once.

For each element we perform approximately:

text

add → average $O(1)$

Therefore:

text

Time: $O(n)$ average

Space: $O(n)$

More precisely, HashSet operations are average/expected $O(1)$, not an unconditional guarantee.

So don't tell the interviewer:

"HashSet is always $O(1)$."

Say:

"HashSet provides expected $O(1)$ insertion and membership behavior under normal hash distribution."

That's technically stronger.

15. Which approach is best?

Let's compare.

Approach	Time	Extra Space	Main Idea
Brute Force	$O(n^2)$	$O(1)$	Compare every pair
Sorting	$O(n \log n)$	Depends	Sort → duplicates become adjacent
HashSet	$O(n)$ average	$O(n)$	Track values already seen

For the **standard problem**, the preferred answer is:

```
text
```

```
HashSet
```

because:

```
text
```

```
 $O(n)$  average time
```

is better than:

```
text
```

```
 $O(n \log n)$ 
```

or:

```
text
```

```
 $O(n^2)$ 
```

16. Follow-up – "Can you solve it without extra space?"

This is where a good candidate should not blindly say yes.

First clarify what "without extra space" means.

If modifying the input array is allowed, then:

```
text
```

```
Sort + scan
```

works.

```
java
```

```

public boolean containsDuplicate(int[] nums) {

    Arrays.sort(nums);

    for (int i = 1; i < nums.length; i++) {

        if (nums[i] == nums[i - 1]) {
            return true;
        }
    }

    return false;
}

```

Complexity:

text

Time: $O(n \log n)$

Extra space: potentially $O(1)$ depending on the sorting implementation

But the input array is modified.

A stronger interview response is:

"Yes, provided modifying the input is allowed. I can sort the array and scan adjacent elements. That reduces auxiliary storage, but the trade-off is $O(n \log n)$ time and mutation of the input array."

That is much better than simply saying:

"Yes, sort it."

17. What if the array must NOT be modified?

Then sorting the original array isn't acceptable.

You could:

text

```

copy array
↓
sort copy
↓
scan

```

but now:

```
text
```

```
O(n)
```

extra space is required.

Or simply use:

```
text
```

```
HashSet
```

which is generally the better choice because it gives:

```
text
```

```
O(n) average
```

time.

18. What if the input is extremely large?

This becomes a trade-off discussion.

Suppose:

```
text
```

```
n = 100 million
```

HashSet may require significant memory.

Then sorting may become attractive if:

```
text
```

```
memory is constrained
```

because you trade memory for time.

This is an important engineering principle:

"The theoretically fastest algorithm is not always the best production choice when memory is constrained."

For a normal interview problem, however, HashSet remains the expected optimal approach.

19. What if values come from a small fixed range?

Another interesting variation.

Suppose the problem says:

text

```
0 <= nums[i] <= 100000
```

Then we could use a boolean array:

java

```
public boolean containsDuplicate(int[] nums) {  
  
    boolean[] seen = new boolean[100001];  
  
    for (int n : nums) {  
  
        if (seen[n]) {  
            return true;  
        }  
  
        seen[n] = true;  
    }  
  
    return false;  
}
```

Complexity:

text

```
Time: O(n)  
Space: O(k)
```

where `k` is the value range.

This can be more memory-efficient or faster than hashing in some circumstances, but only when the value range is sufficiently small and known.

20. Negative numbers with boolean-array approach

The previous technique doesn't directly support negative values.

For example:


```
text
nums = [-5, 2, -5]
```

We can't do:

```
java
seen[-5]
```

So we'd need offsetting if the range is bounded.

For example, with range:

```
text
[-1000, 1000]
```

we could map:

```
text
index = n + 1000
```

But once the range is large or arbitrary, HashSet becomes much more convenient.

This is why `HashSet` is the general solution.

21. Edge Cases

A good candidate should quickly check these.

Empty array

```
text
[]
```

No duplicate:

```
text
false
```

One element

```
text
```

```
[5]
```

No duplicate:

```
text
```

```
false
```

Two equal elements

```
text
```

```
[5, 5]
```

Duplicate:

```
text
```

```
true
```

Negative numbers

```
text
```

```
[-1, -2, -1]
```

Works with HashSet.

All identical

```
text
```

```
[7, 7, 7, 7]
```

The algorithm detects the duplicate on the second element and immediately returns.

This early exit is useful because we don't have to process the remaining elements.

22. Important Optimization — Early Exit

Notice:

```
java
if (!seen.add(n)) {
    return true;
}
```

We return immediately.

Suppose:

```
text

nums = [10, 20, 30, 40, 20, 50, 60, 70, ...]
```

We don't process everything.

As soon as `20` appears again:

```
text

return true
```

This gives excellent practical performance when duplicates occur early.

Of course, the worst-case complexity is still:

```
text

O(n)
```

because an array with no duplicates requires processing all elements.

23. Can we use `contains()` instead?

Yes.

```
java

public boolean containsDuplicate(int[] nums) {

    Set<Integer> seen = new HashSet<>();

    for (int n : nums) {

        if (seen.contains(n)) {
            return true;
        }

        seen.add(n);
    }
}
```

```
}  
  
return false;  
}
```

This is absolutely correct.

But:

```
java  
  
if (!seen.add(n))
```

is more concise because the membership test and insertion are expressed in one logical operation.

Interview answer

If asked:

"Which one is better?"

I'd say:

"Both are correct. `!seen.add(n)` is more concise and idiomatic because `Set.add()` tells us whether the value was newly inserted."

24. Could `contains()` + `add()` be two operations?

Yes, conceptually you're performing two `HashSet` operations:

```
text  
  
contains()  
add()
```

whereas:

```
text  
  
add()
```

alone tells you whether insertion actually happened.

That makes:

```
java  
  
if (!seen.add(n))
```

a cleaner implementation.

It also expresses exactly what we need:

text

```
"Try to add this value; if it was already there, it's a duplicate."
```

25. Could we use a HashMap instead?

Yes, but it would be unnecessary.

For example:

java

```
Map<Integer, Integer> map = new HashMap<>();
```

and maintain counts.

But we're not asked:

text

How many times?

We're only asked:

text

Does it occur more than once?

So storing a value and count is extra information we don't need.

This leads to an important interview principle:

"Choose the simplest data structure that contains exactly the information the problem requires."

Therefore:

text

HashSet > HashMap

for this particular requirement.

26. Pattern Identification

The primary pattern is:

Hashing / Seen Set

Whenever the question contains language like:

text

Have I seen this before?
Has this value appeared?
Is there a repeated element?
Does an element already exist?

you should immediately consider:

text

HashSet

Typical template:

java

```
Set<Integer> seen = new HashSet<>();

for (int x : nums) {

    if (!seen.add(x)) {
        // already existed
    }
}
```

This pattern appears again and again in interviews.

27. Another Pattern – Sort + Neighbor Comparison

There is a second reusable pattern:

text

Sort
↓
Duplicates become adjacent
↓
Compare nums[i] with nums[i - 1]

This is useful for problems involving:

text

duplicate detection
grouping
adjacent relationships
interval processing

But remember the cost:

text

Sorting = $O(n \log n)$

28. Interview Follow-up: "Why not use a nested loop?"

A strong answer:

"Nested loops are the straightforward solution, but they repeatedly search the same information and result in $O(n^2)$ time. Since I only need to know whether an element has occurred before, a HashSet lets me maintain that information and reduce the expected lookup to $O(1)$, giving $O(n)$ overall."

That shows optimization reasoning rather than memorization.

29. Interview Follow-up: "Why HashSet instead of TreeSet?"

Interesting Java-specific question.

HashSet

Expected:

text

add $O(1)$
contains $O(1)$

TreeSet

Typically:

text

add $O(\log n)$
contains $O(\log n)$

because TreeSet is tree-based and maintains sorted order.

We do **not** need sorted order here.

Therefore:

text

HashSet

is the better choice.

Interview answer

"I would prefer HashSet because ordering is unnecessary here. TreeSet would add a logarithmic cost without providing any benefit."

Excellent 2–2.5 YOE answer.

30. Interview Follow-up: "Why not LinkedHashSet?"

LinkedHashSet preserves insertion order.

But this problem doesn't need order.

So:

text

HashSet

is simpler and more appropriate.

The general rule:

text

Need fast membership only



HashSet

Need membership + insertion order



LinkedHashSet

Need membership + sorted order



TreeSet

This is a useful Java collections distinction.

31. Production Example — Duplicate Request IDs

Suppose a backend receives a batch:

```
text

requestIds =
[
  "REQ-1001",
  "REQ-1002",
  "REQ-1003",
  "REQ-1001"
]
```

Before processing the batch, we might want to detect duplicate IDs.

Conceptually:

```
java

Set<String> seen = new HashSet<>();

for (String requestId : requestIds) {

    if (!seen.add(requestId)) {
        // duplicate request ID
        ...
    }
}
```

This can be useful for validating batch input.

However, in a distributed production system, one important distinction arises:

In-memory duplicate detection

A `HashSet` detects duplicates **within the current process/batch**.

It does not guarantee uniqueness across:

```
text

multiple application instances
multiple requests
restarts
multiple data centers
```

For global request-ID uniqueness, we may need something such as:

```
text
```

```
database unique constraint
distributed cache
idempotency store
```

depending on the architecture.

That distinction is particularly valuable in backend interviews because it shows that you understand where a data-structure solution stops being sufficient for a production system.

32. Real Backend Follow-up — Idempotency

An interviewer may extend the production example:

"What if the same request ID comes from two different application instances?"

A local:

```
java
HashSet
```

doesn't solve that.

For example:

```
text

      Load Balancer
      /      \
     /        \
Server A      Server B
HashSet A     HashSet B
```

Request:

```
text

REQ-1001
```

could exist in both independently.

So you'd need shared state, such as a distributed datastore or database constraint, depending on requirements.

This is an important distinction:

```
text
```

Algorithm-level duplicate detection

≠

Distributed-system uniqueness

Don't overcomplicate the standard DSA answer unless the interviewer asks for this extension.

33. Advanced Question – What if memory is limited?

Suppose the interviewer says:

"You have 10 GB of input but only 100 MB of RAM. Can you still use HashSet?"

A naive HashSet may not fit.

Then we need an engineering trade-off.

Possible directions include:

text

External sorting

Bloom filter

Partitioning

External storage

Distributed processing

But these are **not equivalent**.

For example, a Bloom filter can provide probabilistic membership detection, but it can have false positives.

Therefore, whether it is appropriate depends on whether the requirement permits probabilistic answers.

For a normal coding interview, don't jump to Bloom filters unless asked.

34. Advanced Question – Can Bloom Filter solve it exactly?

This is a good senior-ish follow-up.

A Bloom filter can answer:

text

"Definitely not seen"

or:

text

"Possibly seen"

It can produce false positives but not false negatives under its standard assumptions.

So if the requirement is:

"Return true only when I know there is definitely a duplicate."

A plain Bloom filter alone is insufficient because "possibly seen" does not prove a duplicate.

This is usually beyond a 2–2.5 YOE standard round, but knowing the trade-off is valuable.

35. Best Answer for a 2–2.5 YOE Candidate

Suppose the interviewer asks:

"What's your optimal solution?"

I would answer:

"Since we only need to know whether a value has appeared before, I would use a HashSet. I iterate through the array and attempt to add each number to the set. `Set.add()` returns false when the value is already present, which means we've found a duplicate, so we can return true immediately. If we finish the loop, there are no duplicates. This gives $O(n)$ expected time and $O(n)$ space."

Then code:

java

```
public boolean containsDuplicate(int[] nums) {  
  
    Set<Integer> seen = new HashSet<>();  
  
    for (int n : nums) {  
  
        if (!seen.add(n)) {  
            return true;  
        }  
    }  
  
    return false;  
}
```

36. What I would say if the interviewer asks for all approaches

A polished response:

"There are three main approaches.

First, brute force: compare every pair, which is $O(n^2)$ time and $O(1)$ extra space.

Second, sorting: sort the array and compare adjacent elements. That's $O(n \log n)$ time and can use little auxiliary space depending on the sort, but it modifies the input.

Third, HashSet: maintain previously seen values and detect when insertion fails. That's $O(n)$ expected time and $O(n)$ space.

For the standard problem, I would choose HashSet because the requirement is only membership checking and it gives the best asymptotic time."

That is an excellent interview-level progression.

37. Common Mistakes

Mistake 1 — Using HashMap unnecessarily

```
java
Map<Integer, Integer>
```

when no index/count is required.

Better:

```
java
Set<Integer>
```

Mistake 2 — Saying HashSet is guaranteed $O(1)$

Better:

```
text
Expected / average  $O(1)$ 
```

Mistake 3 — Ignoring input mutation

If using:

```
java
```

```
Arrays.sort(nums);
```

remember that the original array is modified.

Mistake 4 – Forgetting early return

Don't process the rest after discovering a duplicate.

```
java  
  
return true;
```

immediately.

Mistake 5 – Not explaining why Set is appropriate

Don't just say:

"I'll use HashSet because it's faster."

Instead say:

"I only need existence information, so a Set directly models the requirement."

That is a much stronger interview explanation.

38. The Pattern You Should Remember

Don't memorize only:

```
java  
  
if (!seen.add(n))
```

Memorize the underlying thought process:

text

```
Question:  
"Have I seen this value before?"  
  ↓  
Need membership lookup  
  ↓  
HashSet  
  ↓  
Process each element once  
  ↓
```

If insertion fails → duplicate

And the alternative:

text

Need minimal extra memory



Can modify array?



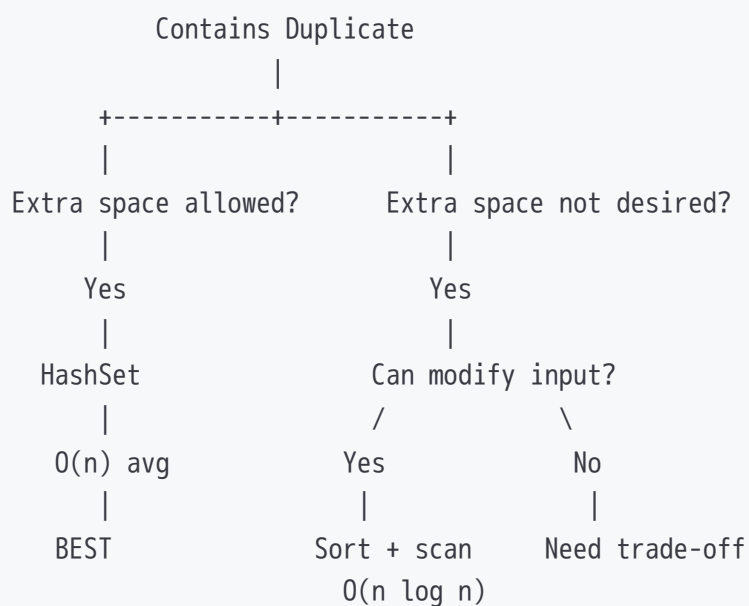
Sort



Compare neighbors

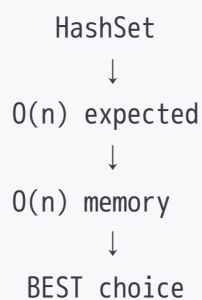
39. Final Comparison – Interview Decision Tree

text



For the standard question:

text



40. Final Interview Cheat Sheet

Basic

text

Brute Force
 $O(n^2)$ time
 $O(1)$ space

Improved

text

Sorting
 $O(n \log n)$ time
low auxiliary space
but mutates input

Optimal standard solution

text

HashSet
 $O(n)$ expected time
 $O(n)$ space

Pattern

text

Seen / Membership
↓
HashSet

Important Java trick

java

```
if (!seen.add(value))
```

means:

text

value already exists



duplicate found

HashSet vs HashMap

text

Need only existence

→ HashSet

Need value + associated information

→ HashMap

Production caveat

text

HashSet

→ local/process-level detection

Database/distributed store

→ global uniqueness/idempotency

For the next DSA question, I would keep this same interview structure: **brute force → better → optimal → pattern recognition → why the data structure works → dry run → edge cases → Java-specific follow-ups → production variation → final interviewer-ready answer.**

2026-09-02 20:15:39 You 

ok as aboved continue for next question, keep this same interview structure: ****brute force → better → optimal → pattern recognition → why the data structure works → dry run → edge cases → Java-specific follow-ups → production variation → final interviewer-ready answer**.**

Q3. Valid Anagram

Question

Determine whether two strings contain the same characters with the same frequencies.

Approach

Count characters.

java

```

public boolean isAnagram(String s, String t) {
    if (s.length() != t.length()) {
        return false;
    }

    int[] count = new int[26];

    for (char c : s.toCharArray()) {
        count[c - 'a']++;
    }

    for (char c : t.toCharArray()) {
        count[c - 'a']--;
    }

    for (int x : count) {
        if (x != 0) {
            return false;
        }
    }

    return true;
}

```

Complexity

text

$O(n)$

Follow-up

What if Unicode is allowed?

Use:

java

Map<Character, Integer>

instead of fixed array.

Production example

Case-insensitive token normalization or duplicate-content detection



ChatGPT gpt-5-6-t-mini

Q3. Valid Anagram

This is a very common interview problem because it looks simple, but the interviewer can use it to test **frequency counting, hashing, sorting, character encoding, Java collections, space optimization, and problem generalization**.

For a 2–2.5 year Java candidate, I would expect you to be able to move from a straightforward comparison to an $O(n)$ solution and explain **why an array is better than a HashMap when the character domain is fixed**.

1. Problem

Given two strings `s` and `t`, determine whether `t` is an anagram of `s`.

Two strings are anagrams when they contain:

- the same characters
- with the same frequencies

Order does not matter.

Example

text

```
s = "anagram"
t = "nagaram"
```

Output:
true

Because both contain:

text

```
a -> 3
n -> 1
g -> 1
r -> 1
m -> 1
```

Another example:

text

```
s = "rat"
t = "car"
```

Output:

```
false
```

because:

```
text
```

```
s: r, a, t
```

```
t: c, a, r
```

They don't contain the same characters.

2. How I would start in a real interview

A strong opening would be:

"We need to determine whether the two strings have identical character frequencies. The simplest solution is to compare their character frequencies after sorting, but that would take $O(n \log n)$. Since the problem is fundamentally about frequency counting, I can maintain counts for characters. If the input is guaranteed to contain only lowercase English letters, an `int[26]` is better than a `HashMap` because the character domain is fixed. That gives $O(n)$ time and $O(1)$ auxiliary space."

That answer already tells the interviewer that you understand the important distinction:

```
text
```

```
Fixed small domain
```



```
Array
```

versus:

```
text
```

```
Unknown / large domain
```



```
HashMap
```

3. Approach 1 – Brute Force

The most naive idea is:

For every character in `s`, find a matching unused character in `t`.

For example:

```
text
s = "aab"
t = "aba"
```

We could:

```
text

take 'a' from s
find 'a' in t
mark it used

take next 'a'
find another 'a'

take 'b'
find 'b'
```

This can be implemented using a visited array.

Java

```
java

public boolean isAnagram(String s, String t) {

    if (s.length() != t.length()) {
        return false;
    }

    boolean[] used = new boolean[t.length()];

    for (int i = 0; i < s.length(); i++) {

        boolean found = false;

        for (int j = 0; j < t.length(); j++) {

            if (!used[j] && s.charAt(i) == t.charAt(j)) {
                used[j] = true;
                found = true;
                break;
            }
        }

        if (!found) {
            return false;
        }
    }
}
```

```
    return true;
}
```

4. Why this is brute force

For each character in `s`, we may scan almost the entire `t`.

Therefore:

```
text

n characters
×
n search
=
O(n²)
```

Complexity

```
text

Time: O(n²)
Space: O(n)
```

The extra `boolean[]` is $O(n)$.

This is correct, but it is clearly not the solution we want for a normal interview.

5. Approach 2 – Sort Both Strings

The next natural improvement is based on a simple observation:

If two strings are anagrams, their sorted forms must be identical.

Example:

```
text

s = "listen"
t = "silent"
```

After sorting:

```
text
```

```
eilnst  
eilnst
```

Therefore they are anagrams.

Java

```
java
```

```
public boolean isAnagram(String s, String t) {  
  
    if (s.length() != t.length()) {  
        return false;  
    }  
  
    char[] sChars = s.toCharArray();  
    char[] tChars = t.toCharArray();  
  
    Arrays.sort(sChars);  
    Arrays.sort(tChars);  
  
    return Arrays.equals(sChars, tChars);  
}
```

6. Dry Run — Sorting Approach

Take:

```
text
```

```
s = "anagram"  
t = "nagaram"
```

Convert:

```
text
```

```
s -> ['a','n','a','g','r','a','m']  
t -> ['n','a','g','a','r','a','m']
```

Sort:

```
text
```

```
s -> ['a','a','a','g','m','n','r']  
t -> ['a','a','a','g','m','n','r']
```

Arrays are equal.

Therefore:

```
text
```

```
true
```

7. Complexity of Sorting Approach

Each string has length `n` .

Sorting:

```
text
```

```
O(n log n)
```

for each string.

Comparison:

```
text
```

```
O(n)
```

Overall:

```
text
```

```
O(n log n)
```

Space

Because strings in Java are immutable, we convert them into `char[]` , so we need additional storage.

At interview level:

```
text
```

```
Time: O(n log n)
```

```
Space: O(n)
```

The exact auxiliary-space behavior of the sorting implementation itself can vary, but converting both strings to arrays already gives $O(n)$ storage.

8. Is sorting good enough?

It is absolutely valid.

A strong candidate should not imply that sorting is "wrong."

Instead:

"Sorting is correct and simple, but since this is a frequency-counting problem and we don't actually need ordering, we can do better with $O(n)$ counting."

This is an important interview habit.

Don't just say:

"Hashing is faster."

Explain **why sorting is doing unnecessary work**.

Sorting establishes complete ordering, but the problem only asks:

```
text
```

```
How many times did each character occur?
```

We don't need ordering at all.

9. Approach 3 — HashMap Frequency Counting

Now we recognize the core pattern:

```
text
```

```
Anagram
```

```
↓
```

```
same frequency of every character
```

```
↓
```

```
frequency counting
```

A general solution is:

```
text
```

```
Map<Character, Integer>
```

We count characters in `s`, then subtract characters from `t`.

Java

java

```
public boolean isAnagram(String s, String t) {  
  
    if (s.length() != t.length()) {  
        return false;  
    }  
  
    Map<Character, Integer> count = new HashMap<>();  
  
    for (char c : s.toCharArray()) {  
        count.put(c, count.getOrDefault(c, 0) + 1);  
    }  
  
    for (char c : t.toCharArray()) {  
        count.put(c, count.getOrDefault(c, 0) - 1);  
    }  
  
    for (int value : count.values()) {  
        if (value != 0) {  
            return false;  
        }  
    }  
  
    return true;  
}
```

10. Why HashMap works

Suppose:

text

```
s = "aabbc"  
t = "bcaab"
```

First count s :

text

```
a -> 2  
b -> 2  
c -> 1
```

Then process t by decrementing:

text

```
b -> 1
c -> 0
a -> 1
a -> 0
b -> 0
```

Everything ends at zero.

Therefore:

```
text
```

```
true
```

If one frequency differs:

```
text
```

```
a -> 1
```

at the end, then:

```
text
```

```
false
```

11. Complexity of HashMap Approach

We process each character once.

For each character:

```
text
```

```
HashMap lookup/update
≈ O(1) average
```

So:

```
text
```

```
Time: O(n) average
```

Space:

```
text
```

```
O(k)
```

where `k` is the number of distinct characters.

If `k` is considered bounded by the character set, this is effectively $O(1)$.

For arbitrary Unicode text, however, `k` can be large, so the distinction matters.

12. Approach 4 — Fixed Character Frequency Array

Now comes the **best standard interview answer** when the problem states:

```
"The strings contain lowercase English letters."
```

There are only:

```
text
```

```
26
```

possible characters.

So why use a HashMap?

We don't need hashing.

We can directly map:

```
text
```

```
'a' -> index 0  
'b' -> index 1  
...  
'z' -> index 25
```

using:

```
java
```

```
int[] count = new int[26];
```

This is simpler and more efficient.

13. Optimal Java Solution

```
java
```

```

public boolean isAnagram(String s, String t) {

    if (s.length() != t.length()) {
        return false;
    }

    int[] count = new int[26];

    for (char c : s.toCharArray()) {
        count[c - 'a']++;
    }

    for (char c : t.toCharArray()) {
        count[c - 'a']--;
    }

    for (int value : count) {
        if (value != 0) {
            return false;
        }
    }

    return true;
}

```

This is the solution I would normally submit in the standard version of the problem.

14. Why `c - 'a'` works

This is a classic Java interview question.

Characters have underlying numeric values.

Because lowercase English letters are contiguous, we can do:

```
java
```

```
c - 'a'
```

Examples:

```
text
```

```

'a' - 'a' = 0
'b' - 'a' = 1
'c' - 'a' = 2
...

```

```
'z' - 'a' = 25
```

Therefore:

```
java  
  
count[c - 'a']++;
```

maps each character to exactly one array position.

This is why `int[26]` works.

15. Dry Run — Optimal Approach

Take:

```
text  
  
s = "anagram"  
t = "nagaram"
```

Initially:

```
text  
  
count = [0, 0, 0, ...]
```

Process `s`

Characters:

```
text  
  
a  
n  
a  
g  
r  
a  
m
```

Counts become:

```
text  
  
a = 3  
g = 1  
m = 1
```

```
n = 1  
r = 1
```

Then process `t` .

```
t = "nagaram"
```

For each character we decrement.

After processing the entire second string:

```
text
```

```
a = 0  
g = 0  
m = 0  
n = 0  
r = 0
```

Every count is zero.

Therefore:

```
text
```

```
true
```

16. Why can we use one array instead of two?

A common beginner implementation is:

```
text
```

```
countS[26]  
countT[26]
```

Then compare them.

That's valid.

But we can do better.

Instead:

```
text
```

```
s → increment  
t → decrement
```

Then only one array is needed.

Why does that work?

Because we are effectively calculating:

```
text
```

```
frequency(s) - frequency(t)
```

For anagrams:

```
text
```

```
frequency(s) = frequency(t)
```

therefore:

```
text
```

```
frequency(s) - frequency(t) = 0
```

for every character.

This is an elegant optimization.

17. Two arrays vs one array

Two arrays

```
java
```

```
int[] countS = new int[26];  
int[] countT = new int[26];
```

Then compare.

Space:

```
text
```

```
O(26) + O(26)
```

Still effectively $O(1)$.

One array

```
java
```



```
int[] count = new int[26];
```

Increment for `s`, decrement for `t`.

Less memory and slightly simpler comparison.

This is the cleaner optimized implementation.

18. Another optimization — Combine the two loops

Because the strings have equal length, we can process corresponding positions together.

java

```
public boolean isAnagram(String s, String t) {  
  
    if (s.length() != t.length()) {  
        return false;  
    }  
  
    int[] count = new int[26];  
  
    for (int i = 0; i < s.length(); i++) {  
        count[s.charAt(i) - 'a']++;  
        count[t.charAt(i) - 'a']--;  
    }  
  
    for (int value : count) {  
        if (value != 0) {  
            return false;  
        }  
    }  
  
    return true;  
}
```

This is slightly more concise.

However, I would generally choose the earlier two-loop version in an interview because it is extremely easy to read:

text

```
first count s  
then subtract t  
then verify
```

Readable code is more important than shaving a tiny constant factor.

19. Is the combined-loop version more time efficient?

Asymptotically, no.

Both are:

```
text
```

```
O(n)
```

The difference is only a constant factor.

Interviewers generally care far more about:

```
text
```

```
O(n²) → O(n log n) → O(n)
```

than whether you reduced the number of loop constructs from two to one.

So don't try to over-optimize prematurely.

20. Approach comparison

Approach	Time	Space	Pattern
Brute-force matching	$O(n^2)$	$O(n)$	Repeated search
Sorting	$O(n \log n)$	$O(n)$	Sort + compare
HashMap	$O(n)$ avg	$O(k)$	Frequency map
Fixed array	$O(n)$	$O(1)^*$	Frequency counting

* assuming a fixed alphabet such as 26 lowercase English letters.

21. Why fixed array is better than HashMap here

This is an important interviewer discussion.

Suppose the input guarantees:

```
text
```

```
'a' to 'z'
```

We know exactly 26 possibilities.

Using:

```
java  
  
HashMap<Character, Integer>
```

introduces:

```
text  
  
hashing  
boxing  
object/reference overhead  
bucket lookup
```

when we don't need any of that.

Instead:

```
java  
  
int[26]
```

gives direct indexed access.

Conceptually:

```
text  
  
character  
  ↓  
c - 'a'  
  ↓  
direct array index  
  ↓  
count
```

So the array is simpler and has lower overhead.

22. Important interview principle

This is the reusable principle:

When the key domain is small and fixed, an array is often preferable to a HashMap.

Examples:

```
text
```

26 lowercase letters

→ `int[26]`

256 possible byte values

→ `int[256]`

digits 0-9

→ `int[10]`

Whereas:

text

arbitrary strings

arbitrary Unicode code points

unknown IDs

large key space

usually suggests:

text

HashMap

This is much more valuable to remember than memorizing this one problem.

23. Follow-up — What if Unicode is allowed?

Now the `int[26]` approach no longer works.

Why?

Because Unicode is much larger than 26 characters.

You may have:

text

English

Hindi

Chinese

Arabic

emoji

etc.

So instead of:

java

```
int[] count = new int[26];
```

we can use:

```
java
```

```
Map<Character, Integer>
```

or, depending on the exact Unicode requirement, a more appropriate representation of Unicode code points.

Basic Java version:

```
java
```

```
public boolean isAnagram(String s, String t) {  
  
    if (s.length() != t.length()) {  
        return false;  
    }  
  
    Map<Character, Integer> count = new HashMap<>();  
  
    for (char c : s.toCharArray()) {  
        count.put(c, count.getOrDefault(c, 0) + 1);  
    }  
  
    for (char c : t.toCharArray()) {  
        count.put(c, count.getOrDefault(c, 0) - 1);  
    }  
  
    for (int value : count.values()) {  
        if (value != 0) {  
            return false;  
        }  
    }  
  
    return true;  
}
```

But there is an important Java/Unicode nuance.

24. Advanced Java Follow-up — Is `char` enough for Unicode?

A strong interviewer may intentionally ask this.

Java's `char` is a **UTF-16 code unit**, not necessarily a complete Unicode code point.

Some Unicode characters are represented using a **surrogate pair**, meaning one logical code point can occupy two `char` s.

So for full Unicode code-point correctness, you may want:

```
java
String.codePoints()
```

rather than simply:

```
java
toCharArray()
```

For example, conceptually:

```
java
Map<Integer, Integer> count = new HashMap<>();
```

and iterate over code points.

```
java
public boolean isAnagramUnicode(String s, String t) {
    Map<Integer, Integer> count = new HashMap<>();

    s.codePoints().forEach(cp ->
        count.merge(cp, 1, Integer::sum)
    );

    t.codePoints().forEach(cp ->
        count.merge(cp, -1, Integer::sum)
    );

    for (int value : count.values()) {
        if (value != 0) {
            return false;
        }
    }

    return true;
}
```

Important interview nuance

Don't introduce this complexity for a problem explicitly constrained to lowercase English letters.

Say:

"For lowercase English letters, `int[26]` is ideal. For arbitrary Unicode code points, I'd use a map keyed by code point rather than assuming a 26-character alphabet."

That is the right level of detail for a 2–2.5 YOE candidate.

25. `char` vs `codePoint`

This is a valuable Java distinction.

`char`

text

UTF-16 code unit

Code point

text

logical Unicode character/code point

So:

java

`s.charAt(i)`

works naturally for ASCII/lowercase English.

For general Unicode-aware processing:

java

`s.codePoints()`

can be more appropriate.

26. Follow-up — What if case should be ignored?

Suppose:

text

```
s = "Listen"
t = "Silent"
```

Should this be an anagram?

That depends on the requirement.

If comparison should be case-insensitive:

```
text

Listen
silent
```

we need normalization.

At a basic level:

```
java

s = s.toLowerCase();
t = t.toLowerCase();
```

Then compare.

But in production-grade internationalized applications, the interviewer may expect you to mention that **case conversion and Unicode normalization are not always trivial**.

The important concept is:

```
text

normalize
  ↓
count/compare
```

27. Follow-up — What if spaces and punctuation should be ignored?

Example:

```
text

s = "rail safety!"
t = "fairy tales"
```

If spaces and punctuation are ignored, we need preprocessing.

Conceptually:


```
text
```

```
normalize input
```



```
remove/ignore unwanted characters
```



```
frequency count
```

The algorithm remains the same.

This is an important insight:

The core algorithm doesn't change; only the normalization step changes.

28. Follow-up — What if the input contains only digits?

Then instead of:

```
java
```

```
int[26]
```

we can use:

```
java
```

```
int[10]
```

with:

```
java
```

```
c - '0'
```

Again:

```
text
```

```
small fixed alphabet
```



```
array
```

29. Follow-up — What if only lowercase English letters are guaranteed?

Then this is the ideal solution:

```
java
```

```
int[] count = new int[26];
```

Don't unnecessarily use:

```
java
```

```
HashMap
```

A good interviewer may actually expect you to notice this constraint and select the array.

30. Follow-up — Can we use HashSet ?

No, not for the full problem.

A HashSet tells us:

```
text
```

```
"Does this character exist?"
```

But anagram comparison needs:

```
text
```

```
"How many times does this character exist?"
```

For example:

```
text
```

```
s = "aab"  
t = "abb"
```

The sets are identical:

```
text
```

```
{a, b}
```

but the strings are not anagrams.

Frequencies differ:

```
text
```

```
s:  
a -> 2
```

```
b -> 1
```

```
t:
```

```
a -> 1
```

```
b -> 2
```

Therefore we need:

```
text
```

```
frequency counting
```

not merely membership checking.

This is a very common interview trap.

31. HashSet vs frequency array

Contains Duplicate

Question:

```
text
```

```
"Have I seen this?"
```

Use:

```
text
```

```
HashSet
```

Valid Anagram

Question:

```
text
```

```
"How many times have I seen this?"
```

Use:

```
text
```

```
frequency array / HashMap
```

That is an extremely useful pattern distinction.

32. Can we compare frequencies while processing?

Yes.

The optimized one-array technique already does exactly that.

Conceptually:

```
text

for s:
    count++

for t:
    count--
```

At the end:

```
text

all zeros
→ anagram
```

This is effectively checking equality of two frequency vectors.

33. Mathematical way to understand the solution

This is a useful way to explain it to a strong interviewer.

Define:

```
text

F_s(c) = frequency of character c in s
F_t(c) = frequency of character c in t
```

The strings are anagrams iff:

```
text

F_s(c) = F_t(c)
```

for every character c .

Our array stores:

```
text

F_s(c) - F_t(c)
```

Therefore, after processing both strings:

```
text

count[c] == 0
```

for every character iff they are anagrams.

This is why the increment/decrement trick is correct.

34. Early Length Check

This line:

```
java

if (s.length() != t.length()) {
    return false;
}
```

is important.

If the lengths are different, they cannot possibly have the same frequencies.

For example:

```
text

"abc"
"abcd"
```

There is no need to perform any counting.

This is a cheap early rejection.

Complexity impact

It doesn't change the Big-O complexity.

It improves practical performance by avoiding unnecessary work in an obvious failure case.

35. Edge Cases

Empty strings

```
text
```

```
s = ""  
t = ""
```

They are anagrams.

The method returns:

```
text
```

```
true
```

One empty, one non-empty

```
text
```

```
s = ""  
t = "a"
```

Lengths differ:

```
text
```

```
false
```

Same character repeated

```
text
```

```
s = "aaaa"  
t = "aaaa"
```

Returns:

```
text
```

```
true
```

Same characters, different frequencies

```
text
```

```
s = "aabb"  
t = "abbb"
```

Returns:

text

false

Case sensitivity

text

```
s = "abc"
t = "ABC"
```

By default:

text

false

unless the requirements specify case-insensitive comparison.

Spaces

text

```
s = "ab c"
t = "c a b"
```

Whether these are valid anagrams depends on the problem's normalization rules.

Never assume spaces should automatically be ignored.

36. Common Mistake – Using `String.contains()`

A candidate might try:

java

```
for (char c : s.toCharArray()) {
    if (!t.contains(String.valueOf(c))) {
        return false;
    }
}
```

This is wrong because it checks **existence**, not frequency.

Example:

```
text

s = "aab"
t = "abb"
```

Each character from `s` appears in `t` , but the frequencies are different.

So:

```
text

contains()
```

is insufficient.

37. Common Mistake – Using a Set

Same problem:

```
text

new HashSet<>(characters)
```

loses frequency information.

This is a core concept:

```
text

Anagram
≠
same distinct characters
```

Instead:

```
text

Anagram
=
same frequency for every character
```

38. Common Mistake – Forgetting constraints

If the problem specifically says:

```
text
```

```
lowercase English letters
```

and you use:

```
java
```

```
HashMap<Character, Integer>
```

your solution may still be correct, but it is not exploiting the strongest constraint.

A stronger interview answer is:

"Because the alphabet is fixed at 26 characters, I can replace the HashMap with a fixed-size array."

That demonstrates optimization.

39. Java-Specific Follow-up — `getOrDefault`

With HashMap:

```
java
```

```
count.put(c, count.getOrDefault(c, 0) + 1);
```

means:

```
text
```

```
if c exists:
    get its current count
else:
    use 0
then add 1
```

For example:

```
text
```

```
count = {a=2}
```

For `a` :

```
text
```

```
getOrDefault('a', 0)
→ 2
```

then:

```
text
```

```
2 + 1
→ 3
```

For `b` :

```
text
```

```
getOrDefault('b', 0)
→ 0
```

then:

```
text
```

```
0 + 1
→ 1
```

40. Java-Specific Follow-up — `merge()`

A more Java-idiomatic alternative is:

```
java

count.merge(c, 1, Integer::sum);
```

For decrement:

```
java

count.merge(c, -1, Integer::sum);
```

This is compact, but for an interview I would still prefer:

```
java

getOrDefault()
```

or the fixed array because they are easier to explain.

For a 2–2.5 YOE interview, **clarity beats showing off API knowledge**.

41. Java-Specific Follow-up — Why not `Arrays.sort()` on `String`?

You cannot directly sort a `String` because Java `String` is immutable.

You need:

```
java

char[] chars = s.toCharArray();
Arrays.sort(chars);
```

This is another small Java detail that interviewers may test.

42. Production Example — Case-Insensitive Token Normalization

Suppose a backend receives text tokens where:

```
text

"ABC"
"abc"
"AbC"
```

should be treated as equivalent.

A reasonable processing pipeline is:

```
text

Raw input
  ↓
normalize case
  ↓
normalize required format
  ↓
frequency / comparison logic
```

For example, if comparing two normalized token strings:

```
text

"Listen"
"Silent"
```

a case-insensitive normalization step can make them equivalent.

But in real production systems, the exact normalization policy matters.

For international text, simply calling:

```
java  
  
toLowerCase()
```

without considering locale and Unicode normalization may not be sufficient for every requirement.

The interview point is:

"Separate normalization from the actual comparison algorithm."

That's good backend engineering.

43. Production Example – Duplicate Content Detection

Suppose a service receives documents and wants to detect whether two pieces of content have the same character multiset.

For small normalized strings:

```
text  
  
normalize  
  ↓  
frequency representation  
  ↓  
compare
```

could work.

However, for large files, building a frequency structure for the entire file may not be the most practical production design.

You might instead consider:

```
text  
  
canonicalization  
+  
hashing
```

depending on the exact requirement.

Important distinction:

```
text
```

```
Anagram algorithm
≠
general duplicate-file detection
```

For exact content equality, cryptographic/content hashes are usually more appropriate than anagram frequency comparison because order matters for normal duplicate content.

44. Important Production Distinction

Suppose:

```
text

file A = "abc"
file B = "cba"
```

Anagram logic says:

```
text

same frequency
→ true
```

But duplicate-content logic normally says:

```
text

contents differ in order
→ not identical
```

So before selecting an algorithm in production, clarify what "duplicate" actually means.

This is exactly the kind of requirement-awareness interviewers like.

45. Advanced Follow-up — Can hashing itself be used?

A candidate may suggest calculating:

```
text

sum of characters
```

or some hash value.

For example:

```
text
"abc"
"bca"
```

can have the same sum.

But a simple sum isn't enough:

```
text

"abb"
"abc"
```

could produce collisions under certain simplistic hash functions.

You could construct more sophisticated fingerprints, but exact anagram checking still needs collision-free frequency information unless you accept probabilistic behavior.

Therefore:

```
text

frequency array
```

is the clean exact solution.

46. Advanced Interview Question — Can you do it in $O(1)$ extra space?

This question needs a careful answer.

If the character alphabet is fixed:

```
text

int[26]
```

is technically:

```
text

 $O(1)$ 
```

extra space because 26 is a constant.

So the optimal solution is:

```
text
```

```
Time:  $O(n)$   
Space:  $O(1)$ 
```

under the lowercase-English-letter constraint.

This is an important distinction from the HashMap solution.

47. Why HashMap is $O(k)$, not automatically $O(n)$

Suppose:

```
text
```

```
s.length() = 1,000,000
```

but only:

```
text
```

```
a  
b  
c
```

appear.

The HashMap may contain only:

```
text
```

```
3 keys
```

So space depends on the number of distinct characters:

```
text
```

```
 $O(k)$ 
```

where:

```
text
```

```
k = distinct characters
```

If the character domain is fixed to 26:

```
text
```

$k \leq 26$

so effectively:

text

$O(1)$

This is a useful precision point.

48. Interview Pattern Recognition

The primary pattern is:

Frequency Counting

When you hear:

text

same characters
same frequency
count occurrences
duplicates by frequency
character frequency

think:

text

Frequency Map / Frequency Array

49. Secondary Pattern — Fixed Domain Mapping

When constraints say:

text

lowercase English letters
digits
small fixed range
ASCII

think:

text

Fixed-size array

instead of:

text

HashMap

50. Pattern Comparison with Previous Questions

Q2 — Contains Duplicate

Question:

text

"Have I seen this value?"

Pattern:

text

HashSet

Q3 — Valid Anagram

Question:

text

"How many times has this character appeared?"

Pattern:

text

Frequency counting

Q1 — Two Sum

Question:

text

"Have I seen the complement, and where?"

Pattern:

text

HashMap

Notice the progression:

text

Existence



HashSet

Existence + associated data



HashMap

Frequency



Frequency array / HashMap

This is how you should start recognizing DSA patterns instead of memorizing individual problems.

51. What if interviewer asks: "Which is better, HashMap or array?"

Answer:

"If the character set is fixed and small, such as lowercase English letters, I'd use an `int[26]` because it gives direct indexing, avoids hashing overhead, and uses constant auxiliary space. If the character domain is not fixed, I'd use a `HashMap`."

That is probably the best concise answer.

52. What if interviewer asks: "Why not use two frequency arrays?"

Answer:

"Two arrays are simpler conceptually and both are $O(1)$ for a 26-character alphabet, but we only need one array. I can increment counts for the first string and decrement them for the second. Then every count must be zero if the strings are anagrams."

That demonstrates optimization without overcomplicating things.

53. What if interviewer asks: "Why length check first?"

Answer:

"Anagrams must have the same total number of characters. If lengths differ, the answer is immediately false, so this avoids unnecessary counting."

54. What if interviewer asks: "Why can't sorting be $O(n)$?"

Because general comparison-based sorting is typically:

text

$O(n \log n)$

For this problem, counting is better because the domain is constrained.

This is another example of:

text

Don't sort when you can count.

when the key space is small.

55. What if interviewer asks: "Could counting be $O(n + k)$?"

Yes.

A more precise complexity formulation is:

text

$O(n + k)$

where:

text

n = string length
 k = alphabet/domain size

For lowercase English:

text

$k = 26$

so:

text

```
0(n + 26)
≈ 0(n)
```

This is technically more precise.

56. What if interviewer asks: "Why is array faster than HashMap?"

A good Java-focused answer:

"The array provides direct indexed access, so there is no hashing, bucket lookup, key object overhead, or map bookkeeping. Since the alphabet is known and small, the array is both simpler and more memory-efficient."

That is a very good 2–2.5 YOE answer.

57. Common Interview Traps

Trap 1 — Compare lengths only

```
text
```

```
s.length() == t.length()
```

does **not** prove anagram.

Example:

```
text
```

```
"abc"
"abd"
```

Same length, not anagrams.

Trap 2 — Compare distinct characters only

```
text
```

```
"abb"
"abc"
```

both have:

```
text
```

a
b

as some common characters, but frequencies differ.

Trap 3 — Use HashSet

Again, Set loses count information.

Trap 4 — Sort without understanding complexity

Sorting is valid, but:

text

$O(n \log n)$

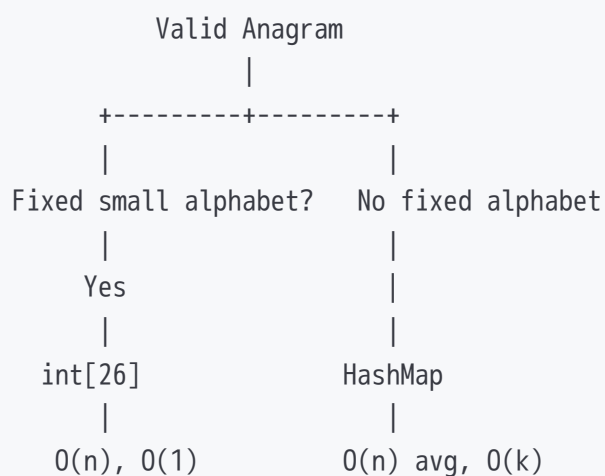
not $O(n)$.

Trap 5 — Use `int[26]` without checking the constraints

If uppercase, punctuation, Unicode, or arbitrary characters are allowed, the fixed array needs to be redesigned.

58. Best Solution Decision Tree

text



If sorting is explicitly acceptable and simplicity is more important:

text

Sort + compare
 $O(n \log n)$

But for the standard constrained problem, the array solution is the best.

59. What I would code in the interview

For the common lowercase-English version:

```
java

public boolean isAnagram(String s, String t) {

    if (s.length() != t.length()) {
        return false;
    }

    int[] count = new int[26];

    for (char c : s.toCharArray()) {
        count[c - 'a']++;
    }

    for (char c : t.toCharArray()) {
        count[c - 'a']--;
    }

    for (int value : count) {
        if (value != 0) {
            return false;
        }
    }

    return true;
}
```

60. Final Interviewer-Expected Answer

If the interviewer asks:

"Give me your optimal approach."

A strong 2–2.5 YOE response would be:

"The brute-force approach tries to match each character with an unused character in the other string, which is $O(n^2)$."

A better approach is to sort both strings and compare them, which is $O(n \log n)$.

The problem is fundamentally about comparing character frequencies, so I can count frequencies instead. If the strings contain only lowercase English letters, I'll use an `int[26]` rather than a `HashMap` because the character domain is fixed. I'll increment the count for every character in `s` and decrement it for every character in `t`. At the end, every count must be zero for the strings to be anagrams.

This gives $O(n)$ time and $O(1)$ auxiliary space because the frequency array always has only 26 entries."

Then write:

```
java

public boolean isAnagram(String s, String t) {

    if (s.length() != t.length()) {
        return false;
    }

    int[] count = new int[26];

    for (char c : s.toCharArray()) {
        count[c - 'a']++;
    }

    for (char c : t.toCharArray()) {
        count[c - 'a']--;
    }

    for (int x : count) {
        if (x != 0) {
            return false;
        }
    }

    return true;
}
```

61. Final Cheat Sheet

Approach 1 — Brute Force

text

Match characters one by one

Time: $O(n^2)$

Space: $O(n)$

Approach 2 – Sorting

text

Sort both strings

Time: $O(n \log n)$

Space: $O(n)$ for character arrays

Approach 3 – HashMap

text

Character $\rightarrow$ frequency

Time: $O(n)$ average

Space: $O(k)$

Approach 4 – Optimal constrained solution

text

int[26]

increment for s

decrement for t

Time: $O(n)$

Space: $O(1)$

Pattern

text

Anagram

↓

Frequency Counting

↓

Fixed alphabet?

|—— Yes $\rightarrow$ Array

|—— No $\rightarrow$ HashMap

Key interview insight

text

Q2:

"Have I seen this?"

→ HashSet

Q3:

"How many times have I seen this?"

→ Frequency Array / HashMap

Q1:

"Have I seen the complement, and where?"

→ HashMap

The **pattern recognition** is more important than memorizing the code. Once you see "same characters with same frequencies," you should immediately think **frequency counting**, then inspect the constraints to decide between a **fixed array** and a **HashMap**.

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ok as aboved continue for next question, keep this same interview structure: ****brute force → better → optimal → pattern recognition → why the data structure works → dry run → edge cases → Java-specific follow-ups → production variation → final interviewer-ready answer**** .

Q4. Group Anagrams

Question

Group strings that are anagrams.

Example:

text

eat

tea

ate

bat

=> [[eat,tea,ate],[bat]]

Approach

Create a canonical key.

Option 1:

text

sort characters

Example:

text

eat → aet

tea → aet

ate → aet

Java

java

```
public List<List<String>> groupAnagrams(String[] strs) {  
    Map<String, List<String>> map = new HashMap<>();
```

```
    for (String s : strs) {  
        char[] chars = s.toCharArray();  
        Arrays.sort(chars);
```

8

```
        String key = new String(chars);
```

```
        map.computeIfAbsent(key, k -> new ArrayList<>())  
            .add(s);  
    }
```

```
    return new ArrayList<>(map.values());  
}
```

Complexity

text

$O(n * k \log k)$

where k is average string length.

Follow-up

Can you do better?

Yes, frequency-based keys can reduce character sorting cost.

Production example

Grouping equivalent product-search keywords.

 ChatGPT gpt-5-6-t-mini

Q4. Group Anagrams

This is an important progression from **Q3 – Valid Anagram**.

In Q3, we asked:

"Are these two strings anagrams?"

Here we ask:

"Which strings belong to the same anagram group?"

The key idea is still **frequency/canonical representation**, but now we need to build a grouping structure.

For a 2–2.5 year Java candidate, I would expect you to recognize that the problem is essentially:

text

```
Many strings
  ↓
Convert each string into a representation
  ↓
Anagrams produce the same representation
  ↓
Use that representation as a HashMap key
  ↓
Group strings having the same key
```

1. Problem

Given an array of strings, group all strings that are anagrams.

Example

text

```
strs = ["eat", "tea", "tan", "ate", "nat", "bat"]
```

Expected result can be:

text

```
[
  ["eat", "tea", "ate"],
  ["tan", "nat"],
  ["bat"]
]
```

The order of groups generally does not matter unless the interviewer explicitly specifies an ordering requirement.

2. What I would say first in the interview

A strong opening:

"The key challenge is deciding how to identify that two strings belong to the same anagram group. Since anagrams have the same character composition, I can create a canonical representation for every string. The simplest representation is the sorted characters, such as `eat -> aet` and `tea -> aet`. Then I can use that canonical representation as a HashMap key and append the original string to the corresponding list. That gives $O(n * k \log k)$, where n is the number of strings and k is the average string length. If the character set is fixed, I can improve this to $O(n * k)$ using a frequency-based key instead of sorting."

That is already a very interview-strong answer because you're showing the interviewer **how you move from identification → key → grouping → optimization**.

3. Approach 1 — Brute Force

Let's start from the basic idea.

We could compare each string against previously created groups.

For every incoming string:

1. Compare it with representatives of existing groups.
2. Check whether the two strings are anagrams.
3. If yes, add it to that group.
4. Otherwise create a new group.

For the anagram check, we can use the frequency-counting technique from Q3.

4. Brute Force Concept

Suppose:

```
text
```

```
["eat", "tea", "bat", "ate"]
```

Start:

```
text
```

```
groups = []
```

Process `"eat"` :

```
text
```

```
groups = [  
    "eat"  
]
```

Process "tea" :

Compare with "eat" .

They are anagrams.

text

```
groups = [  
    "eat", "tea"  
]
```

Process "bat" :

Not an anagram of "eat" .

Create another group:

text

```
[  
    ["eat", "tea"],  
    ["bat"]  
]
```

Process "ate" :

Compare with "eat" .

Anagram → add.

text

```
[  
    ["eat", "tea", "ate"],  
    ["bat"]  
]
```

5. Why brute force is expensive

Suppose there are n strings and average length is k .

For each string, we may compare it against many existing groups.

And every comparison itself requires $O(k)$ work for frequency counting.

So in the worst case, this can approach:

```
text
```

```
O(n² * k)
```

time.

There are also implementation complications around maintaining representatives and checking every group.

The important interview point is:

"This works, but I'm repeatedly asking whether strings are anagrams. We can avoid repeated comparisons by generating a representation that directly identifies the group."

That is the key optimization.

6. Approach 2 — Sort Each String and Use as Key

This is the standard improvement.

Core idea

Anagrams produce the same sorted string.

Example:

```
text
```

```
eat
```

```
↓
```

```
aet
```

```
text
```

```
tea
```

```
↓
```

```
aet
```

```
text
```

```
ate
```

```
↓
```

```
aet
```

But:

```
text
```

```
bat
```

```
↓
```

```
abt
```

So:

```
text
```

```
aet → group 1
```

```
abt → group 2
```

This gives us a canonical key.

7. What is a Canonical Key?

This is a very important concept.

A canonical representation is a standardized form such that:

```
text
```

```
same logical equivalence
```

```
↓
```

```
same representation
```

For anagrams:

```
text
```

```
"eat" → "aet"
```

```
"tea" → "aet"
```

```
"ate" → "aet"
```

So all three map to:

```
text
```

```
"aet"
```

That means we can use:

```
text
```

```
Map<CanonicalKey, List<String>>
```

This is the core design.

8. Java Solution — Sorting Key

```
java

public List<List<String>> groupAnagrams(String[] strs) {

    Map<String, List<String>> map = new HashMap<>();

    for (String s : strs) {

        char[] chars = s.toCharArray();
        Arrays.sort(chars);

        String key = new String(chars);

        map.computeIfAbsent(key, k -> new ArrayList<>())
            .add(s);
    }

    return new ArrayList<>(map.values());
}
```

This is the standard answer most interviewers expect initially.

9. Dry Run

Take:

```
text
```

```
strs = ["eat", "tea", "tan", "ate", "nat", "bat"]
```

Initially:

```
text
```

```
map = {}
```

String = "eat"

Sort:

```
text
```

```
eat → aet
```


Key:

```
text
```

```
"aet"
```

Map:

```
text
```

```
{  
  "aet" : ["eat"]  
}
```

String = "tea"

Sort:

```
text
```

```
tea → aet
```

Same key:

```
text
```

```
"aet"
```

Append:

```
text
```

```
{  
  "aet" : ["eat", "tea"]  
}
```

String = "tan"

Sort:

```
text
```

```
tan → ant
```

New key:

```
text
```

```
{
  "aet" : ["eat", "tea"],
  "ant" : ["tan"]
}
```

String = "ate"

Sort:

text

ate → aet

Append:

text

```
{
  "aet" : ["eat", "tea", "ate"],
  "ant" : ["tan"]
}
```

String = "nat"

Sort:

text

nat → ant

Append:

text

```
{
  "aet" : ["eat", "tea", "ate"],
  "ant" : ["tan", "nat"]
}
```

String = "bat"

Sort:

text

bat → abt

New group:

text

```
{
  "aet" : ["eat", "tea", "ate"],
  "ant" : ["tan", "nat"],
  "abt" : ["bat"]
}
```

Finally:

text

```
[
  ["eat", "tea", "ate"],
  ["tan", "nat"],
  ["bat"]
]
```

10. Why does this work?

The correctness comes from an important property:

Two strings are anagrams if and only if their sorted character sequences are identical.

For example:

text

eat

contains:

text

a, e, t

and:

text

tea

also contains:

```
text
```

```
a, e, t
```

After sorting, both become:

```
text
```

```
aet
```

If the character frequencies differ, the sorted representations differ.

Example:

```
text
```

```
aab
```

```
abb
```

Sorted:

```
text
```

```
aab
```

```
abb
```

Different keys.

Therefore they go into different groups.

11. Complexity

Let:

```
text
```

```
n = number of strings
```

```
k = average string length
```

For each string:

```
text
```

```
toCharArray() →  $O(k)$ 
```

```
sort() →  $O(k \log k)$ 
```

```
new String() →  $O(k)$ 
```

```
HashMap lookup →  $O(1)$  average
```

Therefore:

```
text
```

```
O(k log k)
```

per string.

For `n` strings:

```
text
```

```
O(n * k log k)
```

Space

The map stores:

```
text
```

```
all input strings  
+  
canonical keys
```

The result itself is also stored.

At the auxiliary-data-structure level, we generally describe it as:

```
text
```

```
O(n * k)
```

depending on whether the output is included in the accounting.

For interview purposes:

```
text
```

```
Time: O(n * k log k)  
Space: O(n * k)
```

is a reasonable answer.

12. Why `computeIfAbsent()` ?

This line:

```
java
```

```
map.computeIfAbsent(key, k -> new ArrayList<>())  
    .add(s);
```

is very important Java knowledge.

It means:

"If this key doesn't have a list, create one. Otherwise return the existing list. Then add the current string."

Equivalent longer code:

```
java  
  
if (!map.containsKey(key)) {  
    map.put(key, new ArrayList<>());  
}  
  
map.get(key).add(s);
```

`computeIfAbsent()` makes the grouping logic concise and idiomatic.

13. Java Follow-up — Why `new String(chars)` ?

After:

```
java  
  
char[] chars = s.toCharArray();  
Arrays.sort(chars);
```

we have:

text

char[]

but our `HashMap` key is:

text

String

So:

```
java
```

```
String key = new String(chars);
```

creates the canonical sorted string.

We could also use other representations, but this is simple and readable.

14. Approach 3 — Frequency-Based Key

Now comes the better approach.

The interviewer asks:

"Can you do better than sorting?"

Yes.

From Q3, we already know that anagrams have identical character frequencies.

For lowercase English letters, we can build:

```
text
```

```
count[26]
```

For "eat" :

```
text
```

```
a -> 1
```

```
e -> 1
```

```
t -> 1
```

For "tea" :

```
text
```

```
a -> 1
```

```
e -> 1
```

```
t -> 1
```

They produce the same frequency representation.

So instead of:

```
text
```

```
sort → key
```

we can do:

```
text
```

```
count characters → key
```

This removes the sorting cost.

15. Frequency Key Example

For:

```
text
```

```
"eat"
```

frequency array conceptually:

```
text
```

```
a = 1  
b = 0  
c = 0  
d = 0  
e = 1  
...  
t = 1  
...  
z = 0
```

For:

```
text
```

```
"tea"
```

the same array is produced.

Therefore they belong to the same group.

16. How do we turn an array into a HashMap key?

Java arrays do not have value-based equality/hash behavior suitable for directly using:

```
java
```

```
int[]
```

as a `HashMap` key in the naive way.

This is an important Java-specific issue.

Don't do:

```
java

Map<int[], List<String>> map = new HashMap<>();
```

and assume two equal-content arrays will match.

Two separate arrays are different objects.

Instead, we need a canonical immutable representation.

Several options exist.

17. Frequency Key Option 1 — String Builder

We can encode:

```
text

character + count
```

into a String.

For example:

```
text

a1#b0#c0#...#e1#...#t1
```

The separators are important to avoid ambiguity.

Java

```
java

public List<List<String>> groupAnagrams(String[] strs) {

    Map<String, List<String>> map = new HashMap<>();

    for (String s : strs) {

        int[] count = new int[26];

        for (char c : s.toCharArray()) {
            count[c - 'a']++;
        }
    }
}
```

```

    StringBuilder keyBuilder = new StringBuilder();

    for (int i = 0; i < 26; i++) {
        keyBuilder.append('#');
        keyBuilder.append(count[i]);
    }

    String key = keyBuilder.toString();

    map.computeIfAbsent(key, k -> new ArrayList<>())
        .add(s);
}

return new ArrayList<>(map.values());
}

```

18. Why use separators?

This is a subtle but valuable interview detail.

Suppose you concatenate counts without separators.

For example:

text

[1, 23]

could produce:

text

"123"

while:

text

[12, 3]

could also produce:

text

"123"

That creates an ambiguous key.

Using separators:

text

"1#23"

versus:

text

"12#3"

keeps the representation unambiguous.

This is exactly the type of detail a good interviewer may appreciate.

19. Frequency-Key Complexity

For every string of length k :

Counting:

text

$O(k)$

Building a fixed 26-entry key:

text

$O(26)$

Therefore:

text

$O(k + 26)$

which simplifies to:

text

$O(k)$

For n strings:

text

```
0(n * k)
```

This improves over:

```
text
```

```
0(n * k log k)
```

from the sorting solution.

20. Is Frequency Key always better?

No.

This is an important interview nuance.

For the restricted problem:

```
text
```

```
lowercase English letters
```

yes, the frequency key is asymptotically better.

But the sorting solution is:

- simpler
- easy to understand
- easy to debug
- generalizable to larger character domains

So a strong candidate can say:

"I'd start with sorting because it's straightforward and has a clean canonical representation. If the interviewer wants better asymptotic performance and the character domain is fixed, I'd switch to a frequency-based key."

That demonstrates engineering judgment.

21. Approach 4 — Frequency Key with an Immutable List

Another approach is to convert the frequency array into something with value-based equality.

For example:

```
java
```

```
List<Integer> key = Arrays.stream(count)
    .boxed()
    .toList();
```

Then:

```
java
```

```
Map<List<Integer>, List<String>> map = new HashMap<>();
```

Because `List` implements value-based `equals()` and `hashCode()` .

But this creates boxing and additional objects.

So it is usually less efficient than building a compact String key.

This is more of an alternative technique than the preferred solution.

22. Could we use `Arrays.toString(count)` ?

Yes.

For example:

```
java
```

```
String key = Arrays.toString(count);
```

This provides a canonical representation such as:

```
text
```

```
[1, 0, 0, 0, 1, ...]
```

Then:

```
java
```

```
map.computeIfAbsent(key, k -> new ArrayList<>())
    .add(s);
```

This is correct.

But it involves formatting the whole array into a String, so:

```
text
```

```
StringBuilder
```

is often a cleaner demonstration of intentionally constructing the key.

In an interview, `Arrays.toString(count)` is perfectly acceptable if you explain the complexity.

23. Very Important – Why not use `int[]` directly as the key?

This is a classic Java trap.

Consider:

```
java

int[] a = {1, 2, 3};
int[] b = {1, 2, 3};
```

Although:

```
text

a and b
```

contain the same values, they are different array objects.

`HashMap` uses:

```
text

equals()
hashCode()
```

for key comparison.

Arrays inherit identity-based `equals()` / `hashCode()` behavior rather than content-based behavior.

So:

```
java

a.equals(b)
```

is false.

For array content comparison, Java provides:

```
java

Arrays.equals(a, b)
Arrays.hashCode(a)
```

but those aren't automatically used by a `HashMap` when the array itself is the key.

This is why we convert the frequency array into a value-based key such as a `String` or immutable `List`.

That is an excellent Java interview point.

24. Best Standard Solution

If the interviewer says:

"Give me the optimal solution assuming lowercase English letters."

I would use:

text

frequency count
+
canonical frequency key
+
`HashMap`

Java

java

```
public List<List<String>> groupAnagrams(String[] strs) {  
  
    Map<String, List<String>> map = new HashMap<>();  
  
    for (String s : strs) {  
  
        int[] count = new int[26];  
  
        for (char c : s.toCharArray()) {  
            count[c - 'a']++;  
        }  
  
        StringBuilder keyBuilder = new StringBuilder();  
  
        for (int i = 0; i < 26; i++) {  
            keyBuilder.append('#')  
                .append(count[i]);  
        }  
  
        String key = keyBuilder.toString();  
  
        map.computeIfAbsent(key, k -> new ArrayList<>())
```

```

        .add(s);
    }

    return new ArrayList<>(map.values());
}

```

25. Complexity of Optimal Solution

Let:

text

n = number of strings
k = average string length

For each string:

text

count characters = $O(k)$
build 26-count key = $O(26)$

So:

text

$O(k + 26)$
=
 $O(k)$

Across all strings:

text

Time = $O(n * k)$

Space:

text

$O(n * k)$

for stored input/output/group information, with the frequency array itself being only $O(1)$ per string.

If discussing auxiliary space excluding the output, the exact accounting depends on how keys and grouping storage are treated, but the main improvement is clearly:


```
text
O(n * k log k)
  ↓
O(n * k)
```

26. Pattern Recognition

This problem has several related patterns.

Pattern 1 — Canonical Representation

Whenever multiple inputs are logically equivalent, ask:

"Can I convert each input to a common canonical form?"

For anagrams:

```
text

eat → aet
tea → aet
ate → aet
```

Then:

```
text

canonical key
  ↓
HashMap grouping
```

This is a very powerful general pattern.

27. Pattern 2 — Group By Key

The broader pattern is:

```
text

Input
  ↓
derive key
  ↓
Map<Key, List<Value>>
```

Examples:

```
text
```

```
Group anagrams  
→ sorted/frequency key
```

```
Group employees by department  
→ department ID
```

```
Group words by length  
→ length
```

```
Group transactions by date  
→ date
```

So the reusable structure is:

```
java
```

```
Map<Key, List<Value>>
```

This is one of the most important patterns in backend Java programming as well.

28. Pattern 3 – Hashing for Equivalence Classes

Conceptually, all anagrams belong to the same equivalence class.

For example:

```
text
```

```
eat ~ tea ~ ate
```

because they have the same character multiset.

We represent that equivalence class using:

```
text
```

```
"aet"
```

Then all objects with the same key automatically get grouped.

This is the deeper idea behind the algorithm.

29. Why HashMap Works Here

We don't want to compare each string with every other string.

Instead:

```
text

string
  ↓
canonical key
  ↓
HashMap lookup
  ↓
existing group?
```

So instead of potentially doing:

```
text

string vs string
string vs string
string vs string
...
```

we perform approximately:

```
text

key → group
```

with expected $O(1)$ HashMap lookup.

This is a huge reduction in repeated comparison.

30. Dry Run — Frequency-Based Optimal Approach

Let's run:

```
text

["eat", "tea", "tan", "ate", "nat", "bat"]
```

"eat"

Counts:

```
text

a = 1
e = 1
```

t = 1

Key:

text

#1#0#0#0#1#...#1...

Map:

text

key1 → ["eat"]

"tea"

Counts:

text

a = 1

e = 1

t = 1

Same key:

text

key1

Map:

text

key1 → ["eat", "tea"]

"tan"

Counts:

text

a = 1

n = 1

t = 1

Different key:

```
text
```

```
key2
```

Map:

```
text
```

```
key1 → ["eat", "tea"]
```

```
key2 → ["tan"]
```

"ate"

Counts:

```
text
```

```
a = 1
```

```
e = 1
```

```
t = 1
```

Again:

```
text
```

```
key1
```

Map:

```
text
```

```
key1 → ["eat", "tea", "ate"]
```

```
key2 → ["tan"]
```

And so on.

Final result:

```
text
```

```
[  
  ["eat", "tea", "ate"],  
  ["tan", "nat"],  
  ["bat"]  
]
```

31. Edge Case — Empty String

Suppose:

```
text
strs = ["", "a", ""]
```

For:

```
text
""
```

the frequency key contains all zeros.

Therefore both empty strings receive the same key.

Result:

```
text
[
  ["", ""],
  ["a"]
]
```

The algorithm handles this naturally.

32. Edge Case — One String

```
text
["hello"]
```

Result:

```
text
[["hello"]]
```

Nothing special is required.

33. Edge Case — All Strings Identical

```
text
```

```
["abc", "abc", "abc"]
```

They all produce the same key:

```
text
```

```
abc
```

and therefore:

```
text
```

```
[["abc", "abc", "abc"]]
```

34. Edge Case — No Two Strings Are Anagrams

```
text
```

```
["abc", "def", "ghi"]
```

Each produces a different key:

```
text
```

```
abc  
def  
ghi
```

So each gets its own group.

35. Edge Case — Duplicate Identical Strings

This is worth distinguishing from anagrams.

```
text
```

```
["eat", "eat"]
```

They obviously have the same key.

Output:

```
text
```

```
[["eat", "eat"]]
```

The requirement usually says to preserve all input strings, so we do.

36. Edge Case — Case Sensitivity

Suppose:

```
text
```

```
["Eat", "tea"]
```

If comparison is case-sensitive:

```
text
```

```
E != e
```

They are not anagrams.

If the requirement says case-insensitive grouping, normalize first:

```
text
```

```
toLowerCase()
```

before generating the key.

Again:

```
text
```

```
normalization
```

```
↓
```

```
canonicalization
```

```
↓
```

```
grouping
```

37. Edge Case — Unicode

If arbitrary Unicode is allowed, this implementation:

```
java
```

```
count[c - 'a']
```


is invalid because it assumes:

```
text
```

```
a-z
```

A frequency map is more appropriate:

```
java
```

```
Map<Integer, Integer>
```

when dealing with Unicode code points.

But there is another question:

What exactly does "character" mean?

In Java, `char` represents a UTF-16 code unit, while some Unicode code points require surrogate pairs.

So for complete Unicode-aware grouping, consider code points via:

```
java
```

```
String.codePoints()
```

rather than blindly using:

```
java
```

```
charAt()
```

This is an advanced but very good follow-up.

38. Java Follow-up — `computeIfAbsent`

Interviewer:

"What does `computeIfAbsent()` do?"

Answer:

"It returns the existing value for the key if present. If the key is absent, it computes and stores a new value using the supplied function, and returns it."

Here:

```
java
```

```
map.computeIfAbsent(key, k -> new ArrayList<>())  
    .add(s);
```

means:

text

```
key exists?  
  ↓ yes  
use existing list  
  ↓  
add s  
  
key exists?  
  ↓ no  
create ArrayList  
  ↓  
store it  
  ↓  
add s
```

39. Java Follow-up – Why `HashMap<String, List<String>>` ?

Because we're solving:

text

```
canonical key  
  ↓  
all original strings belonging to it
```

Therefore:

text

```
Map<Key, List<String>>
```

is exactly the data structure matching the problem.

This is more than a coding trick.

It's a direct representation of the requirement.

40. Java Follow-up – Why not `HashSet<List<String>>` ?

Because we need:

```
text
```

```
key → collection of strings
```

not just a set of groups.

A HashSet doesn't naturally provide lookup by canonical key.

HashMap does:

```
text
```

```
Map<Key, List<String>>
```

41. Java Follow-up — Could the key be a custom class?

Absolutely.

We could define something like:

```
java
```

```
record AnagramKey(int[] counts) {}
```

But there is a catch.

An array inside the record does not automatically get value-based equality/hash semantics that compare array contents the way we need.

We would need to implement appropriate `equals()` and `hashCode()` using:

```
java
```

```
Arrays.equals()  
Arrays.hashCode()
```

For example, conceptually:

```
java
```

```
final class AnagramKey {  
  
    private final int[] counts;  
  
    AnagramKey(int[] counts) {  
        this.counts = counts.clone();  
    }  
}
```

```

@Override
public boolean equals(Object o) {
    if (this == o) return true;
    if (!(o instanceof AnagramKey other)) return false;

    return Arrays.equals(counts, other.counts);
}

@Override
public int hashCode() {
    return Arrays.hashCode(counts);
}
}

```

This works, but it is unnecessarily complicated for a standard interview.

A String key is simpler.

42. Important Java Question – Why clone the array in a key?

Suppose a mutable array is used inside a hash key.

If we later modify the array:

```

text

hashCode changes

```

after it has already been inserted into the map.

That can break HashMap lookup behavior.

So if using an array-backed custom key, you'd want immutability or a defensive copy.

This is a valuable general Java principle:

HashMap keys should be effectively immutable while they are stored in the map.

For our String key, this is naturally satisfied because String is immutable.

That's another reason String makes a very convenient canonical key.

43. Sorting vs Frequency Key

This is the key comparison an interviewer may expect.

Approach	Time	Space	Main Advantage
Compare each group	$O(n^2 \cdot k)$ worst-case	$O(n \cdot k)$	Basic

Approach	Time	Space	Main Advantage
Sort each string	$O(n \cdot k \log k)$	$O(n \cdot k)$	Simple canonical key
Frequency key	$O(n \cdot k)$	$O(n \cdot k)$	Best for fixed alphabet

44. When Would I Choose Sorting?

Use sorting when:

```
text
```

```
character domain is not tightly constrained
```

or:

```
text
```

```
simplicity is more important
```

For example:

```
text
```

```
word length = small
```

and the interviewer wants a quick clean implementation.

Sorting gives a beautifully obvious key:

```
text
```

```
eat → aet
tea → aet
```

It's difficult to misunderstand.

45. When Would I Choose Frequency Counting?

Use frequency counting when:

```
text
```

```
alphabet is small and fixed
```

such as:

text

a-z

Then:

text

sort

does unnecessary work.

Instead of:

text

$k \log k$

we just count:

text

k

This is the optimized answer.

46. Which one would I actually write first?

For a real interview, I would likely do this:

"I'll first implement the sorted-key approach because it gives a straightforward canonical representation and is $O(n \cdot k \log k)$. If the interviewer wants the best asymptotic solution under the lowercase-English constraint, I'll switch the key generation to a 26-frequency vector, making it $O(n \cdot k)$."

Then depending on time, implement the optimized one.

This gives the interviewer both:

text

correctness

+

optimization awareness

47. Production Variation — Grouping Equivalent Search Keywords

Suppose a product-search system receives:

```
text
```

```
["laptop", "aptopl", "potlap", ...]
```

and wants to group anagram-equivalent strings.

The same architecture is:

```
text
```

```
raw keyword
```



```
normalize
```



```
canonical key
```



```
HashMap<key, group>
```

But in production, normalization matters.

We might need to decide:

```
text
```

```
case sensitivity
```

```
spaces
```

```
punctuation
```

```
Unicode normalization
```

```
locale
```

```
stemming/tokenization
```

For example:

```
text
```

```
"Wireless Mouse"
```

```
"wireless mouse"
```

might or might not be considered equivalent depending on business rules.

The key interview lesson is:

Don't confuse the grouping algorithm with the business definition of equivalence.

First define what makes two search terms equivalent.

Then create a canonical representation for that equivalence relation.

48. Production Variation — Database Perspective

Suppose we want to identify equivalent product-search keywords.

We might generate a canonical key:

```
text  
  
canonical_key
```

and store:

```
text  
  
canonical_key → group/document
```

in Redis, a database, or a search index.

For example:

```
text  
  
"eet" → canonical key
```

could map to a collection of related terms.

But production systems need to consider:

```
text  
  
concurrency  
memory  
cache eviction  
normalization versioning  
data persistence
```

The DSA solution remains the same conceptual building block:

```
text  
  
canonicalization + hash-based grouping
```

49. Production Question — What if the dataset is huge?

Suppose:

```
text
```


100 million strings

and each string is long.

Building all groups in memory may not be appropriate.

Possible system-level strategies include:

text

partition by canonical key hash
distributed processing
external storage
MapReduce-style grouping
database aggregation

For example:

text

String
↓
canonical key
↓
 $\text{hash}(\text{key}) \% \text{partitionCount}$
↓
worker responsible for that partition

This allows strings with the same key to reach the same worker.

This is the distributed version of the exact same concept:

text

local HashMap grouping

becomes:

text

distributed key-based grouping

That's a nice backend interview connection.

50. Advanced Production Concern – HashMap Memory

The canonical-key approach stores:

```
text

key
+
list
+
references to strings
```

For millions of records, object overhead can become substantial.

A production implementation may need:

```
text

compact key representations
stream processing
database-backed grouping
partitioning
batching
```

But don't bring these up during the normal coding portion unless the interviewer starts discussing scale.

51. Important Pattern – Canonicalization

This pattern appears far beyond anagrams.

Example 1

Email addresses:

```
text

normalize(email)
  ↓
canonical email
```

Example 2

URLs:

```
text

normalize URL
  ↓
canonical URL
```

Example 3

Product search:

```
text

normalize query
  ↓
canonical query
```

Example 4

Anagrams:

```
text

sort/count characters
  ↓
canonical anagram key
```

The generic architecture is:

```
text

Different representations
  ↓
same logical entity
  ↓
canonical representation
  ↓
HashMap / HashSet
```

This is a very useful pattern for interviews.

52. Common Mistake – Using String Hash Code Alone

A candidate might try:

```
java

int key = s.hashCode();
```

and group by that.

This is unsafe because different strings can have the same hash code.

Hash codes are not guaranteed to be unique.

For example, Java's `String.hashCode()` can have collisions.

So:

```
text
```

```
hashCode
```

is useful for hash-table indexing but is **not itself a unique canonical identifier**.

We need a representation with proper equality semantics.

This is a very important interview distinction.

53. Could we use a custom numeric hash for the frequency counts?

You could construct one, but unless you can guarantee collision-free encoding within the allowed limits, it can produce incorrect grouping.

For exact grouping:

```
text
```

```
canonical representation
```

should preserve equality rather than merely relying on a probabilistic fingerprint.

A String encoding of the full frequency vector is straightforward and exact.

54. Common Mistake – Mutating the Original String

Strings are immutable, so:

```
java
```

```
Arrays.sort(s)
```

isn't possible.

You must create:

```
java
```

```
char[] chars = s.toCharArray();
```

then sort that.

This is an important Java-specific implementation detail.

55. Common Mistake – Returning the Map Directly

The method requires:

```
java  
  
List<List<String>>
```

while our structure is:

```
java  
  
Map<String, List<String>>
```

So:

```
java  
  
return new ArrayList<>(map.values());
```

extracts the grouped lists.

56. Common Mistake — Assuming Group Order

The `HashMap` does not guarantee iteration order.

So:

```
text  
  
[  
  ["eat", "tea"],  
  ["bat"]  
]
```

and:

```
text  
  
[  
  ["bat"],  
  ["eat", "tea"]  
]
```

may both be valid unless the question explicitly requires a particular ordering.

If deterministic order is required, then you need an explicit ordering strategy.

For example:

```
text
LinkedHashMap
```

can preserve insertion order of keys.

But don't use `LinkedHashMap` unnecessarily.

57. HashMap vs LinkedHashMap vs TreeMap

Potential Java follow-up:

HashMap

```
text

Expected O(1) key operations
No ordering guarantee
```

Best for standard grouping.

LinkedHashMap

```
text

Expected O(1)
Preserves insertion order
```

Use if group order matters.

TreeMap

```
text

O(log n)
Keys sorted
```

Only use if sorted key order is required.

Since anagram grouping doesn't require key ordering:

```
text

HashMap
```

is the correct default.

58. Could we use `Collectors.groupingBy()` ?

Yes.

With sorting keys:

```
java

public List<List<String>> groupAnagrams(String[] strs) {

    return new ArrayList<>()
        .stream(strs)
        .collect(Collectors.groupingBy(s -> {
            char[] chars = s.toCharArray();
            Arrays.sort(chars);
            return new String(chars);
        }))
        .values()
    );
}
```

This is concise, but in an interview I would usually write the explicit loop first.

Why?

Because it clearly demonstrates:

```
text

key generation
+
map lookup
+
group creation
```

rather than hiding the logic inside Streams.

For a DSA round, clarity is often preferable.

59. Streams Follow-up

If interviewer asks:

"Can you write it using streams?"

Yes.

But don't jump to Streams initially just because you're interviewing for Java.

The interviewer is testing:

text

algorithm
data structure
complexity
problem-solving

not whether you can make the code as functional-looking as possible.

60. Final Pattern Recognition

When you see:

text

"Group items that are equivalent"

ask:

text

What makes two items equivalent?

Then:

text

Define canonical key
↓
Map<Key, List<Item>>

For this problem:

text

Anagram equivalence
↓
same character frequencies
↓
canonical key
↓
HashMap

And two ways to construct the key:

text

sort characters



$O(k \log k)$

or:

text

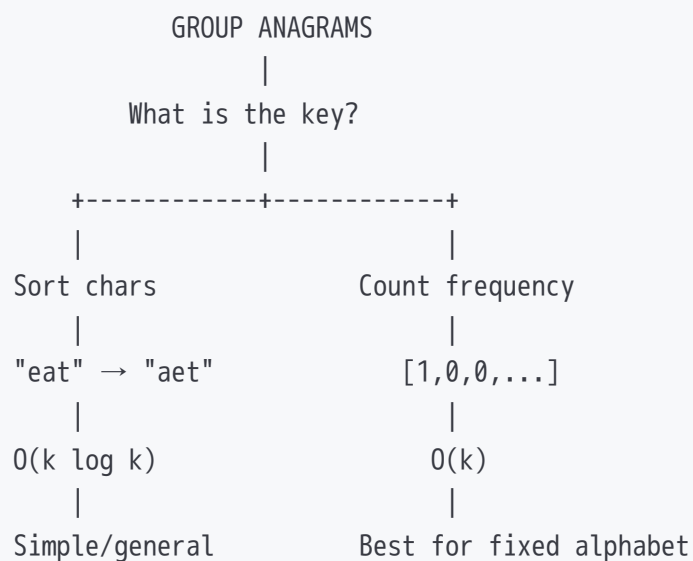
frequency counting



$O(k)$

61. Best Approach Decision Tree

text



Then:

text

canonical key



HashMap<Key, List<String>>



group

62. The Interview Conversation I Recommend

An excellent sequence would be:

Interviewer: "Solve Group Anagrams."

You:

"I'll first define how to recognize two strings as belonging to the same group. Since anagrams have the same characters, I can sort each string and use the sorted string as a canonical key."

Then explain:

text

```
eat → aet
tea → aet
ate → aet
bat → abt
```

Then:

text

```
Map<String, List<String>>
```

Then give:

text

```
O(n * k log k)
```

Then interviewer asks:

"Can you improve it?"

You answer:

"Yes. Since anagrams are defined by character frequencies, and assuming lowercase English letters, I can use a 26-element frequency array instead of sorting. That makes key generation $O(k)$, giving $O(n*k)$ overall."

That is exactly the progression I'd want from a 2–2.5 YOE candidate.

63. Final Interview-Ready Optimal Code

For the standard lowercase-English-letter version:

java

```
public List<List<String>> groupAnagrams(String[] strs) {

    Map<String, List<String>> map = new HashMap<>();
```

```

for (String s : strs) {

    int[] count = new int[26];

    for (char c : s.toCharArray()) {
        count[c - 'a']++;
    }

    StringBuilder keyBuilder = new StringBuilder();

    for (int i = 0; i < 26; i++) {
        keyBuilder.append('#').append(count[i]);
    }

    String key = keyBuilder.toString();

    map.computeIfAbsent(key, k -> new ArrayList<>())
        .add(s);
}

return new ArrayList<>(map.values());
}

```

Complexity

text

Let:

n = number of strings

k = average string length

Time:

$O(n * k)$

Space:

$O(n * k)$ for stored groups/keys

Under the fixed 26-character alphabet, the frequency array itself is:

text

$O(1)$

per string.

64. Final Interviewer-Expected Answer

If the interviewer asks:

"What is your best solution and why?"

Say:

"The basic approach would be to compare each string with existing groups using an anagram check, but that can become $O(n^2 \cdot k)$.

A better approach is to create a canonical representation for each string. Sorting the characters gives an $O(n \cdot k \log k)$ solution because anagrams produce the same sorted representation.

We can optimize this further because anagrams are actually defined by character frequencies. Since the problem uses lowercase English letters, I can build a 26-element frequency array for each string and convert that frequency vector into an immutable canonical key. Then I use a `HashMap<String, List<String>>` to group strings with the same key.

This reduces the complexity to $O(n \cdot k)$, where n is the number of strings and k is the average string length. I prefer this approach when the alphabet is fixed because it avoids the sorting cost."

65. Final Cheat Sheet

text

Q4 – Group Anagrams

Basic

text

Compare each string against existing groups
 $O(n^2 * k)$ worst case

Better

text

Sort each string
"eat" → "aet"
"tea" → "aet"

 $O(n * k \log k)$

Optimal for fixed alphabet

```
text
Frequency count
"a/e/t" → same 26-frequency vector
```

```
O(n * k)
```

Core data structure

```
text
```

```
Map<CanonicalKey, List<String>>
```

Core pattern

```
text
```

```
Equivalence
```

```
↓
```

```
Canonicalization
```

```
↓
```

```
HashMap grouping
```

Java pattern

```
java
```

```
map.computeIfAbsent(key, k -> new ArrayList<>())
    .add(s);
```

Important Java trap

```
text
```

```
int[] as HashMap key
```

does **not** automatically provide content-based equality.

Use a value-based immutable key such as:

```
text
```

```
String
```

or a properly implemented immutable custom key.

Production thinking

text

normalize



canonicalize



group by key

Most important connection to Q3

text

Q3:

Are these two strings anagrams?



Compare frequencies

Q4:

Which strings are anagrams?



Generate frequency/canonical key



HashMap<Key, List<String>>

That is the progression the interviewer is looking for: **Q3 teaches frequency comparison; Q4 turns that same idea into canonicalization + hash-based grouping.**

2026-09-02 21:00:12 You 

ok as aboved continue for next question, keep this same interview structure: ****brute force → better → optimal → pattern recognition → why the data structure works → dry run → edge cases → Java-specific follow-ups → production variation → final interviewer-ready answer****.

Q5. Top K Frequent Elements

Question

Return the k most frequent elements.

Approach

1. Count using HashMap.

2. Maintain a min-heap of size k.

java

```
public int[] topKFrequent(int[] nums, int k) {  
    Map<Integer, Integer> freq = new HashMap<>();
```

```
for (int n : nums) {
    freq.merge(n, 1, Integer::sum);
}
```

9

```
PriorityQueue<int[]> pq =
    new PriorityQueue<>(Comparator.comparingInt(a -> a[1]));
```

```
for (var entry : freq.entrySet()) {
    pq.offer(new int[]{entry.getKey(), entry.getValue()});
```

```
    if (pq.size() > k) {
        pq.poll();
    }
}
```

```
int[] result = new int[k];
```

```
for (int i = k - 1; i >= 0; i--) {
    result[i] = pq.poll()[0];
}
```

```
    return result;
}
```

Complexity

text

$O(n \log k)$

Follow-up

Why min-heap rather than max-heap?

Because we want to efficiently remove the least frequent among the current top K.

Backend example

Top-K most frequently requested products/API endpoints.

Q5. Top K Frequent Elements

This is a very important interview problem because it combines **HashMap + Heap/PriorityQueue + Top-K pattern**.

It also builds naturally on the previous problems:

text

Q2 Contains Duplicate



HashSet / membership

Q3 Valid Anagram



Frequency counting

Q4 Group Anagrams



Canonical key + HashMap grouping

Q5 Top K Frequent Elements



Frequency counting + Top-K selection

For a **2–2.5 year Java candidate**, the interviewer will usually care about whether you can recognize that after counting frequencies, you **do not need to sort every element**. You only need the best `k`.

1. Problem

Given an integer array `nums` and an integer `k`, return the `k` most frequent elements.

Example

text

```
nums = [1,1,1,2,2,3]
```

```
k = 2
```

Frequencies:

text

```
1 -> 3
```

```
2 -> 2
```

```
3 -> 1
```

Top 2 frequent elements are:


```
text
[1, 2]
```

The order is usually not important unless the interviewer explicitly says otherwise.

2. How I would start in the interview

A strong opening:

"First, I need the frequency of every distinct element, so I'll use a HashMap. Then I need to select only the top **k** frequencies. A straightforward way is to sort all distinct elements by frequency, but that costs $O(m \log m)$, where **m** is the number of distinct elements. We can do better using a min-heap of size **k**: while processing each frequency, keep only the best **k** elements in the heap. That gives $O(n + m \log k)$, which is $O(n \log k)$ in the worst case."

That is the answer the interviewer wants to hear.

3. Approach 1 — Brute Force

Let's begin with the most basic approach.

We could, for every element, count how many times it appears in the whole array.

For example:

```
text

nums = [1,1,1,2,2,3]
```

For **1** :

```
text

count = 3
```

For **2** :

```
text

count = 2
```

For **3** :

```
text

count = 1
```

But if we do this by scanning the entire array for every element, we repeat the same work.

Java

java

```
public int[] topKFrequent(int[] nums, int k) {

    List<Integer> distinct = new ArrayList<>();

    for (int num : nums) {
        if (!distinct.contains(num)) {
            distinct.add(num);
        }
    }

    List<int[]> frequencies = new ArrayList<>();

    for (int num : distinct) {

        int count = 0;

        for (int x : nums) {
            if (x == num) {
                count++;
            }
        }

        frequencies.add(new int[]{num, count});
    }

    frequencies.sort(Comparator.comparingInt(a -> a[1]));

    int[] result = new int[k];

    for (int i = 0; i < k; i++) {
        result[i] = frequencies
            .get(frequencies.size() - 1 - i)[0];
    }

    return result;
}
```

This implementation is intentionally basic. It repeats work and can be very inefficient.

4. Complexity of Brute Force

Suppose:

```
text
```

```
n = number of elements  
m = number of distinct elements
```

Finding distinct values naively can already cost $O(n^2)$ because of:

```
java
```

```
distinct.contains(num)
```

Then counting each distinct number scans the array:

```
text
```

```
 $O(m * n)$ 
```

So the solution can approach:

```
text
```

```
 $O(n^2)$ 
```

or:

```
text
```

```
 $O(n * m)$ 
```

depending on how we implement it.

This is not what we want.

The interviewer expects us to notice:

"I am repeatedly counting the same values. I should count each value once."

That leads to HashMap.

5. Approach 2 — HashMap + Sort by Frequency

This is the most straightforward optimized solution.

Step 1

Build:

```
text
```

```
value -> frequency
```

using a HashMap.

Example:

```
text
```

```
nums = [1,1,1,2,2,3]
```

gives:

```
text
```

```
1 -> 3
```

```
2 -> 2
```

```
3 -> 1
```

Step 2

Sort the entries by frequency descending.

Then take the first `k` .

6. Java — HashMap + Sorting

```
java
```

```
public int[] topKFrequent(int[] nums, int k) {  
  
    Map<Integer, Integer> freq = new HashMap<>();  
  
    for (int num : nums) {  
        freq.merge(num, 1, Integer::sum);  
    }  
  
    List<Map.Entry<Integer, Integer>> entries =  
        new ArrayList<>(freq.entrySet());  
  
    entries.sort(  
        (a, b) -> Integer.compare(b.getValue(), a.getValue())  
    );  
  
    int[] result = new int[k];  
  
    for (int i = 0; i < k; i++) {
```

```
        result[i] = entries.get(i).getKey();
    }

    return result;
}
```

7. Dry Run — Sorting Approach

Input:

text

nums = [1,1,1,2,2,3]

k = 2

Frequency map:

text

1 -> 3

2 -> 2

3 -> 1

Sort descending:

text

1 -> 3

2 -> 2

3 -> 1

Take first two:

text

[1, 2]

8. Complexity

Let:

text

n = number of array elements

m = number of distinct elements

Frequency counting:

text

$O(n)$

Sorting distinct elements:

text

$O(m \log m)$

Therefore:

text

$O(n + m \log m)$

Since:

text

$m \leq n$

worst case:

text

$O(n \log n)$

Space:

text

$O(m)$

plus the result.

9. Is this a good answer?

Yes.

It is much better than brute force and is easy to understand.

But there is an important observation:

Suppose:

```
text
m = 1,000,000
k = 5
```

Do we really need to fully sort all one million distinct elements?

No.

We only need:

```
text

top 5
```

This is the key insight behind the Top-K pattern.

When you only need the best K elements, you often don't need to sort everything.

That leads to the heap solution.

10. Approach 3 — HashMap + Min-Heap

This is the standard interview solution.

The algorithm has two stages.

Stage 1 — Count frequencies

```
text

value -> frequency
```

using:

```
java

HashMap
```

Stage 2 — Maintain top K

Use:

```
java

PriorityQueue
```

as a **min-heap**.

The heap stores:

```
text
```

```
(element, frequency)
```

and the smallest frequency stays at the top.

We keep the heap size at:

```
text
```

```
k
```

Whenever it becomes:

```
text
```

```
k + 1
```

we remove the smallest-frequency element.

So at every point:

```
text
```

```
heap = best k elements seen so far
```

11. Why a Min-Heap?

This is the most important question in this problem.

You might initially think:

"We want the largest frequencies, so why not use a max-heap?"

Because we need to efficiently remove the **worst element among the current top K**.

Suppose:

```
text
```

```
k = 3
```

and heap currently contains:

```
text
```



```
A -> 10
B -> 8
C -> 7
```

New element:

```
text
```

```
D -> 12
```

Now we have four candidates:

```
text
```

```
A -> 10
B -> 8
C -> 7
D -> 12
```

We need to remove:

```
text
```

```
C -> 7
```

because it is the least frequent.

A min-heap gives us exactly that:

```
text
```

```
peek/poll -> smallest frequency
```

So:

```
text
```

```
Min-heap
  ↓
keep top K
  ↓
remove the weakest
```

That's the key pattern.

12. Why not Max-Heap?

A max-heap gives:

text

largest frequency at top

But when we have:

text

$k + 1$

candidates, we need the **smallest** one to remove.

A max-heap gives us the wrong candidate for efficient removal.

We could still use a max-heap with additional logic, but it is not the natural structure for this problem.

Therefore:

text

Top K largest

+

need to remove smallest among selected

↓

Min-Heap of size K

This is one of the most important heap patterns to remember.

13. Java – Optimal Heap Solution

java

```
public int[] topKFrequent(int[] nums, int k) {  
  
    Map<Integer, Integer> freq = new HashMap<>();  
  
    // Step 1: count frequencies  
    for (int num : nums) {  
        freq.merge(num, 1, Integer::sum);  
    }  
  
    // Step 2: min-heap based on frequency  
    PriorityQueue<int[]> pq =  
        new PriorityQueue<>(  
            Comparator.comparingInt(a -> a[1])  
        );
```

```

// Keep only top k frequent elements
for (Map.Entry<Integer, Integer> entry : freq.entrySet()) {

    pq.offer(new int[]{
        entry.getKey(),
        entry.getValue()
    });

    if (pq.size() > k) {
        pq.poll();
    }
}

// Step 3: build result
int[] result = new int[k];

for (int i = k - 1; i >= 0; i--) {
    result[i] = pq.poll()[0];
}

return result;
}

```

14. Dry Run — Heap Approach

Input:

text

```

nums = [1,1,1,2,2,3]
k = 2

```

Frequency:

text

```

1 -> 3
2 -> 2
3 -> 1

```

Now process frequency entries.

Process 1 -> 3

Heap:

text

[(1,3)]

Size:

text

1

No removal.

Process 2 -> 2

Heap:

text

[(2,2), (1,3)]

Size:

text

2

No removal.

Current top 2:

text

1 -> 3
2 -> 2

Process 3 -> 1

Heap becomes:

text

[(3,1), (1,3), (2,2)]

Size:

text

```
3
```

But:

```
text
```

```
k = 2
```

So:

```
java
```

```
pq.poll();
```

removes:

```
text
```

```
3 -> 1
```

Heap becomes:

```
text
```

```
[(2,2), (1,3)]
```

Exactly the top two.

15. Why keeping heap size K is enough

This is a very important correctness idea.

At any point, suppose the heap contains the best k elements processed so far.

When a new candidate arrives:

Case 1 — New candidate is not better

It enters the heap, making size $k + 1$.

The smallest-frequency element gets removed.

So the heap again contains the best k .

Case 2 — New candidate is better

It also enters.

The weakest of the $k + 1$ gets removed.

The new stronger candidate survives.

Therefore the invariant remains:

```
text
```

```
heap contains the top K candidates seen so far
```

This is why the algorithm is correct.

16. Complexity

Let:

```
text
```

```
n = number of elements
```

```
m = number of distinct elements
```

Frequency counting

```
text
```

```
 $O(n)$ 
```

Heap processing

Each of the m distinct elements is inserted into a heap of size at most k .

Each heap operation costs:

```
text
```

```
 $O(\log k)$ 
```

Therefore:

```
text
```

```
 $O(m \log k)$ 
```

Total:

```
text
```

```
 $O(n + m \log k)$ 
```

Since:

text

$m \leq n$

worst case:

text

$O(n \log k)$

Space

Frequency map:

text

$O(m)$

Heap:

text

$O(k)$

Therefore:

text

$O(m + k)$

Since $k \leq m$ in the usual valid setting:

text

$O(m)$

plus output.

17. Important Complexity Precision

You may see solutions simply write:

text

$O(n \log k)$

That's acceptable for interview purposes.

But the more precise explanation is:

```
text
```

```
O(n + m log k)
```

where:

```
text
```

```
m = number of distinct elements
```

This is a stronger answer.

Then explain:

"Because $m \leq n$, the worst-case upper bound is $O(n \log k)$."

That is the kind of precision a 2–2.5 YOE candidate should be comfortable with.

18. Approach 4 – Bucket Sort

Now comes another important approach.

The frequency of an element can be at most:

```
text
```

```
n
```

because an element cannot appear more than n times.

So frequency itself has a bounded range:

```
text
```

```
1 ... n
```

We can use **buckets indexed by frequency**.

For example:

```
text
```

```
bucket[1] = elements occurring once  
bucket[2] = elements occurring twice  
bucket[3] = elements occurring three times  
...
```



```
bucket[n] = elements occurring n times
```

Then iterate frequencies from:

```
text
```

```
n → 1
```

and collect elements until we have `k`.

This is a very important alternative.

19. Bucket Sort Example

Input:

```
text
```

```
nums = [1,1,1,2,2,3]
```

Frequency:

```
text
```

```
1 -> 3
```

```
2 -> 2
```

```
3 -> 1
```

Buckets:

```
text
```

```
bucket[3] = [1]
```

```
bucket[2] = [2]
```

```
bucket[1] = [3]
```

Walk from high frequency:

```
text
```

```
frequency 3 → 1
```

```
frequency 2 → 2
```

We already have:

```
text
```

```
k = 2
```

Return:

text

[1, 2]

20. Java – Bucket Approach

java

```
public int[] topKFrequent(int[] nums, int k) {  
  
    Map<Integer, Integer> freq = new HashMap<>();  
  
    for (int num : nums) {  
        freq.merge(num, 1, Integer::sum);  
    }  
  
    List<Integer>[] buckets = new List[nums.length + 1];  
  
    for (Map.Entry<Integer, Integer> entry : freq.entrySet()) {  
  
        int number = entry.getKey();  
        int count = entry.getValue();  
  
        if (buckets[count] == null) {  
            buckets[count] = new ArrayList<>();  
        }  
  
        buckets[count].add(number);  
    }  
  
    int[] result = new int[k];  
    int index = 0;  
  
    for (int frequency = buckets.length - 1;  
        frequency >= 1 && index < k;  
        frequency--) {  
  
        if (buckets[frequency] == null) {  
            continue;  
        }  
  
        for (int number : buckets[frequency]) {  
  
            result[index++] = number;  
        }  
    }  
}
```

```

        if (index == k) {
            break;
        }
    }

    return result;
}

```

This is more code than the heap solution, but it has excellent asymptotic performance.

21. Bucket Complexity

Frequency counting:

text

$O(n)$

Creating buckets:

text

$O(m)$

Scanning possible frequencies:

text

$O(n)$

Overall:

text

$O(n + m)$

Since:

text

$m \leq n$

this is:

text

$O(n)$

time.

Space:

text

$O(n)$

because buckets can have size proportional to the input.

So asymptotically:

text

Bucket

Time: $O(n)$

Space: $O(n)$

This can beat the heap's:

text

$O(n \log k)$

but uses more explicit bucket storage and is somewhat more cumbersome.

22. Which is the "best" approach?

This needs a nuanced answer.

For standard interview discussion

The **min-heap** is often the expected answer because it teaches the Top-K pattern and works naturally when **k** is small.

text

HashMap + Min-Heap

For strict asymptotic optimization

If frequencies are bounded from **1** to **n**, bucket grouping can achieve:

text

$O(n)$

time.

So:

text

Min-Heap:
 $O(n \log k)$

Bucket:
 $O(n)$

23. What would I actually tell the interviewer?

I would say:

"The standard optimal Top-K approach is a frequency HashMap plus a min-heap of size k , giving $O(n \log k)$ worst-case time. Since the frequency of an element is bounded by n , there is also a bucket approach that can achieve $O(n)$ time. For a general Top-K problem, though, I prefer the min-heap because it's more reusable and directly applies when the score range isn't naturally bounded."

That is a **very strong 2–2.5 YOE answer**.

It shows you know:

text

practical reusable optimization
+
problem-specific stronger optimization

24. Pattern Recognition — Top K

This is one of the most important patterns from the problem.

Whenever you see:

text

Top K largest
Top K smallest
K most frequent
K highest scores
K closest

K best candidates

immediately consider:

text

Top-K pattern



Heap / PriorityQueue

Especially:

text

Top K largest



Min-heap of size K

and:

text

Top K smallest



Max-heap of size K

25. Why the Heap Direction Changes

This is worth memorizing carefully.

Want K largest

Example:

text

[100, 90, 80, 70, 60]

k = 3

Keep:

text

100, 90, 80

The weakest member among these is:

text
80

So we need quick access to the **smallest among selected**.

Therefore:

text
Min-heap

Want K smallest

Example:

text
[10,20,30,40,50]
k = 3

Keep:

text
10,20,30

The weakest member is:

text
30

which is the largest among selected.

So use:

text
Max-heap

because we need quick access to the largest.

Rule

text

Top K largest
↓
Min-heap of size K

Top K smallest
↓
Max-heap of size K

This is one of the most useful heap rules for interviews.

26. Why not sort?

If:

```
text

m = 1,000,000
k = 10
```

sorting all one million candidates does:

```
text

O(m log m)
```

work.

A heap only needs:

```
text

O(m log k)
```

and:

```
text

log 10
```

is tiny compared with:

```
text

log 1,000,000
```

So the heap is especially valuable when:


```
text
k << m
```

27. Why HashMap + Heap?

Each data structure has a separate responsibility.

HashMap

Answers:

```
text

"What is the frequency of this number?"
```

Min-Heap

Answers:

```
text

"Among my current top K candidates,
which one is the weakest?"
```

This decomposition is important.

```
text

HashMap
  ↓
count

Heap
  ↓
select
```

This is a general DSA design pattern:

Use one structure to compute the required score and another to maintain the desired ordering/selection.

28. Dry Run — Full Optimal Approach

Let's perform the complete algorithm.

```
text
```

```
nums = [1,1,1,2,2,3]
k = 2
```

Step 1 – Frequency

text

```
1 → 3
2 → 2
3 → 1
```

Step 2 – Min-heap

Insert:

text

```
(1,3)
```

Heap:

text

```
[(1,3)]
```

Insert:

text

```
(2,2)
```

Heap:

text

```
[(2,2), (1,3)]
```

Insert:

text

```
(3,1)
```

Heap:

text

```
[(3,1), (1,3), (2,2)]
```

Size is 3.

But:

```
text
```

```
k = 2
```

So remove minimum:

```
text
```

```
(3,1)
```

Final heap:

```
text
```

```
(2,2)
```

```
(1,3)
```

Then extract:

```
text
```

```
[2,1]
```

or another valid ordering such as:

```
text
```

```
[1,2]
```

depending on extraction/output ordering.

29. Why Does the Result Extraction Use Reverse Index?

Your supplied code does:

```
java
```

```
for (int i = k - 1; i >= 0; i--) {  
    result[i] = pq.poll()[0];  
}
```

Why?

Because a min-heap gives elements from:

text

smallest frequency $\rightarrow$ largest frequency

Suppose heap has:

text

2 $\rightarrow$ 2

1 $\rightarrow$ 3

Polling gives:

text

2

1

If we place them from the back:

text

result[1] = 2

result[0] = 1

we get:

text

[1,2]

which happens to be descending frequency.

But unless the problem requires a specific ordering, this reversal isn't actually necessary.

We could simply do:

java

```
for (int i = 0; i < k; i++) {  
    result[i] = pq.poll()[0];  
}
```

and return:

```
text  
[2,1]
```

which is still valid in the usual problem specification.

This is a nice interview detail:

"I'm reversing the extraction only if I want the result ordered by increasing/decreasing frequency; otherwise, result order is irrelevant."

30. Java-Specific Follow-up — Why `PriorityQueue` ?

In Java:

```
java  
  
PriorityQueue
```

is a heap implementation.

By default:

```
text  
  
min-heap
```

for natural ordering.

For example:

```
java  
  
PriorityQueue<Integer> pq = new PriorityQueue<>();
```

means the smallest integer is at the top.

```
java  
  
pq.peek()
```

returns the smallest.

```
java  
  
pq.poll()
```

removes and returns the smallest.

31. Java-Specific Follow-up — How do we make a Max-Heap?

Use a reverse comparator:

```
java

PriorityQueue<Integer> maxHeap =
    new PriorityQueue<>(Comparator.reverseOrder());
```

or:

```
java

PriorityQueue<Integer> maxHeap =
    new PriorityQueue<>((a, b) -> Integer.compare(b, a));
```

For objects:

```
java

PriorityQueue<int[]> pq =
    new PriorityQueue<>(
        Comparator.comparingInt((int[] a) -> a[1]).reversed()
    );
```

32. Java-Specific Follow-up — Why `int[]` in the heap?

The code stores:

```
java

new int[]{number, frequency}
```

So:

```
text

a[0] = element
a[1] = frequency
```

This avoids creating a dedicated class.

However, in production code, a small domain object or record can often be more readable:

```
java

record Entry(int value, int frequency) {}
```

Then:

```
java

PriorityQueue<Entry> pq =
    new PriorityQueue<>(
        Comparator.comparingInt(Entry::frequency)
    );
```

This is much easier to read.

33. Java Record Version

```
java

record Entry(int value, int frequency) {}
```

Then:

```
java

PriorityQueue<Entry> pq =
    new PriorityQueue<>(
        Comparator.comparingInt(Entry::frequency)
    );
```

And:

```
java

pq.offer(new Entry(entry.getKey(), entry.getValue()));
```

This is elegant in modern Java.

For an interview, both are fine.

If the interviewer is focused on DSA, `int[]` is concise.

If they care about maintainability, a `record` is cleaner.

34. Java-Specific Follow-up — `merge()`

This line:

```
java

freq.merge(n, 1, Integer::sum);
```

means:

```
text

if key doesn't exist:
    insert 1

if key exists:
    oldValue + 1
```

Equivalent:

```
java

freq.put(
    n,
    freq.getOrDefault(n, 0) + 1
);
```

Both are correct.

For interview explanation, say:

"merge() lets me update the frequency concisely: initialize to 1 when absent and add 1 when present."

35. Java-Specific Follow-up — Comparator

Your code:

```
java

Comparator.comparingInt(a -> a[1])
```

means:

"Compare heap entries using frequency stored at index 1."

Therefore the smallest frequency becomes the head.

This produces a min-heap.

A descending comparator:

```
java

Comparator.comparingInt((int[] a) -> a[1]).reversed()
```


would produce a max-heap.

36. Tricky Question — What if frequencies are equal?

Suppose:

```
text

1 -> 3
2 -> 3
3 -> 2
```

and:

```
text

k = 2
```

Both 1 and 2 have the same frequency.

If the problem allows any top-K answer, either is valid where tied candidates are involved.

If the interviewer says:

"Break ties by smaller numeric value."

then the comparator must include a secondary condition.

For example, for a min-heap where lower frequency should be removed first:

```
java

PriorityQueue<int[]> pq = new PriorityQueue<>(
    (a, b) -> {
        int byFreq = Integer.compare(a[1], b[1]);

        if (byFreq != 0) {
            return byFreq;
        }

        return Integer.compare(a[0], b[0]);
    }
);
```

Now:

```
text
```

```
frequency first  
then value
```

37. Important Comparator Rule

Never write:

```
java  
  
(a, b) -> a[1] - b[1]
```

for arbitrary integers.

Why?

Because subtraction can overflow.

Prefer:

```
java  
  
Integer.compare(a[1], b[1])
```

This is a good Java interview habit.

38. Edge Case — **k = 1**

Example:

```
text  
  
nums = [1,1,2,2,2]  
k = 1
```

Frequency:

```
text  
  
2 -> 3  
1 -> 2
```

Answer:

```
text  
  
[2]
```

Heap size is maintained at:

```
text
```

```
1
```

Very efficient.

39. Edge Case — `k` equals number of distinct elements

Example:

```
text
```

```
nums = [1,1,2,3]
```

```
k = 3
```

Distinct:

```
text
```

```
1
```

```
2
```

```
3
```

We need all of them.

The heap never needs to discard a candidate after it reaches exactly `k`.

40. Edge Case — Negative numbers

No problem.

Example:

```
text
```

```
nums = [-1,-1,-2,-2,-2,3]
```

HashMap keys can be negative.

The heap is comparing:

```
text
```

```
frequency
```

not the actual number.

So negative values do not affect the algorithm.

41. Edge Case — Large frequencies

Suppose:

```
text
```

```
n = 1,000,000
```

and one value occurs nearly a million times.

`int` is sufficient as long as array length itself is within Java's `int` indexing limits, which is the normal case.

If dealing with external counts that can exceed `int`, a `long` frequency may be appropriate.

For standard array problems, `int` is expected.

42. Edge Case — All elements identical

```
text
```

```
nums = [5,5,5,5,5]
```

```
k = 1
```

Frequency:

```
text
```

```
5 -> 5
```

Return:

```
text
```

```
[5]
```

Easy.

43. Edge Case — `k = distinct count`

Then every distinct element is returned.

The heap becomes a complete set of all candidates.

At that point, sorting or heap optimization gives less practical benefit, but the algorithm remains correct.

44. What if $k > \text{number of distinct elements}$?

A good production-quality function should define this.

For standard interview constraints, usually:

```
text

1 <= k <= number of distinct elements
```

is guaranteed.

If not guaranteed, you must decide whether to:

```
text

throw IllegalArgumentException
```

or:

```
text

return all distinct elements
```

depending on API requirements.

Don't silently assume unspecified behavior in a production API.

45. Production Variation — Top K Most Requested Products

Suppose an e-commerce backend receives events:

```
text

productId
```

and wants the 10 most requested products.

Input:

```
text
```

```
P100
P101
P100
P102
P100
P101
...
```

The algorithm is exactly:

```
text

events
  ↓
count product requests
  ↓
HashMap<ProductId, Count>
  ↓
Min-heap size 10
  ↓
top 10
```

This is the DSA version of a real backend analytics problem.

46. Production Variation — Top K API Endpoints

Suppose we have access logs:

```
text

GET /users
GET /orders
GET /users
GET /products
GET /users
...
```

We want:

Top 5 most frequently requested endpoints.

Again:

```
text

request logs
  ↓
count endpoint frequency
```

```
↓  
HashMap  
↓  
min-heap size 5  
↓  
Top 5
```

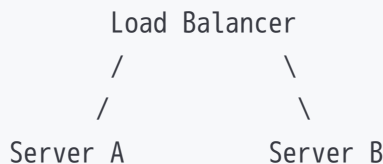
This is a direct production use of the same algorithm.

47. Important Production Concern – Can we use this across multiple servers?

Not directly with a local HashMap.

Imagine:

text



Each server sees only part of the traffic.

A local:

java

```
Map<String, Integer>
```

would produce only a local top-K.

For a global top-K system, you may need:

text

```
centralized aggregation  
distributed counters  
Kafka/stream processing  
Redis  
database aggregation
```

depending on throughput and latency requirements.

This becomes a **distributed Top-K problem**.

48. Distributed Top-K Thinking

Conceptually:

text

Events



Partition



Multiple workers



local frequency counts



merge counts



global Top K

The same algorithmic idea survives.

We simply move from:

text

single HashMap

to:

text

distributed aggregation

This is a useful backend interview connection.

49. Production Concern — Streaming Data

What if the data never ends?

For example:

text

millions of requests per second

You cannot keep every historical event in memory forever.

You might use:


```
text
sliding windows
time buckets
stream processors
approximate counting
```

For example:

```
text

Top 10 products in the last 5 minutes
```

is very different from:

```
text

Top 10 products since the beginning of time
```

A good backend engineer asks:

```
"Top K over what time window?"
```

That's an excellent production follow-up.

50. Production Concern — Approximate Top-K

At very large scale, exact counting can become expensive.

Systems may use approximate algorithms such as:

```
text

Count-Min Sketch
Heavy Hitters algorithms
approximate frequency structures
```

when small errors are acceptable.

But for a standard coding interview:

```
text

HashMap + Heap
```

is the expected exact solution.

Don't introduce approximate algorithms unless the interviewer asks about huge-scale streaming.

51. Pattern Recognition – "Top K"

This is the reusable decision rule:

text

Question says:

Top K...



Think Heap

Then ask:

text

Want K largest?



Min-heap size K

Want K smallest?



Max-heap size K

Then:

text

Need frequency?



HashMap first

So:

text

Top K Frequent



HashMap + Min-Heap

52. Another Important Pattern – "Keep Only K"

The deeper idea isn't just "use a PriorityQueue."

It is:

When K is small, don't maintain all candidates in sorted order. Maintain only the current best K.

For every new candidate:



text

candidate



heap



size > k?



remove weakest

This is broadly applicable.

53. Example — Top K Largest Numbers

Given:

text

[8, 1, 9, 4, 7, 2]

k = 3

Use:

text

Min-heap size 3

Process:

text

8 → heap [8]

1 → [1,8]

9 → [1,8,9]

4 → [1,8,9,4]



remove 1



[4,8,9]

7 → [4,7,8,9]



remove 4



[7,8,9]



```
2 → [2,7,8,9]
  ↓
remove 2
  ↓
[7,8,9]
```

Result:

text

[7,8,9]

Same pattern.

54. Why PriorityQueue Is So Useful in Interviews

It gives:

text

```
peek → O(1)
offer → O(log k)
poll → O(log k)
```

When the heap is restricted to size `k` .

That's exactly why it is useful for Top-K.

The interviewer may ask:

"Why not ArrayList?"

Because keeping the list sorted after every insertion would involve expensive shifting/sorting.

The heap is specifically designed to efficiently maintain the extreme element.

55. Heap vs Full Sorting

Suppose:

text

```
m = 1,000,000
k = 10
```

Full sorting:

```
text
```

```
 $O(m \log m)$ 
```

Min-heap:

```
text
```

```
 $O(m \log 10)$ 
```

This difference can be substantial.

When:

```
text
```

```
 $k \ll m$ 
```

the heap approach is usually preferable.

56. Heap vs Bucket

Now the deeper trade-off:

Min-Heap

```
text
```

```
 $O(n \log k)$ 
```

Advantages:

```
text
```

```
general Top-K pattern  
works with arbitrary comparable scores  
small memory proportional to k
```

Bucket

```
text
```

```
 $O(n)$ 
```

Advantages:

```
text
```

faster asymptotically
takes advantage of bounded frequency range

Disadvantages:

text

more implementation complexity
extra bucket memory
specific to bounded scores/frequencies

Thus:

Heap is the better general-purpose Top-K pattern; bucket is a problem-specific linear-time optimization.

That's a very strong interview statement.

57. Can we use a Max-Heap and stop after K?

Yes, but then we'd generally insert all candidates:

text

max-heap of size m

and poll k times.

Complexity becomes approximately:

text

$O(m + k \log m)$

depending on construction.

It can be correct, but if:

text

$k \ll m$

the min-heap of size k is generally more memory-efficient and is the standard Top-K approach.

58. Common Mistake – Using Min-Heap of All Elements

A candidate may correctly say:

"I'll use a min-heap."

but then insert **all** candidates.

That misses the main optimization.

The key is:

```
text
```

```
heap size <= k
```

not:

```
text
```

```
heap size = number of distinct elements
```

The entire Top-K optimization comes from limiting the heap to `k`.

59. Common Mistake – Sorting Full Input Instead of Frequencies

You don't want to sort:

```
text
```

```
nums
```

directly because the problem asks for **frequency**, not numerical value.

You first need:

```
text
```

```
number → frequency
```

Then select by frequency.

The correct pipeline is:

```
text
```

```
nums
```

```
↓
```

```
frequency map
```

```
↓
```

```
top-k by frequency
```

60. Common Mistake – Using a HashSet

A HashSet only tells:

```
text
```

```
"is this element present?"
```

It cannot tell:

```
text
```

```
"how many times does this element occur?"
```

So:

```
text
```

```
HashSet
```

is insufficient.

This is directly connected to the distinction from Q3 and Q4.

61. Common Mistake – Using TreeMap

A TreeMap sorts by key:

```
text
```

```
number
```

but we need to select based on:

```
text
```

```
frequency
```

So simply using:

```
java
```

```
TreeMap<Integer, Integer>
```

does not solve the main problem.

You could invert the mapping:

text

frequency -> elements

which leads toward the bucket approach.

But a heap is the more natural dynamic Top-K structure.

62. Common Mistake – Assuming HashMap Iteration Order

Don't assume:

java

```
for (var entry : freq.entrySet())
```

returns numbers in frequency order.

HashMap does not guarantee that.

That is exactly why we need the heap or sorting stage.

63. Java-Specific Follow-up – Map.Entry

This:

java

```
for (Map.Entry<Integer, Integer> entry : freq.entrySet())
```

means:

text

```
entry.getKey()
```

↓

number

```
entry.getValue()
```

↓

frequency

This is useful when iterating through a map.

You can also use:

java

```
for (var entry : freq.entrySet())
```

with modern Java.

64. Java-Specific Follow-up — Should we use Integer or int ?

HashMap uses objects:

```
java  
  
Map<Integer, Integer>
```

because Java's generic collections store reference types.

Primitive `int` cannot be used directly as a generic type parameter.

Java uses autoboxing:

```
text  
  
int  
↓  
Integer
```

when putting values into the collection.

For `PriorityQueue<int[]>`, the array keeps primitive `int`, avoiding separate Integer objects for the pair values.

65. Java-Specific Follow-up — record vs int[]

For readability:

```
java  
  
record Entry(int value, int frequency) {}
```

is preferable.

For compact DSA interview code:

```
java  
  
int[]
```

is common.

I would not spend a large amount of interview time debating this.

The algorithm matters more.

66. Java-Specific Follow-up — PriorityQueue and comparator consistency

The comparator should be consistent with the ordering you want.

For example:

```
java  
  
Comparator.comparingInt(Entry::frequency)
```

means:

```
text  
  
smallest frequency first
```

which is exactly what the min-heap needs.

For tie-breaking, add a second comparison only if the problem requires deterministic ordering.

67. Final Interview Strategy

If asked:

"Solve Top K Frequent Elements."

I would proceed like this.

First

State brute force:

```
text  
  
Repeated counting  
O(n2)
```

Then improve

"I'll count frequencies with a HashMap."

Now:

```
text
```

$O(n)$

for counting.

Then

"I could sort the distinct elements by frequency."

text

$O(n + m \log m)$

Then optimize

"But because I only need K elements, I don't need to sort every candidate. I can maintain a min-heap of size K."

Final:

text

$O(n + m \log k)$

or:

text

$O(n \log k)$

worst case.

Then mention advanced alternative

"Because frequency is bounded by n, bucket grouping can achieve $O(n)$, although the heap solution is more general."

That progression is extremely strong in an interview.

68. Final Interviewer-Ready Answer

If the interviewer asks:

"What is your optimal approach and why?"

Say:

"First, I'll count the frequency of each distinct number using a HashMap. Then I need only the top **k** frequencies, so I don't want to sort all distinct elements because that would cost $O(m \log m)$."

Instead, I'll maintain a min-heap containing at most k elements, ordered by frequency.

For every $(\text{number}, \text{frequency})$ pair, I add it to the heap. If the heap size becomes greater than k , I remove the minimum-frequency element. Therefore the heap always contains the top k frequent elements seen so far.

The frequency map takes $O(n)$, and each of the m distinct elements costs $O(\log k)$ for heap operations, giving $O(n + m \log k)$, which is $O(n \log k)$ in the worst case. Space is $O(m)$.

I use a min-heap, not a max-heap, because when the heap exceeds size k , I need to remove the least frequent element among the current top k ."

Then write:

```
java

public int[] topKFrequent(int[] nums, int k) {

    Map<Integer, Integer> freq = new HashMap<>();

    for (int n : nums) {
        freq.merge(n, 1, Integer::sum);
    }

    PriorityQueue<int[]> pq = new PriorityQueue<>(
        Comparator.comparingInt(a -> a[1])
    );

    for (var entry : freq.entrySet()) {

        pq.offer(new int[]{
            entry.getKey(),
            entry.getValue()
        });

        if (pq.size() > k) {
            pq.poll();
        }
    }

    int[] result = new int[k];

    for (int i = 0; i < k; i++) {
        result[i] = pq.poll()[0];
    }

    return result;
}
```

Notice that I don't actually need to reverse the result unless the output order is required.

69. Final Cheat Sheet

text

Q5 – Top K Frequent Elements

Approach 1 – Brute Force

text

Repeatedly count frequencies
 $O(n^2)$ worst case

Approach 2 – HashMap + Sort

text

Count
↓
Sort by frequency

$O(n + m \log m)$

Approach 3 – HashMap + Min-Heap

text

Count
↓
Min-heap size K
↓
remove weakest whenever size > K

$O(n + m \log k)$
 $\approx O(n \log k)$

Approach 4 – Bucket

text

frequency → bucket
↓
scan from n down to 1

$O(n)$

Best general interview pattern

text

Top K

↓

Heap

↓

K largest → Min-Heap

K smallest → Max-Heap

Why?

text

K largest:

need to remove smallest among selected

→ Min-Heap

K smallest:

need to remove largest among selected

→ Max-Heap

Data structure responsibilities

text

HashMap

→ frequency

PriorityQueue

→ maintain current Top K

Java

java

```
freq.merge(n, 1, Integer::sum);
```

and:

java

```
PriorityQueue<int[]> pq =  
    new PriorityQueue<>(Comparator.comparingInt(a -> a[1]));
```

Advanced optimization

text

Frequency range is bounded by n



Bucket approach



$O(n)$

Production connection

text

API/product request logs



frequency aggregation



Top-K



HashMap + Heap

Most important thing to remember

Don't memorize:

java

```
PriorityQueue<int[]>
```

Memorize the reasoning:

text

Need Top K



Don't fully sort if K is small



Keep only K candidates



Need weakest among selected



Min-Heap

ok as aboved continue for next question, keep this same interview structure: ****brute force → better → optimal → pattern recognition → why the data structure works → dry run → edge cases → Java-specific follow-ups → production variation → final interviewer-ready answer****.

Q6. Product of Array Except Self

Question

Return an array where:

text

answer[i] = product of every element except nums[i]

10

without division.

Key idea

Prefix × suffix.

text

nums = [1,2,3,4]

prefix:

[1,1,2,6]

suffix:

[24,12,4,1]

Java

java

```
public int[] productExceptSelf(int[] nums) {  
    int n = nums.length;  
    int[] result = new int[n];  
  
    result[0] = 1;  
  
    for (int i = 1; i < n; i++) {  
        result[i] = result[i - 1] * nums[i - 1];  
    }  
  
    int suffix = 1;
```

```
for (int i = n - 1; i >= 0; i--) {  
    result[i] *= suffix;  
    suffix *= nums[i];  
}
```

```
return result;  
}
```

Complexity

11

text

Time: $O(n)$

Space: $O(1)$ extra

Tricky question

Why not division?

Zero makes division-based solutions incorrect and the problem explicitly disallows it.



ChatGPT gpt-5-6-t-mini

Q6. Product of Array Except Self

This is one of the most important array problems for product-company interviews because it tests whether you can recognize a **prefix/suffix pattern**, avoid division, reason about edge cases with zeros, and optimize auxiliary space.

For a 2–2.5 year Java candidate, I would expect you to be able to progress through:

text

Brute Force

↓

Prefix/Suffix arrays

↓

Prefix output + running suffix

↓

$O(n)$ time, $O(1)$ extra space

The important thing is not to memorize the final two-loop code. You should understand **why the answer can be decomposed into "everything before i" × "everything after i."**

1. Problem

Given an integer array `nums` , return an array `answer` where:

text

`answer[i]` = product of all elements of `nums` except `nums[i]`

Constraint

Division is not allowed.

Example

text

`nums = [1, 2, 3, 4]`

For index `0` :

text

$2 \times 3 \times 4 = 24$

For index `1` :

text

$1 \times 3 \times 4 = 12$

For index `2` :

text

$1 \times 2 \times 4 = 8$

For index `3` :

text

$1 \times 2 \times 3 = 6$

So:

text

`[24, 12, 8, 6]`

2. How I would start in the interview

A strong opening answer:

"For each index, I need the product of all elements to its left and all elements to its right. A brute-force solution would recompute the product for every index and take $O(n^2)$. A better approach is to precompute prefix and suffix products. We can optimize the space further by storing the prefix product directly in the result array, then scanning from right to left with one running suffix product. That gives $O(n)$ time and $O(1)$ extra space, excluding the output array."

That is the exact reasoning path I would want an interviewer to hear.

3. Approach 1 — Brute Force

The most straightforward solution is:

For every `i` :

1. Initialize `product = 1` .
2. Iterate through the whole array.
3. Skip `i` .
4. Multiply everything else.

Java

```
java

public int[] productExceptSelf(int[] nums) {

    int n = nums.length;
    int[] result = new int[n];

    for (int i = 0; i < n; i++) {

        int product = 1;

        for (int j = 0; j < n; j++) {

            if (i != j) {
                product *= nums[j];
            }
        }

        result[i] = product;
    }
}
```

```
    return result;  
}
```

4. Dry Run — Brute Force

Input:

text

[1, 2, 3, 4]

For **i = 0** :

text

$2 \times 3 \times 4 = 24$

For **i = 1** :

text

$1 \times 3 \times 4 = 12$

For **i = 2** :

text

$1 \times 2 \times 4 = 8$

For **i = 3** :

text

$1 \times 2 \times 3 = 6$

Result:

text

[24, 12, 8, 6]

5. Complexity of Brute Force

Outer loop:

text

```
n
```

Inner loop:

```
text
```

```
n
```

Therefore:

```
text
```

```
Time:  $O(n^2)$ 
```

```
Space:  $O(1)$  extra
```

excluding the required output array.

This is correct but wastes a huge amount of repeated multiplication.

For example, when computing:

```
text
```

```
answer[2]
```

we recompute products that we already calculated for other positions.

That repeated work is exactly what we want to eliminate.

6. Key Observation

For a position `i` :

```
text
```

```
answer[i]
```

```
=
```

```
product of elements before i
```

```
×
```

```
product of elements after i
```

Example:

```
text
```

```
nums = [1, 2, 3, 4]
```

For index `2` :

```
text
```

```
elements before:
```

```
[1, 2]
```

```
product before:
```

```
 $1 \times 2 = 2$ 
```

```
elements after:
```

```
[4]
```

```
product after:
```

```
4
```

Therefore:

```
text
```

```
answer[2] =  $2 \times 4 = 8$ 
```

This is the central observation.

7. Approach 2 – Prefix and Suffix Arrays

We can explicitly calculate:

```
text
```

```
prefix[i] = product of elements before i
```

and:

```
text
```

```
suffix[i] = product of elements after i
```

Then:

```
text
```

```
answer[i] = prefix[i]  $\times$  suffix[i]
```

8. Prefix Array

For:

```
text
nums = [1, 2, 3, 4]
```

the prefix products are:

```
text
prefix[0] = 1
```

because there is nothing before index 0 .

Then:

```
text
prefix[1] = 1
prefix[2] = 1 × 2 = 2
prefix[3] = 1 × 2 × 3 = 6
```

So:

```
text
prefix = [1, 1, 2, 6]
```

9. Suffix Array

Similarly:

```
text
suffix[3] = 1
```

because nothing is after index 3 .

Then:

```
text
suffix[2] = 4
suffix[1] = 3 × 4 = 12
suffix[0] = 2 × 3 × 4 = 24
```

So:

```
text
```



```
suffix = [24, 12, 4, 1]
```

10. Combine Prefix × Suffix

Now:

```
text
```

```
answer[i] = prefix[i] * suffix[i]
```

For index 0 :

```
text
```

```
1 × 24 = 24
```

Index 1 :

```
text
```

```
1 × 12 = 12
```

Index 2 :

```
text
```

```
2 × 4 = 8
```

Index 3 :

```
text
```

```
6 × 1 = 6
```

Result:

```
text
```

```
[24, 12, 8, 6]
```

11. Java — Prefix + Suffix Arrays

```
java
```

```

public int[] productExceptSelf(int[] nums) {

    int n = nums.length;

    int[] prefix = new int[n];
    int[] suffix = new int[n];
    int[] result = new int[n];

    prefix[0] = 1;

    for (int i = 1; i < n; i++) {
        prefix[i] = prefix[i - 1] * nums[i - 1];
    }

    suffix[n - 1] = 1;

    for (int i = n - 2; i >= 0; i--) {
        suffix[i] = suffix[i + 1] * nums[i + 1];
    }

    for (int i = 0; i < n; i++) {
        result[i] = prefix[i] * suffix[i];
    }

    return result;
}

```

12. Complexity

We have:

text

prefix calculation $\rightarrow O(n)$
 suffix calculation $\rightarrow O(n)$
 result calculation $\rightarrow O(n)$

Total:

text

Time: $O(n)$

But we use:

text

```
prefix[]
suffix[]
result[]
```

So extra space is:

```
text
```

```
O(n)
```

excluding the output itself, there are still two additional arrays.

This is much better in time, but we can improve the space.

13. Approach 3 – Prefix Stored in Result Array

Now comes the key optimization.

Do we really need a separate `prefix[]` array?

No.

We can store the prefix product directly inside `result[i]` .

For:

```
text
```

```
nums = [1, 2, 3, 4]
```

after the left-to-right pass:

```
text
```

```
result = [1, 1, 2, 6]
```

This is exactly our prefix array.

So `result` becomes the prefix array.

14. First Pass – Build Prefix Products

Code:

```
java
```

```
result[0] = 1;
```

```
for (int i = 1; i < n; i++) {  
    result[i] = result[i - 1] * nums[i - 1];  
}
```

Let's dry run.

Initially:

text

```
result[0] = 1
```

So:

text

```
result = [1, 0, 0, 0]
```

At **i = 1** :

text

```
result[1] = result[0] * nums[0]  
          = 1 * 1  
          = 1
```

Result:

text

```
[1, 1, 0, 0]
```

At **i = 2** :

text

```
result[2] = result[1] * nums[1]  
          = 1 * 2  
          = 2
```

Result:

text

```
[1, 1, 2, 0]
```

At **i = 3** :

text

```
result[3] = result[2] * nums[2]
          = 2 * 3
          = 6
```

Final:

```
text
```

```
[1, 1, 2, 6]
```

Exactly the prefix products.

15. We Still Need the Suffix Product

Now we need:

```
text
```

```
suffix[i]
```

But we don't need a whole suffix array.

We only need the **running suffix product**.

Start from the right:

```
text
```

```
suffix = 1
```

Why **1** ?

Because **1** is the multiplicative identity:

```
text
```

```
 $x \times 1 = x$ 
```

At each index:

```
text
```

```
result[i] *= suffix
```

Then:

```
text
```

```
suffix *= nums[i]
```

This is the critical trick.

16. Final Optimal Code

java

```
public int[] productExceptSelf(int[] nums) {  
  
    int n = nums.length;  
    int[] result = new int[n];  
  
    // Prefix product  
    result[0] = 1;  
  
    for (int i = 1; i < n; i++) {  
        result[i] = result[i - 1] * nums[i - 1];  
    }  
  
    // Suffix product  
    int suffix = 1;  
  
    for (int i = n - 1; i >= 0; i--) {  
  
        result[i] *= suffix;  
  
        suffix *= nums[i];  
    }  
  
    return result;  
}
```

This is the standard optimal solution.

17. Deep Dry Run — Optimal Approach

Let's fully trace:

text

```
nums = [1, 2, 3, 4]
```

After prefix pass

text

```
result = [1, 1, 2, 6]
```

Now:

text

```
suffix = 1
```

i = 3

Current:

text

```
result[3] = 6  
suffix = 1
```

Update:

text

```
result[3] = 6 × 1 = 6
```

Then:

text

```
suffix = 1 × nums[3]  
        = 4
```

State:

text

```
result = [1, 1, 2, 6]  
suffix = 4
```

i = 2

Current:

text

```
result[2] = 2  
suffix = 4
```

Update:

text

```
result[2] = 2 × 4 = 8
```

Then:

text

```
suffix = 4 × nums[2]  
        = 4 × 3  
        = 12
```

State:

text

```
result = [1, 1, 8, 6]  
suffix = 12
```

i = 1

Current:

text

```
result[1] = 1  
suffix = 12
```

Update:

text

```
result[1] = 1 × 12  
          = 12
```

Then:

text

```
suffix = 12 × 2  
        = 24
```

State:


```
text
result = [1, 12, 8, 6]
suffix = 24
```

i = 0

Current:

```
text

result[0] = 1
suffix = 24
```

Update:

```
text

result[0] = 1 × 24
           = 24
```

Then:

```
text

suffix = 24 × 1
        = 24
```

Final:

```
text

[24, 12, 8, 6]
```

18. Why Must We Multiply Before Updating `suffix` ?

This is a classic tricky question.

The code does:

```
java

result[i] *= suffix;
suffix *= nums[i];
```

and not:

```
java
suffix *= nums[i];
result[i] *= suffix;
```

Why?

Because `result[i]` should include the product of elements **after** `i`.

It must **not include** `nums[i]` itself.

At index `i`:

text

suffix

must represent:

text

$\text{nums}[i + 1] \times \text{nums}[i + 2] \times \dots$

So we first use the current suffix:

```
java
```

```
result[i] *= suffix;
```

Then we include `nums[i]` for future positions:

```
java
```

```
suffix *= nums[i];
```

That's a crucial invariant.

19. The Core Invariant

This is a very strong interview explanation.

During the right-to-left pass:

text

suffix

represents:

text

```
product of nums[i+1 ... n-1]
```

before processing index `i`.

Therefore:

```
text
```

```
result[i]
```

already contains:

```
text
```

```
product of nums[0 ... i-1]
```

from the first pass.

So:

```
text
```

```
result[i] × suffix
```

becomes:

```
text
```

```
left product × right product
```

which is exactly:

```
text
```

```
product of every element except nums[i]
```

This explains the algorithm rigorously.

20. Why Does the Prefix Pass Use `nums[i - 1]` ?

This is another common question.

We want:

```
text
```

```
result[i]
```

```
=
```

```
nums[0] × nums[1] × ... × nums[i-1]
```

So:

```
java
```

```
result[i] = result[i - 1] * nums[i - 1];
```

For example, at:

```
text
```

```
i = 3
```

we want:

```
text
```

```
nums[0] × nums[1] × nums[2]
```

so we multiply the previous prefix by:

```
text
```

```
nums[2]
```

which is:

```
text
```

```
nums[i - 1]
```

Again, we deliberately exclude:

```
text
```

```
nums[i]
```

21. Why Start with 1 ?

Both directions start with:

```
java
```

```
1
```

This is not arbitrary.

For multiplication:

```
text
```

```
1
```

is the identity element.

So:

```
text
```

```
1 × x = x
```

At the first position there are no elements on the left.

The empty product is naturally treated as:

```
text
```

```
1
```

Likewise, there are no elements to the right of the final position.

So:

```
text
```

```
suffix = 1
```

is correct.

22. The "Empty Prefix / Empty Suffix" Idea

At index 0 :

```
text
```

```
left side = nothing
```

so:

```
text
```

```
prefix = 1
```

At index n - 1 :

```
text
right side = nothing
```

so:

```
text

suffix = 1
```

This idea occurs in many prefix/suffix problems.

23. Why Not Use Division?

The interviewer will very likely ask this.

A tempting approach is:

```
text

totalProduct = product of all elements
answer[i] = totalProduct / nums[i]
```

This works only when there are no zeroes.

24. Division Problem – Zero

Consider:

```
text

nums = [1, 2, 0, 4]
```

Total product:

```
text

0
```

Now:

```
text

answer[0] = 0 / 1 = 0
answer[1] = 0 / 2 = 0
```

But:

```
text
```

```
answer[2]
```

should be:

```
text
```

```
1 × 2 × 4 = 8
```

and:

```
text
```

```
0 / 0
```

is undefined.

So the division solution breaks.

25. Could We Handle Zero Separately?

Yes.

You could build a division-based algorithm with special cases:

```
text
```

```
number of zeros = 0  
number of zeros = 1  
number of zeros > 1
```

For example:

Zero count = 0

Normal division works.

Zero count = 1

Only the position containing zero gets the product of all non-zero elements.

Every other position gets zero.

Zero count > 1

Every answer is zero.

This can be made correct.

But the problem explicitly says:

```
text
```

```
without division
```

so it is not the desired solution.

The better interview response is:

"Even though I could handle zero with special cases, division is explicitly prohibited, so the prefix/suffix approach is the correct general solution."

26. Approach Comparison

Approach	Time	Extra Space	Problems
Brute force	$O(n^2)$	$O(1)$	Repeats work
Total product + division	$O(n)$	$O(1)$	Division prohibited; zero problem
Prefix + suffix arrays	$O(n)$	$O(n)$	Extra arrays
Prefix in result + running suffix	$O(n)$	$O(1)$	Best standard solution

This progression is exactly what I would explain to the interviewer.

27. Pattern Recognition

This problem is primarily:

Prefix + Suffix Pattern

Whenever a problem asks for something like:

```
text
```

```
Everything before i  
+  
Everything after i
```

think:

```
text
```

```
prefix + suffix
```

For multiplication:


```
text
leftProduct × rightProduct
```

For sums:

```
text

leftSum + rightSum
```

This pattern appears frequently.

28. More General Prefix/Suffix Pattern

Suppose:

```
text

nums = [a, b, c, d, e]
```

For index 2 :

```
text

left:
a × b

right:
d × e
```

So:

```
text

answer[2] =
(a × b) × (d × e)
```

The current element `c` is naturally excluded.

This is why prefix/suffix decomposition is so powerful.

29. Another Pattern — Running Aggregate

The optimal solution also uses:

```
text

running suffix product
```

instead of storing every suffix.

This is a broader pattern:

If you only need the previous aggregate while traversing, you may not need an entire auxiliary array.

Examples:

```
text

running sum
running max
running min
running product
prefix XOR
```

This is a reusable optimization technique.

30. Why the Result Array Can Be Reused

At first, we think we need:

```
text

prefix[]
suffix[]
result[]
```

Then ask:

"Does the final result array need to exist separately from prefix?"

No.

The final result is literally:

```
text

prefix × suffix
```

So we can store the prefix inside:

```
text

result[]
```

and apply the suffix later.

This is a common **in-place reuse of output storage** optimization.

31. Is Space Really $O(1)$?

This is a frequent interview trick.

We return:

```
java  
  
int[] result
```

which is $O(n)$.

So technically, the algorithm uses $O(n)$ memory because the output itself has size n .

But when interviewers say:

" $O(1)$ extra space"

they normally mean:

"Apart from the output array, no additional $O(n)$ data structure is used."

So the precise answer is:

```
text  
  
Output space:  $O(n)$   
Extra/auxiliary space:  $O(1)$ 
```

This is a very important distinction.

Don't say:

"The entire algorithm uses $O(1)$ space."

Better:

"It uses $O(1)$ auxiliary space excluding the required output array."

32. Java-Specific Follow-up — Integer Overflow

This is an important practical issue.

The implementation uses:

```
java  
  
int
```

and multiplication:

```
java  
  
result[i] * nums[j]
```

can overflow `int`.

Example:

```
text  
  
2,000,000,000 × 2
```

does not fit in Java's 32-bit signed integer range.

Java integer overflow wraps around.

So the actual expected type depends on the problem's constraints.

If products can exceed `int`, use:

```
java  
  
long[]
```

and:

```
java  
  
long suffix = 1L;
```

Example:

```
java  
  
public long[] productExceptSelf(long[] nums) {  
  
    int n = nums.length;  
    long[] result = new long[n];  
  
    result[0] = 1L;  
  
    for (int i = 1; i < n; i++) {  
        result[i] = result[i - 1] * nums[i - 1];  
    }  
  
    long suffix = 1L;
```

```
for (int i = n - 1; i >= 0; i--) {  
    result[i] *= suffix;  
    suffix *= nums[i];  
}  
  
return result;  
}
```

For standard coding-platform constraints, `int` is often what the problem expects.

33. Java-Specific Follow-up — What about `BigInteger` ?

If this were a production problem where arbitrary-precision products are required, Java provides:

```
java
```

```
BigInteger
```

But:

```
java
```

```
BigInteger
```

is much more expensive than primitive arithmetic.

So choose based on numeric bounds.

For normal DSA questions:

```
text
```

```
int
```

or:

```
text
```

```
long
```

is expected.

34. Edge Case — Single Element

Consider:

```
text
```

```
nums = [5]
```

What is the product of every element except 5 ?

There are no other elements.

The standard empty product is:

```
text
```

```
1
```

So:

```
text
```

```
[1]
```

Our algorithm handles it.

35. Edge Case – Two Elements

```
text
```

```
nums = [2, 3]
```

Answer:

```
text
```

```
[3, 2]
```

Prefix pass:

```
text
```

```
[1, 2]
```

Suffix pass:

```
text
```

```
[3, 2]
```

Correct.

36. Edge Case — One Zero

Consider:

```
text
```

```
nums = [1, 2, 0, 4]
```

Expected:

```
text
```

```
[0, 0, 8, 0]
```

Let's see why the algorithm handles it naturally.

Prefix:

```
text
```

```
[1, 1, 2, 0]
```

Suffix processing eventually gives:

```
text
```

```
[0, 0, 8, 0]
```

No special zero handling is required.

This is one of the biggest advantages over division.

37. Edge Case — Two Zeros

```
text
```

```
nums = [1, 0, 3, 0]
```

Every answer should be:

```
text
```

```
[0, 0, 0, 0]
```

because after removing any one element, there is still another zero in the remaining array.

The prefix/suffix algorithm naturally produces this.

38. Edge Case — Negative Numbers

Consider:

```
text
```

```
nums = [-1, 2, -3, 4]
```

The same multiplication logic works.

There is nothing special about negative values.

The answer is determined by normal signed multiplication.

39. Edge Case — All Ones

```
text
```

```
[1, 1, 1, 1]
```

Every product except self is:

```
text
```

```
1
```

Result:

```
text
```

```
[1, 1, 1, 1]
```

The prefix/suffix method handles this trivially.

40. Edge Case — Large Array

For:

```
text
```

```
n = 1,000,000
```

the optimal algorithm still performs only:

```
text
```


approximately $2n$

main-pass operations.

That's why:

```
text
```

```
O(n)
```

is dramatically better than:

```
text
```

```
O(n2)
```

41. Java-Specific Follow-up — Why `int[]` for result?

Because the input is:

```
java
```

```
int[] nums
```

and the typical problem constraints expect the products to fit into `int`.

If the constraints say the answer fits in a 32-bit signed integer, `int[]` is appropriate.

Always read the numeric constraints before selecting the arithmetic type.

42. Java-Specific Follow-up — Do we need `Arrays.fill()` ?

No.

Because we explicitly initialize:

```
java
```

```
result[0] = 1;
```

and calculate every subsequent value.

Java automatically initializes the array to zero, but we overwrite every required position during the passes.

43. Java-Specific Follow-up – Can I write one loop?

You could attempt some more complicated formulation, but there is no real benefit.

Two directional passes are natural:

```
text

left → right
right → left
```

The important optimization is space, not minimizing the number of loops.

A clean two-pass $O(n)$ solution is preferable.

44. Why Not Use Streams?

You could technically construct solutions using Streams, but this is not a good place to prioritize them.

This problem is about:

```
text

prefix state
+
reverse traversal
+
constant auxiliary space
```

A traditional loop makes the state transitions obvious.

For DSA interviews:

```
text

clarity > unnecessary abstraction
```

45. Tricky Question – Can we modify `nums` ?

The standard optimal solution does not need to modify `nums` .

That's good because:

```
text

input remains unchanged
```

while:

```
text  
  
result
```

stores the prefix and then final product.

If an interviewer permits mutation, there may be other approaches, but there is no need to mutate the input here.

46. Tricky Question — Can we calculate suffix first?

Yes.

You could build suffix information into the result and then run a prefix multiplication pass.

The direction doesn't matter as long as the invariant is maintained.

For example:

```
text  
  
right pass:  
result[i] = suffix  
  
left pass:  
result[i] *= prefix
```

The important concept is:

```
text  
  
one directional aggregate  
+  
opposite directional aggregate
```

not the specific order.

47. Alternative Optimal Formulation

Another common implementation builds suffix into the result first:

```
java  
  
public int[] productExceptSelf(int[] nums) {  
  
    int n = nums.length;
```

```

int[] result = new int[n];

result[n - 1] = 1;

for (int i = n - 2; i >= 0; i--) {
    result[i] = result[i + 1] * nums[i + 1];
}

int prefix = 1;

for (int i = 0; i < n; i++) {
    result[i] *= prefix;
    prefix *= nums[i];
}

return result;
}

```

This is equally valid.

Here:

text

result

initially stores suffix products.

Then:

text

prefix

is applied left to right.

48. Which Direction Is Better?

Neither asymptotically.

Both are:

text

Time: $O(n)$

Extra space: $O(1)$

I prefer the version you supplied:

text

prefix → result
suffix → running variable

because it is particularly intuitive:

text

result[i]
already has left product
suffix
has right product

Then multiply them.

49. Pattern Recognition — Running Prefix/Suffix

A useful interview mental checklist:

When a problem says:

text

except self
everything before i
everything after i
left contribution + right contribution

immediately think:

text

Prefix
+
Suffix

Then ask:

"Can I reuse the output array to store one side?"

If yes:

text

$O(n)$ time
 $O(1)$ auxiliary space

This is the key optimization.

50. Production Variation — Price Basket / Inventory Calculation

Suppose a service processes:

```
text
```

```
prices = [2, 3, 5, 7]
```

and needs a transformed array where each position represents a product of all other factors for some analytical model.

The same prefix/suffix decomposition can be used whenever:

```
text
```

```
output[i]
```

depends on all values except one.

This can appear in:

```
text
```

```
feature computation  
risk models  
scoring systems  
batch analytics
```

The DSA pattern is more general than literal product calculations.

51. Production Variation — API Metrics

Imagine a monitoring pipeline has independent multiplicative weights:

```
text
```

```
weights = [...]
```

and each metric needs:

```
text
```

```
product of all weights except the current service
```

Naively recomputing every product is expensive.

Instead:

```
text

left product
+
right product
```

allows linear-time batch processing.

Again, the production value is the general prefix/suffix technique.

52. Production Concern – Overflow

This becomes much more important in real systems.

A production service should not blindly use:

```
java

int
```

for potentially huge multiplicative calculations.

Depending on requirements, you may need:

```
text

long
BigInteger
checked arithmetic
domain-specific bounds
```

The algorithm remains the same, but the numeric representation changes.

53. Production Concern – Very Large Inputs

Suppose the array has hundreds of millions of elements.

The algorithm is still $O(n)$, but:

```
text

result = new int[n]
```

itself requires substantial memory.

You may need to consider:

```
text
```

```
streaming  
chunking  
off-heap storage  
distributed processing
```

depending on whether the entire answer must be materialized.

For a standard coding interview, however, return the array directly.

54. Deep Interview Question — Is There Any Way to Do $O(1)$ Total Space?

No, not if the function must return `n` separate answers.

The output itself is:

```
text
```

```
 $O(n)$ 
```

So what we mean by " $O(1)$ extra space" is:

```
text
```

```
required output:  $O(n)$   
auxiliary workspace:  $O(1)$ 
```

This distinction is often tested.

55. Common Mistake — Division

```
java
```

```
total / nums[i]
```

Fails because:

```
text
```

```
zero
```

can make division undefined and the problem explicitly prohibits division.

56. Common Mistake – Recomputing Prefix/Suffix

Some candidates do:

```
text

for each i:
    calculate left product
    calculate right product
```

This goes back toward:

```
text

O(n2)
```

The whole point is to **reuse previously calculated aggregate information**.

57. Common Mistake – Using Separate Prefix and Suffix Arrays Without Asking About Space

That solution is correct:

```
text

O(n)
```

time.

But if the interviewer asks:

"Can you reduce space?"

you should immediately realize:

```
text

prefix array
```

can be stored in:

```
text

result
```

and:

```
text
suffix array
```

can become:

```
text

one running variable
```

That is the optimal transformation.

58. Common Mistake – Updating Suffix Too Early

Wrong:

```
java

suffix *= nums[i];
result[i] *= suffix;
```

because now `suffix` includes:

```
text

nums[i]
```

which must be excluded.

Correct:

```
java

result[i] *= suffix;
suffix *= nums[i];
```

The order matters.

59. Common Mistake – Incorrect Prefix Initialization

Wrong:

```
java

result[0] = nums[0];
```

Then `result[0]` would incorrectly include the current element.

Correct:

```
java  
  
result[0] = 1;
```

because index 0 has no elements to its left.

60. The Core Invariant — Interview-Level Explanation

This is perhaps the strongest explanation you can give.

First pass invariant

After processing index i :

```
text  
  
result[i]  
=  
product of nums[0 ... i-1]
```

So `result[i]` contains everything to the left.

Second pass invariant

Before processing index i :

```
text  
  
suffix  
=  
product of nums[i+1 ... n-1]
```

So `suffix` contains everything to the right.

Therefore:

```
text  
  
result[i] × suffix
```

is:

```
text  
  
product of everything except nums[i]
```

Then we update:

text

```
suffix *= nums[i]
```

so that the invariant becomes correct for the next index to the left.

This is the cleanest proof of correctness.

61. Mathematical View

For each i :

text

```
answer[i]  
=  
 $\prod$  nums[j]  
where  $j \neq i$ 
```

Break the set of indices into:

text

```
 $j < i$ 
```

and:

text

```
 $j > i$ 
```

Therefore:

text

```
answer[i]  
=  
( $\prod_{j < i}$  nums[j])  
×  
( $\prod_{j > i}$  nums[j])
```

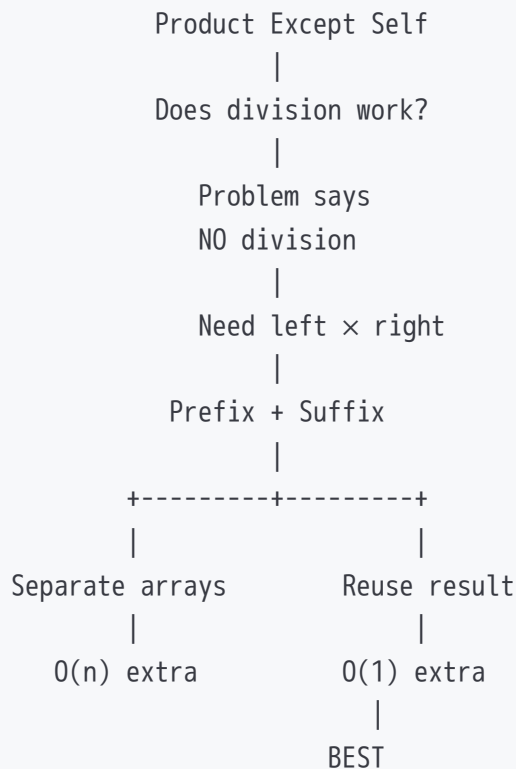
which is exactly:

text

```
prefix[i] × suffix[i]
```

62. Final Decision Tree

text



63. Final Interview Conversation

Interviewer: "Solve Product of Array Except Self."

You:

"The brute-force solution calculates the product of all other elements for every index, which is $O(n^2)$."

Then:

"I can observe that each answer is the product of everything to the left and everything to the right, so I can use prefix and suffix products. That gives $O(n)$ time but $O(n)$ auxiliary space."

Then interviewer asks:

"Can you optimize space?"

You:

"Yes. I can store the prefix product directly in the result array and maintain only a single running suffix product during a reverse traversal. Then the result at every index is **prefix × suffix**."

Then:

"This gives $O(n)$ time and $O(1)$ auxiliary space, excluding the required output array."

That's an excellent progression.

64. Final Interviewer-Ready Optimal Code

java

```
public int[] productExceptSelf(int[] nums) {  
  
    int n = nums.length;  
    int[] result = new int[n];  
  
    // result[i] = product of all elements before i  
    result[0] = 1;  
  
    for (int i = 1; i < n; i++) {  
        result[i] = result[i - 1] * nums[i - 1];  
    }  
  
    // suffix = product of all elements after i  
    int suffix = 1;  
  
    for (int i = n - 1; i >= 0; i--) {  
  
        result[i] *= suffix;  
  
        // Include nums[i] for the next position to the left  
        suffix *= nums[i];  
    }  
  
    return result;  
}
```

Complexity

text

Time: $O(n)$

Auxiliary space: $O(1)$

Output space: $O(n)$

65. Final Interview-Ready Answer

If the interviewer asks:

"Explain your optimal solution."

Say:

"For every index i , the required product can be split into two independent parts: the product of all elements before i and the product of all elements after i .

The brute-force solution recomputes these products for every index, giving $O(n^2)$. A prefix/suffix-array solution reduces this to $O(n)$, but uses $O(n)$ additional space.

To optimize the space, I'll store the prefix product directly in the output array. Then I'll traverse from right to left while maintaining a running suffix product. Before updating the suffix with `nums[i]`, I multiply the current result by the existing suffix, because that suffix represents only the elements to the right of i .

This gives $O(n)$ time and $O(1)$ auxiliary space, excluding the output array. It also handles zeros naturally, which is another reason not to use division."

66. Final Cheat Sheet

Approach 1 — Brute Force

text

```
For every i:  
    multiply all j != i
```

Time: $O(n^2)$

Space: $O(1)$ extra

Approach 2 — Prefix + Suffix Arrays

text

```
prefix[i] = product before i  
suffix[i] = product after i
```

```
answer[i] = prefix[i] * suffix[i]
```

Time: $O(n)$

Space: $O(n)$

Approach 3 — Optimal

text

result[] stores prefix
+
running suffix variable

Time: $O(n)$

Auxiliary Space: $O(1)$

Core Pattern

text

Everything before i
+
Everything after i
↓
Prefix + Suffix

Optimization Pattern

text

Need an entire prefix array?
↓
Reuse output array

Need an entire suffix array?
↓
Use running aggregate

Critical Invariant

text

result[i]
= product before i

suffix
= product after i

Therefore:

text

```
result[i] *= suffix
```

Then:

text

```
suffix *= nums[i]
```

Key Interview Traps

text

Division

→ zero problem + prohibited

```
result[0] = nums[0]
```

→ wrong, current element gets included

suffix update before result

→ wrong, `nums[i]` gets included

"Space = $O(1)$ "

→ say " $O(1)$ auxiliary space excluding output"

Most Important Connection

This problem is a major **prefix/suffix template**.

Once you recognize:

text

`answer[i]` depends on

all elements before `i`

+

all elements after `i`

you should immediately consider:

text

PREFIX + SUFFIX

And then ask the second optimization question:

text

Can one side be stored in the output
and the other maintained with one running variable?

For this problem, that gives the **interviewer-expected best solution: $O(n)$ time and $O(1)$ auxiliary space**.

2026-09-02 21:09:15 You 

ok as aboved continue for next question, keep this same interview structure: ****brute force → better → optimal → pattern recognition → why the data structure works → dry run → edge cases → Java-specific follow-ups → production variation → final interviewer-ready answer****.

Q7. Best Time to Buy and Sell Stock

Question

Find the maximum profit from one transaction.

Core idea

Maintain minimum price seen so far.

java

```
public int maxProfit(int[] prices) {  
    int minPrice = Integer.MAX_VALUE;  
    int maxProfit = 0;  
  
    for (int price : prices) {  
        minPrice = Math.min(minPrice, price);  
        maxProfit = Math.max(maxProfit, price - minPrice);  
    }  
  
    return maxProfit;  
}
```

Complexity

text

$O(n)$

Interview explanation

At day i:

text

profit = todayPrice - cheapestPreviousPrice

12

Follow-up

What if multiple transactions are allowed?

That becomes a different problem:

text

sum every positive difference



ChatGPT gpt-5-6-t-mini

Q7. Best Time to Buy and Sell Stock

This is a classic **one-pass array optimization** problem and is very common in product-company interviews.

It looks like a buy/sell problem, but the real DSA pattern is:

text

For every current value:
compare it with the best previous value

The important progression is:

text

Brute Force
↓
Try every buy/sell pair
↓
Notice that for selling today,
we only care about the minimum price before today
↓
Track minimum-so-far
↓
O(n) one-pass solution

This is closely related to the prefix-style reasoning from Q6, but here instead of maintaining a product, we're maintaining a **minimum prefix value**.

1. Problem

Given an array `prices`, where:

text

```
prices[i] = stock price on day i
```

You may:

- buy exactly once
- sell exactly once
- buy before selling

Return the maximum possible profit.

If no profit is possible, return `0`.

2. Example

```
text
```

```
prices = [7, 1, 5, 3, 6, 4]
```

Best transaction:

```
text
```

```
Buy  at 1  
Sell at 6
```

Profit:

```
text
```

```
6 - 1 = 5
```

So:

```
text
```

```
5
```

3. How I would Start in a Real Interview

A strong opening would be:

"For a single transaction, I need to choose a buying day before the selling day and maximize `sellPrice - buyPrice`. The brute-force approach checks every possible pair, which is $O(n^2)$. But if I'm considering selling on day `i`, the only thing I need to know about previous days is the

minimum price seen so far. So I can scan once, maintain the minimum price, calculate today's possible profit, and update the maximum profit. That gives $O(n)$ time and $O(1)$ extra space."

This is exactly the kind of explanation expected from a 2–2.5 YOE candidate.

4. Approach 1 — Brute Force

The most straightforward approach is to try every possible buy day and sell day.

For every:

```
text
```

```
i
```

try every later day:

```
text
```

```
j > i
```

and calculate:

```
text
```

```
prices[j] - prices[i]
```

Keep the maximum.

5. Java — Brute Force

```
java
```

```
public int maxProfit(int[] prices) {  
  
    int maxProfit = 0;  
  
    for (int buy = 0; buy < prices.length; buy++) {  
  
        for (int sell = buy + 1; sell < prices.length; sell++) {  
  
            int profit = prices[sell] - prices[buy];  
  
            maxProfit = Math.max(maxProfit, profit);  
        }  
    }  
}
```

```
    return maxProfit;  
}
```

6. Dry Run — Brute Force

Take:

text

```
prices = [7, 1, 5, 3, 6, 4]
```

Buy at 7 :

text

```
sell at 1 → -6  
sell at 5 → -2  
sell at 3 → -4  
sell at 6 → -1  
sell at 4 → -3
```

No profit.

Buy at 1 :

text

```
sell at 5 → 4  
sell at 3 → 2  
sell at 6 → 5  
sell at 4 → 3
```

Maximum so far:

text

5

Buy at 5 :

text

```
sell at 3 → -2  
sell at 6 → 1  
sell at 4 → -1
```

And so on.

Final:

```
text
```

```
5
```

7. Complexity – Brute Force

There are potentially:

```
text
```

```
n(n - 1) / 2
```

buy/sell pairs.

Therefore:

```
text
```

```
Time:  $O(n^2)$ 
```

```
Space:  $O(1)$ 
```

Correct, but unnecessarily expensive.

8. First Important Observation

Now ask:

"If I decide to sell on day `i`, which previous buying day gives me the maximum profit?"

Suppose today's price is:

```
text
```

```
prices[i] = 10
```

We want:

```
text
```

```
10 - buyPrice
```

To maximize that, we obviously want:

```
text
smallest previous buyPrice
```

So:

```
text

Best profit for today
=
today's price
-
minimum price seen before today
```

That is the key insight.

9. Approach 2 — Track Minimum Price So Far

Instead of comparing today's price with every previous day, maintain:

```
text

minPrice
```

which represents:

```
text

minimum price seen so far
```

Then for every price:

```
text

profit = price - minPrice
```

and update:

```
text

maxProfit
```

10. Java — Optimal One-Pass Solution

```
java
```



```
public int maxProfit(int[] prices) {  
  
    int minPrice = Integer.MAX_VALUE;  
    int maxProfit = 0;  
  
    for (int price : prices) {  
  
        minPrice = Math.min(minPrice, price);  
  
        maxProfit = Math.max(  
            maxProfit,  
            price - minPrice  
        );  
    }  
  
    return maxProfit;  
}
```

This is the standard optimal solution.

But let's understand it deeply.

11. Dry Run — Optimal Solution

Input:

text

```
prices = [7, 1, 5, 3, 6, 4]
```

Initially:

text

```
minPrice = Integer.MAX_VALUE  
maxProfit = 0
```

Day 1 — Price 7

Update minimum:

text

```
minPrice = min(MAX_VALUE, 7)  
          = 7
```

Profit:

text

$$7 - 7 = 0$$

Maximum:

text

$$\text{maxProfit} = 0$$

State:

text

$$\begin{aligned}\text{minPrice} &= 7 \\ \text{maxProfit} &= 0\end{aligned}$$

Day 2 — Price 1

Minimum:

text

$$\begin{aligned}\text{minPrice} &= \min(7, 1) \\ &= 1\end{aligned}$$

Profit:

text

$$1 - 1 = 0$$

State:

text

$$\begin{aligned}\text{minPrice} &= 1 \\ \text{maxProfit} &= 0\end{aligned}$$

This is now the best buying price seen so far.

Day 3 — Price 5

Minimum:

text

```
minPrice = min(1, 5)
          = 1
```

Potential profit:

text

$5 - 1 = 4$

Update:

text

$\text{maxProfit} = 4$

Day 4 — Price 3

Minimum stays:

text

1

Profit:

text

$3 - 1 = 2$

Maximum remains:

text

4

Day 5 — Price 6

Minimum:

text

1

Profit:

text

$$6 - 1 = 5$$

Update:

text

$$\text{maxProfit} = 5$$

Day 6 — Price 4

Profit:

text

$$4 - 1 = 3$$

Maximum stays:

text

5

Final:

text

5

12. The Most Important Interview Detail — Why Is This Correct?

The interviewer may ask:

"How do you know your minimum price is a valid buying day?"

Because we process the array from left to right.

At day `i`, `minPrice` is computed only from:

text

`prices[0 ... i]`

or, more precisely, after updating it with today's price, the profit:

```
text
price - minPrice
```

cannot create an invalid buy-after-sell transaction because the current price itself gives zero profit.

For a positive profit:

```
text
price > minPrice
```

the minimum necessarily came from an earlier day.

This maintains the ordering requirement:

```
text
buy day < sell day
```

That's an important correctness argument.

13. A Better Invariant

A very strong interview explanation is:

"Before processing each price, `minPrice` represents the minimum stock price seen so far, and `maxProfit` represents the maximum profit achievable using a buy and sell within the processed prefix."

After processing today's price:

```
text

minPrice
  = minimum price in processed prefix

maxProfit
  = maximum valid profit in processed prefix
```

This invariant makes the entire algorithm easy to reason about.

14. Why Do We Track Minimum Instead of Maximum?

Because we're trying to maximize:

```
text
```

```
sell - buy
```

For a fixed selling price:

```
text
```

```
sell
```

we want the smallest possible:

```
text
```

```
buy
```

So:

```
text
```

```
maximize(sell - buy)
```

↓

```
minimize buy
```

↓

```
track minimum prefix price
```

This is the mathematical heart of the solution.

15. Alternative Interpretation

We can also rewrite profit as:

```
text
```

```
profit = today's price - cheapest previous price
```

So every day asks:

"If I sold today, how much could I have made using the best possible purchase before today?"

Then take the maximum among all days.

This converts the problem from:

```
text
```

```
all buy/sell pairs
```

to:

```
text
```

```
best buy for today's sell
```

That's the optimization.

16. Why Doesn't This Need a HashMap?

Notice the difference from the previous problems.

We don't need:

```
text
```

```
membership  
frequency  
lookup by key
```

We only need:

```
text
```

```
minimum value seen so far  
maximum answer seen so far
```

So two scalar variables are enough.

This is a good example of an important optimization principle:

Before reaching for a data structure, ask whether a simple running state is sufficient.

17. Pattern Recognition

This problem belongs to the:

Prefix Minimum / Running Minimum Pattern

Whenever you see:

```
text
```

```
current value compared against  
best previous value
```

consider:

```
text
```

```
maintain best-so-far
```

Here:

```
text
```

```
best previous = minimum price
```

Other examples include:

```
text
```

```
maximum difference  
minimum difference under ordering constraint  
maximum subarray variants  
best previous score  
stock problems
```

18. Another Way to Think About the Pattern

Suppose the question is:

```
text
```

```
maximize:  
currentValue - previousValue
```

If `previousValue` must occur earlier, then:

```
text
```

```
track minimum previous value
```

Similarly, if the problem were:

```
text
```

```
maximize:  
previousValue - currentValue
```

you might instead track:

```
text
```

```
maximum previous value
```


This is a highly reusable mental pattern.

19. Approach Comparison

Approach	Time	Extra Space	Idea
Brute force	$O(n^2)$	$O(1)$	Every buy/sell pair
Minimum-so-far	$O(n)$	$O(1)$	Best previous buy
Two pointers	Not natural	—	No sorted structure
HashMap	$O(n)$	$O(n)$	Unnecessary

The one-pass solution is both asymptotically optimal and very simple.

20. Is $O(n)$ the Best Possible?

Yes.

You must inspect the prices because any unseen day could potentially contain:

text

new minimum

or:

text

new maximum profit

Therefore, in the general case, you cannot determine the answer without considering the elements.

So:

text

$\Omega(n)$

is effectively required.

Our algorithm is:

text

$O(n)$

Therefore it is asymptotically optimal.

This is a nice point to mention if the interviewer asks:

"Can we do better?"

Answer:

"Not asymptotically. We need to inspect the array, so $O(n)$ is optimal."

21. Edge Case — Prices Always Decrease

Example:

```
text
```

```
prices = [7, 6, 4, 3, 1]
```

There is no profitable transaction.

The algorithm calculates:

```
text
```

```
all potential profits <= 0
```

but starts with:

```
java
```

```
maxProfit = 0;
```

so it returns:

```
text
```

```
0
```

This correctly means:

"Don't make a transaction."

22. Edge Case — Prices Always Increase

```
text
```

```
prices = [1, 2, 3, 4, 5]
```

The algorithm sees:

text

```
1 → minimum  
2 → profit 1  
3 → profit 2  
4 → profit 3  
5 → profit 4
```

Answer:

text

4

Buy once at the first day and sell at the last day.

23. Edge Case — One Price

text

```
prices = [5]
```

No possible transaction.

Return:

text

0

24. Edge Case — Empty Array

Depending on the problem constraints, empty input may or may not be allowed.

A defensive implementation could say:

java

```
if (prices == null || prices.length < 2) {  
    return 0;  
}
```

But on coding platforms, if the constraints guarantee at least one element, this isn't necessary.

For production API code, validating input may be appropriate.

25. Edge Case — Two Prices

```
text
```

```
prices = [2, 5]
```

Profit:

```
text
```

```
5 - 2 = 3
```

Return:

```
text
```

```
3
```

26. Edge Case — Two Prices in Wrong Order

```
text
```

```
prices = [5, 2]
```

We cannot buy at 2 and sell at 5 because the 5 happened first.

Correct answer:

```
text
```

```
0
```

This demonstrates why order matters.

27. Tricky Follow-Up — Why Can't We Just Find Global Min and Global Max?

This is a very common trap.

Suppose:

```
text
```

```
prices = [7, 6, 4, 10, 2, 8]
```

Global minimum:

```
text
```

```
2
```

Global maximum:

```
text
```

```
10
```

If we blindly calculate:

```
text
```

```
10 - 2 = 8
```

that's invalid because:

```
text
```

```
10 occurs before 2
```

You cannot sell before buying.

So we need:

```
text
```

```
minimum price seen BEFORE today's selling day
```

not simply the global minimum.

This is one of the most important conceptual points.

28. Example Showing Why Prefix Minimum Matters

```
text
```

```
prices = [8, 2, 6, 1, 5]
```

Global:

```
text
min = 1
max = 8
```

Global difference:

```
text

8 - 1 = 7
```

But invalid because 8 comes before 1 .

Valid best:

```
text

Buy at 2
Sell at 6
profit = 4
```

The one-pass algorithm correctly returns:

```
text

4
```

29. Java-Specific Follow-Up – Why Integer.MAX_VALUE ?

The code starts with:

```
java

int minPrice = Integer.MAX_VALUE;
```

Why?

Because any valid stock price will be smaller than:

```
text

Integer.MAX_VALUE
```

so the first price automatically becomes the initial minimum.

Equivalent approach:

```
java
```

```
int minPrice = prices[0];
```

and start looping from index `1` .

For example:

```
java
```

```
public int maxProfit(int[] prices) {  
  
    if (prices.length < 2) {  
        return 0;  
    }  
  
    int minPrice = prices[0];  
    int maxProfit = 0;  
  
    for (int i = 1; i < prices.length; i++) {  
  
        int profit = prices[i] - minPrice;  
  
        maxProfit = Math.max(maxProfit, profit);  
  
        minPrice = Math.min(minPrice, prices[i]);  
    }  
  
    return maxProfit;  
}
```

Both are correct.

30. Which Update Order Is Better?

You may notice two styles.

Style A

```
java
```

```
minPrice = Math.min(minPrice, price);  
maxProfit = Math.max(maxProfit, price - minPrice);
```

Style B

```
java
```

```
maxProfit = Math.max(maxProfit, price - minPrice);
minPrice = Math.min(minPrice, price);
```

At first glance, Style B looks more logically aligned with "buy before sell."

But Style A is also correct because if today's price is the new minimum, then:

text

```
price - minPrice = 0
```

so it does not create a positive invalid profit.

Still, for explanation clarity, many candidates prefer:

java

```
int profit = price - minPrice;

maxProfit = Math.max(maxProfit, profit);

minPrice = Math.min(minPrice, price);
```

because it makes the temporal relationship more explicit:

text

```
use previous minimum
    ↓
calculate today's sell profit
    ↓
update minimum for future days
```

I actually prefer this version when explaining the algorithm in an interview.

31. Recommended Interview Code

java

```
public int maxProfit(int[] prices) {

    if (prices == null || prices.length < 2) {
        return 0;
    }

    int minPrice = prices[0];
    int maxProfit = 0;
```



```

    for (int i = 1; i < prices.length; i++) {

        int profit = prices[i] - minPrice;

        maxProfit = Math.max(maxProfit, profit);

        minPrice = Math.min(minPrice, prices[i]);
    }

    return maxProfit;
}

```

This version makes the ordering particularly easy to explain.

32. Why This Version May Be Easier to Explain

At index `i`:

text

minPrice

represents:

text

cheapest price from days 0 to i-1

Then:

text

profit = prices[i] - minPrice

is guaranteed to be a valid transaction because the buying day occurs before day `i`.

After evaluating selling today:

text

minPrice = min(minPrice, prices[i])

prepares the state for future selling days.

This is a very clean invariant.

33. Java-Specific Follow-Up — Why Not Use `Math.max()` with Everything?

You could do:

```
java

maxProfit = Math.max(
    maxProfit,
    prices[i] - minPrice
);
```

That's fine.

But use variables if they make the logic clearer:

```
java

int profit = prices[i] - minPrice;
maxProfit = Math.max(maxProfit, profit);
```

In an interview, readable code is preferable.

34. Java-Specific Follow-Up — Integer Overflow

If prices are constrained to normal `int` values, subtraction is typically fine under the problem's stated bounds.

But in general:

```
java

prices[i] - minPrice
```

could theoretically overflow if values span the complete integer range.

If the domain requires it, use `long`.

For example:

```
java

long minPrice = prices[0];
long maxProfit = 0;
```

and return `long`.

Again, the numeric constraints determine the correct type.

35. Follow-Up – "What if Multiple Transactions Are Allowed?"

Now the problem changes.

This is no longer:

```
text

one buy + one sell
```

Instead, you can perform multiple transactions.

The common version is:

Buy and sell multiple times, but hold at most one stock at a time.

Then the standard greedy solution is:

```
text

add every positive difference
```

For:

```
text

prices = [7,1,5,3,6,4]
```

positive increases:

```
text

5 - 1 = 4
6 - 3 = 3
```

Total:

```
text

7
```

36. Java – Multiple Transactions

```
java

public int maxProfitMultipleTransactions(int[] prices) {
```

```

int profit = 0;

for (int i = 1; i < prices.length; i++) {

    if (prices[i] > prices[i - 1]) {
        profit += prices[i] - prices[i - 1];
    }
}

return profit;
}

```

Why?

Every increasing segment can be captured.

Example:

text

1 → 5

instead of:

text

1 → 2 → 3 → 4 → 5

both produce the same total profit:

text

4

This is a different problem and a different greedy strategy.

37. Important Interview Distinction

One transaction

text

track minimum so far

Unlimited transactions

text

```
sum positive consecutive differences
```

Do not accidentally use the multiple-transaction logic for the one-transaction problem.

For:

```
text
```

```
[1, 5, 2, 10]
```

multiple transactions:

```
text
```

```
(5 - 1) + (10 - 2)  
= 12
```

But one transaction:

```
text
```

```
10 - 1  
= 9
```

Different answers.

38. Follow-Up – What if You Must Return Buy and Sell Indices?

Very common.

Instead of only returning profit, track:

```
text
```

```
minPrice  
minIndex  
bestBuy  
bestSell  
maxProfit
```

Java

```
java
```

```
public int[] maxProfitIndices(int[] prices) {  
  
    if (prices == null || prices.length < 2) {
```

```

        return new int[]{-1, -1};
    }

    int minPrice = prices[0];
    int minIndex = 0;

    int maxProfit = 0;

    int bestBuy = -1;
    int bestSell = -1;

    for (int i = 1; i < prices.length; i++) {

        int profit = prices[i] - minPrice;

        if (profit > maxProfit) {
            maxProfit = profit;
            bestBuy = minIndex;
            bestSell = i;
        }

        if (prices[i] < minPrice) {
            minPrice = prices[i];
            minIndex = i;
        }
    }

    return new int[]{bestBuy, bestSell};
}

```

For:

text

[7,1,5,3,6,4]

returns:

text

[1,4]

meaning:

text

buy on day 1
sell on day 4

Profit:

```
text
```

```
6 - 1 = 5
```

39. Follow-Up — What if Tie-Breaking Matters?

Suppose several transactions give the same maximum profit.

Example:

```
text
```

```
[1, 3, 2, 4]
```

Maximum profit is:

```
text
```

```
2
```

both:

```
text
```

```
1 → 3
```

and:

```
text
```

```
2 → 4
```

produce profit 2.

If interviewer says:

```
"Return the earliest buy/sell pair."
```

then update only when:

```
java
```

```
profit > maxProfit
```

If they want a different tie rule, modify the condition accordingly.

This is a useful reminder:

Correct algorithm + unspecified tie behavior does not necessarily imply one unique output.

40. Follow-Up – What If Same-Day Buy/Sell Is Allowed?

It doesn't affect the maximum because:

```
text
```

```
price - price = 0
```

and the problem typically allows zero profit.

The important restriction remains:

```
text
```

```
buy <= sell
```

or, under a strict transaction interpretation:

```
text
```

```
buy day < sell day
```

Our algorithm naturally handles the standard requirement.

41. Follow-Up – Can We Use Prefix Minimum Array?

Yes.

You could build:

```
text
```

```
minPrefix[i]
```

where:

```
text
```

```
minPrefix[i]  
=  
minimum prices from 0 to i
```

Then calculate profit.

But that would use:

text

$O(n)$

extra space.

There's no need because we only need the current minimum.

This is exactly the same optimization concept as Q6:

text

Whole prefix array



Can I keep only the running aggregate?

Yes.

So:

text

one variable

is enough.

42. Pattern Connection to Q6

Q6:

text

Product Except Self
→ running prefix/suffix product

Q7:

text

Stock Profit
→ running prefix minimum

The deeper pattern is:

text

```
Process array once
+
maintain useful aggregate/state
+
derive answer for current index
```

The aggregate can be:

```
text

sum
minimum
maximum
product
count
frequency
```

This is one of the most important DSA habits to develop.

43. Why Not Two Pointers?

Two pointers are usually useful when:

```
text

array has useful ordering
```

or the problem has a left/right boundary structure.

The stock array is not sorted.

The crucial relationship is not:

```
text

left + right
```

but:

```
text

current price vs best earlier price
```

So a running minimum is the natural pattern.

44. Production Variation – Financial Service

Suppose a backend receives daily stock prices and needs to compute:

"What is the maximum profit an investor could have made from one buy and one sell during this period?"

The same algorithm applies:

```
text

prices
  ↓
minimum seen so far
  ↓
current possible profit
  ↓
maximum profit
```

It's $O(n)$ and requires only constant state.

45. Production Variation — Monitoring Metrics

The same logic can apply to a metric stream.

Suppose:

```
text

latencies = [...]
```

and we want:

"Maximum increase from an earlier observation to a later observation."

We can use exactly the same pattern:

```
text

track minimum earlier value
calculate current - minimum
track maximum
```

This demonstrates that the "stock" problem is really an instance of a more general **maximum difference under ordering constraint** problem.

46. Production Variation — Real-Time Streaming

This algorithm becomes especially attractive for streaming data.

Suppose prices arrive one at a time:

```
text
```

```
7  
1  
5  
3  
6  
4  
...
```

You don't need to keep the entire history.

You only need:

```
text
```

```
minimumPriceSeen  
maximumProfit
```

That means the core state is:

```
text
```

```
O(1)
```

per stream.

This is a major practical advantage.

47. Streaming Interpretation

Imagine values arrive continuously.

For every new price:

```
text
```

```
receive price  
  ↓  
calculate price - minPrice  
  ↓  
update maxProfit  
  ↓  
update minPrice
```

We never need to store the whole sequence.

So the algorithm has excellent streaming characteristics:

```
text
```

```
Time per item:  $O(1)$ 
```

```
State:  $O(1)$ 
```

This is a nice production connection.

48. Production Caveat — Historical Window

Real monitoring/financial systems may ask:

```
"Maximum profit over the last 30 days."
```

Now the entire historical state cannot be retained forever.

You'd need:

```
text
```

```
sliding window
```

```
expiration
```

```
timestamped values
```

Then the problem becomes more complicated.

The standard coding problem assumes the entire array is the relevant period.

49. Common Mistake — Updating Minimum After Calculating Incorrectly

A candidate may write:

```
java
```

```
minPrice = Math.min(minPrice, prices[i]);  
maxProfit = Math.max(  
    maxProfit,  
    prices[i] - minPrice  
);
```

This is actually still correct because the new minimum produces zero profit.

But if you're explaining the algorithm, the clearer formulation is:

```
java
```

```
int profit = prices[i] - minPrice;

maxProfit = Math.max(maxProfit, profit);

minPrice = Math.min(minPrice, prices[i]);
```

This explicitly says:

text

first evaluate selling today against previous buys
then allow today to become a buy price for future days

I would use this form in an interview.

50. Common Mistake — Global Min/Max

As discussed:

text

global max - global min

is not necessarily valid because:

text

minimum might occur after maximum

Always respect chronological order.

51. Common Mistake — Allowing Multiple Transactions

For:

text

[1,5,2,10]

one transaction:

text

9

Multiple transactions:

```
text
```

```
12
```

So always confirm:

```
text
```

```
one transaction
```

before applying the greedy stock algorithm.

52. Common Mistake – Starting `maxProfit` at Negative Infinity

Usually:

```
java
```

```
int maxProfit = 0;
```

is correct because the problem says you can choose not to transact and therefore zero is the minimum valid answer.

If the problem instead required:

"You must buy and sell exactly once."

then negative profits might matter, and initialization would need reconsideration.

This is an important requirement-driven distinction.

53. Tricky Follow-Up – "What if loss is allowed and transaction is mandatory?"

Then:

```
text
```

```
maxProfit = 0
```

would be wrong if all prices decrease.

For:

```
text
```

```
[5,4,3]
```

mandatory transaction means the best profit is:

text

$$3 - 5 = -2$$

whereas optional transaction gives:

text

0

So always check whether:

text

transaction is optional

The standard problem assumes zero is allowed.

54. Can We Prove $O(n)$ Is Optimal?

Yes.

Consider two arrays that differ only in the final element:

text

[5,4,3,2,1]

versus:

text

[5,4,3,2,100]

The final element can dramatically change the answer.

Without reading the relevant elements, you cannot know the maximum.

Therefore we need linear inspection in the general comparison model.

So:

text

$O(n)$

is asymptotically optimal.

55. Java-Specific Follow-Up – Can We Make It Functional?

A candidate might try:

```
java  
  
Arrays.stream(prices)...
```

But this problem is stateful because:

```
text  
  
minimum seen so far
```

depends on previous elements.

You can force it into Streams, but it generally becomes less readable.

For an interview, a simple loop is much better.

56. Java-Specific Follow-Up – Is `Math.min()` / `Math.max()` $O(1)$?

Yes.

They perform constant-time primitive comparisons.

So:

```
java  
  
Math.min(...)  
Math.max(...)
```

do not change the algorithm's $O(n)$ complexity.

57. Java-Specific Follow-Up – Why enhanced for-loop?

The version:

```
java  
  
for (int price : prices)
```

is concise and appropriate because we don't need the index to compute the profit.

However, if we're returning the buy/sell indices, use:

```
java  
  
for (int i = 1; i < prices.length; i++)
```

because we need the position.

This is a good example of choosing the loop style based on what information the algorithm requires.

58. Pattern Recognition – Maximum Difference

There is a more general formulation:

Given:

```
text  
  
a[0 ... n-1]
```

find:

```
text  
  
max(a[j] - a[i])
```

subject to:

```
text  
  
i < j
```

The solution is:

```
text  
  
For each j:  
    track minimum a[i] for i < j
```

This is the actual mathematical pattern behind the stock problem.

Once you recognize this, many seemingly different questions become easy.

59. Example of the General Pattern

```
text
```

```
nums = [8, 2, 10, 4, 7]
```

We want:

```
text
```

```
nums[j] - nums[i]
```

with:

```
text
```

```
i < j
```

At 10 :

```
text
```

```
10 - min(8,2)  
= 8
```

At 7 :

```
text
```

```
7 - 2  
= 5
```

Answer:

```
text
```

```
8
```

Exactly the same algorithm.

60. Best Approach Decision Tree

```
text
```

```
Best Time to Buy/Sell  
|  
One transaction?  
|  
Yes  
|
```

```
Can buy/sell order matter?
|
Yes
|
Track minimum prefix price
|
current price - minPrice
|
track maximum profit
|
O(n)
O(1)
```

61. Final Interview Sequence

Here's how I would actually conduct the solution on a whiteboard.

Step 1 — Brute force

"Try every pair `i < j`."

text

$O(n^2)$

Step 2 — Ask what matters when selling today

"For a fixed selling day, I only care about the cheapest earlier buying day."

Step 3 — Maintain state

text

minPrice
maxProfit

Step 4 — Scan once

text

```
profit = currentPrice - minPrice
maxProfit = max(maxProfit, profit)
minPrice = min(minPrice, currentPrice)
```

Step 5 — Complexity

```
text
O(n) time
O(1) auxiliary space
```

Step 6 — Edge cases

Mention:

```
text

decreasing array → 0
single element → 0
two elements
```

This is enough for the standard round.

62. Final Interview-Ready Code

```
java

public int maxProfit(int[] prices) {

    if (prices == null || prices.length < 2) {
        return 0;
    }

    int minPrice = prices[0];
    int maxProfit = 0;

    for (int i = 1; i < prices.length; i++) {

        // Best profit if we sell today
        int profit = prices[i] - minPrice;

        // Best profit seen so far
        maxProfit = Math.max(maxProfit, profit);

        // Best buying price for future days
        minPrice = Math.min(minPrice, prices[i]);
    }

    return maxProfit;
}
```

63. Final Interviewer-Expected Answer

If the interviewer asks:

"Explain your optimal approach."

A strong 2–2.5 YOE response would be:

"The brute-force solution checks every possible buy and sell pair, so it takes $O(n^2)$. We can optimize because for any fixed selling day, the best transaction is always obtained by buying at the minimum price seen before that day.

So I scan the array once and maintain two variables: `minPrice`, which is the cheapest price seen so far, and `maxProfit`, which is the best profit found so far.

For each current price, I first calculate `price - minPrice` as the profit if I sell today. I update `maxProfit`, then update `minPrice` so that today's price can be used as a buying price for future days.

This respects the requirement that buying happens before selling and gives $O(n)$ time with $O(1)$ auxiliary space. We cannot asymptotically improve the time because we need to inspect the prices."

64. Final Cheat Sheet

Approach 1 — Brute Force

text

```
For every buy day:
    for every later sell day:
        calculate profit
```

Time: $O(n^2)$

Space: $O(1)$

Approach 2 — Prefix Minimum

text

Track minimum price seen so far

`profit = currentPrice - minPrice`

Time: $O(n)$

Space: $O(1)$

Core Pattern

```
text

current value
-
best previous value
    ↓
running minimum / maximum
```

Key Formula

```
text

profit today
=
today's price
-
cheapest previous price
```

Important Invariant

```
text

minPrice
= minimum price before current day

maxProfit
= maximum valid profit found so far
```

Critical Trap

```
text

global max - global min
❌

Why?
minimum may appear after maximum.
```

Multiple transactions

```
text
```

```
sum positive differences
```

but that's a **different problem**.

Return indices

Track:

```
text
```

```
minIndex  
bestBuy  
bestSell
```

Production / Streaming

```
text
```

```
new price  
  ↓  
current profit  
  ↓  
update max  
  ↓  
update minimum
```

Only:

```
text
```

```
O(1)
```

state is required.

The most important pattern to remember

```
text
```

```
Find maximum of:
```

```
current - previous
```

```
where previous must occur earlier
```

```
↓
```


Track the minimum previous value



One-pass $O(n)$

This is the real lesson from Q7. It is not just a stock problem; it is a **"best previous value + current value"** pattern that appears in many product-company DSA questions.

2026-09-02 21:16:18 You 

ok as aboved continue for next question,i want the solutions as interviewer expcted in the interview of 2-2.5 year experience candidate and the solutions first is basic approach then other approach then next another better approach as like that possible approaches which patterns used and why that patterns used and at last best optimize approach as per the interviewer. keep this same interview structure: ****brute force → better → optimal → pattern recognition → why the data structure works → dry run → edge cases → Java-specific follow-ups → production variation → final interviewer-ready answer****.

Q8. Maximum Subarray

Question

Find the contiguous subarray with maximum sum.

Pattern

Kadane's Algorithm

java

```
public int maxSubArray(int[] nums) {
    int current = nums[0];
    int best = nums[0];

    for (int i = 1; i < nums.length; i++) {
        current = Math.max(nums[i], current + nums[i]);
        best = Math.max(best, current);
    }

    return best;
}
```

Complexity

text

Time: $O(n)$

Space: $O(1)$

Interview insight

At each position:

text

Should I:

start a new subarray?

13

OR

extend the old one?

Follow-up

How do you return the actual subarray?

Track:

text

start

bestStart

bestEnd



ChatGPT gpt-5-6-t-mini

Q8. Maximum Subarray

This is a **must-know product-company DSA problem** because it tests much more than Kadane's Algorithm.

For a 2–2.5 year Java candidate, the interviewer is usually checking whether you can move from:

text

Brute Force

↓

Prefix Sum

↓

Better $O(n \log n)$ divide-and-conquer

↓

Kadane's $O(n)$

and, more importantly, whether you understand the decision:

text

At index i:

Should I:

1. start a new subarray at `nums[i]`?

OR

2. extend the previous best subarray?

That decision is the heart of Kadane's Algorithm.

1. Problem

Given an integer array `nums`, find the **contiguous subarray** having the largest possible sum, and return that maximum sum.

Example

text

```
nums = [-2,1,-3,4,-1,2,1,-5,4]
```

The maximum-sum contiguous subarray is:

text

```
[4,-1,2,1]
```

Its sum is:

text

$$4 + (-1) + 2 + 1 = 6$$

Therefore:

text

6

2. The First Thing I Would Clarify

A very important word is:

contiguous

For example:

text

```
[4, -1, 2, 1]
```

is contiguous.

But:

```
text
```

```
[4, 2, 1]
```

is not a contiguous subarray because `-1` was skipped.

This problem is **not** asking for the maximum subsequence/subset sum.

That distinction is important because the solution would be different.

3. How I Would Start in a Real Interview

A strong opening:

"The brute-force solution considers every contiguous subarray and calculates its sum, which can be $O(n^3)$ if we sum each subarray separately. We can improve that to $O(n^2)$ using prefix sums. There is also an $O(n \log n)$ divide-and-conquer approach. But we can do better with Kadane's Algorithm. For each position, I only need to decide whether the current element is better taken as a new subarray or appended to the previous maximum subarray ending at the previous position. That gives an $O(n)$ time and $O(1)$ space solution."

That is exactly the progression I would recommend.

4. Approach 1 – Brute Force

The most direct idea:

Generate every possible subarray:

```
text
```

```
i ... j
```

and calculate its sum.

For example:

```
text
```

```
[1, -2, 3]
```

Possible subarrays:

```
text
```

```
[1]
[1,-2]
[1,-2,3]

[-2]
[-2,3]

[3]
```

Take the largest sum.

5. Brute Force – Very Basic Implementation

The most naive implementation recalculates the sum for every subarray.

```
java

public int maxSubArray(int[] nums) {

    int best = Integer.MIN_VALUE;

    for (int start = 0; start < nums.length; start++) {

        for (int end = start; end < nums.length; end++) {

            int sum = 0;

            for (int i = start; i <= end; i++) {
                sum += nums[i];
            }

            best = Math.max(best, sum);
        }
    }

    return best;
}
```

6. Why Is This $O(n^3)$?

There are:

text

$O(n^2)$

possible subarrays.

And for each one, we may calculate a sum in:

```
text
```

```
0(n)
```

So:

```
text
```

```
0(n2) × 0(n)
```

```
=
```

```
0(n3)
```

Complexity

```
text
```

```
Time: 0(n3)
```

```
Space: 0(1)
```

This is correct but obviously not acceptable for a large input.

7. Approach 2 – Brute Force With Running Sum

We can notice something.

Suppose we have:

```
text
```

```
start = 2
```

Instead of recalculating:

```
text
```

```
nums[2] + nums[3]
```

```
nums[2] + nums[3] + nums[4]
```

```
...
```

we can maintain a running sum.

```
java
```

```

public int maxSubArray(int[] nums) {

    int best = Integer.MIN_VALUE;

    for (int start = 0; start < nums.length; start++) {

        int sum = 0;

        for (int end = start; end < nums.length; end++) {

            sum += nums[end];

            best = Math.max(best, sum);
        }
    }

    return best;
}

```

Now the inner sum is $O(1)$ rather than $O(n)$.

8. Complexity of Running-Sum Approach

We have:

text

$O(n^2)$

pairs of (start, end) .

Each pair now takes only $O(1)$ additional work.

So:

text

Time: $O(n^2)$

Space: $O(1)$

This is a meaningful improvement:

text

$O(n^3)$

↓

$O(n^2)$

9. Approach 3 – Prefix Sum

There is another standard way.

Define:

```
text

prefix[i]
=
sum of elements before i
```

Then the sum of subarray:

```
text

[l, r]
```

can be calculated in $O(1)$:

```
text

prefix[r + 1] - prefix[l]
```

For example:

```
text

[1, -2, 3, 4]
```

prefix:

```
text

[0, 1, -1, 2, 6]
```

Sum from index 1 to 3 :

```
text

prefix[4] - prefix[1]
=
6 - 1
=
5
```

which corresponds to:

text

[-2, 3, 4]

10. Java – Prefix Sum Approach

java

```
public int maxSubArray(int[] nums) {  
  
    int n = nums.length;  
  
    int[] prefix = new int[n + 1];  
  
    for (int i = 0; i < n; i++) {  
        prefix[i + 1] = prefix[i] + nums[i];  
    }  
  
    int best = Integer.MIN_VALUE;  
  
    for (int start = 0; start < n; start++) {  
  
        for (int end = start; end < n; end++) {  
  
            int sum = prefix[end + 1] - prefix[start];  
  
            best = Math.max(best, sum);  
        }  
    }  
  
    return best;  
}
```

11. Complexity of Prefix Sum

Building prefix:

text

$O(n)$

Enumerating all subarrays:

text

$O(n^2)$

Each sum lookup:

text

$O(1)$

Overall:

text

Time: $O(n^2)$

Space: $O(n)$

So prefix sums improve the repeated-sum calculation, but they do **not** solve the real problem.

Why?

Because we are still considering every possible subarray.

That leads to a deeper question:

"Do we really need to consider every subarray?"

No.

12. Approach 4 — Divide and Conquer

This is another classic solution.

The maximum subarray can be in exactly one of three places:

text

1. Entirely in the left half

2. Entirely in the right half

3. Crosses the midpoint

So recursively solve left and right halves, then calculate the best crossing subarray.

13. Divide-and-Conquer Structure

Suppose:

text

```
[ -2, 1, -3, 4, -1, 2, 1, -5, 4 ]
```

Split around the middle.

Conceptually:

```
text
```

```
left half      |      right half  
-----+-----
```

The answer must be:

```
text
```

```
max(  
    best left,  
    best right,  
    best crossing  
)
```

The crossing case requires:

```
text
```

```
best suffix of left  
+  
best prefix of right
```

14. Complexity – Divide and Conquer

The recurrence is:

```
text
```

```
 $T(n) = 2T(n/2) + O(n)$ 
```

Therefore:

```
text
```

```
 $O(n \log n)$ 
```

Space due to recursion:

```
text
```

```
O(log n)
```

assuming recursion stack only.

This is better than $O(n^2)$, but still not as good as Kadane's:

```
text
```

```
O(n)
```

15. Why Do We Need Kadane's Algorithm?

At this point we've reached:

```
text
```

Brute force	$O(n^3)$
Running sum	$O(n^2)$
Prefix sum	$O(n^2)$
Divide and conquer	$O(n \log n)$

Now we ask:

"Can we make a decision locally while scanning once?"

Yes.

This is the central insight.

16. Kadane's Algorithm – Core Idea

At index `i`, suppose:

```
text
```

```
nums[i] = x
```

We want the maximum-sum subarray that **ends at index `i`**.

There are only two possibilities.

Option 1 – Start a new subarray

```
text
```

```
[x]
```

sum:

```
text
```

```
x
```

Option 2 – Extend the previous subarray

If the best subarray ending at `i - 1` has sum:

```
text
```

```
current
```

then extending it gives:

```
text
```

```
current + x
```

So:

```
text
```

```
best ending at i  
=  
max(  
    x,  
    current + x  
)
```

That's Kadane's Algorithm.

17. The Famous Kadane Formula

```
java
```

```
current = Math.max(nums[i], current + nums[i]);
```

Then:

```
java
```

```
best = Math.max(best, current);
```

So:

text

```
current
=
best subarray sum ending at current index

best
=
best subarray sum found anywhere so far
```

This distinction is extremely important.

18. Optimal Java Solution

java

```
public int maxSubArray(int[] nums) {

    int current = nums[0];
    int best = nums[0];

    for (int i = 1; i < nums.length; i++) {

        current = Math.max(
            nums[i],
            current + nums[i]
        );

        best = Math.max(best, current);
    }

    return best;
}
```

19. Deep Dry Run

Use the classic:

text

```
nums = [-2,1,-3,4,-1,2,1,-5,4]
```

Initialize:

text

```
current = -2  
best = -2
```

i = 1, value = 1

Two choices:

text

```
start new:  
1
```

```
extend:  
-2 + 1 = -1
```

Choose:

text

1

So:

text

```
current = 1  
best = 1
```

i = 2, value = -3

Choices:

text

```
start:  
-3
```

```
extend:  
1 + (-3) = -2
```

Choose:

text

-2

State:

text

```
current = -2  
best = 1
```

i = 3, value = 4

Choices:

text

```
start:  
4
```

```
extend:  
-2 + 4 = 2
```

Choose:

text

4

State:

text

```
current = 4  
best = 4
```

Notice something important.

We throw away:

text

[-3]

and start fresh at 4 .

Why?

Because carrying a negative sum would make every future extension worse.

i = 4, value = -1

Choices:

text

start:

-1

extend:

$4 + (-1) = 3$

Choose:

text

3

State:

text

current = 3

best = 4

Current subarray:

text

[4, -1]

i = 5, value = 2

Choices:

text

start:

2

extend:

$3 + 2 = 5$

Choose:

text

5

State:

text

current = 5
best = 5

Subarray:

text

[4,-1,2]

i = 6, value = 1

Choices:

text

start:
1

extend:
 $5 + 1 = 6$

Choose:

text

6

State:

text

current = 6
best = 6

Subarray:

text

[4,-1,2,1]

This is the answer.

i = 7, value = -5

Choices:

text

start:
-5

extend:
 $6 - 5 = 1$

Choose:

text

1

State:

text

current = 1
best = 6

i = 8, value = 4

Choices:

text

start:
4

extend:
 $1 + 4 = 5$

Choose:

text

5

Best remains:

text

6

Final answer:

text

6

20. The Most Important Insight

Why can we discard a negative running sum?

Suppose the current subarray has sum:

text

-5

and the next element is:

text

10

Extending gives:

text

$-5 + 10 = 5$

Starting fresh gives:

text

10

So carrying a negative sum can never help maximize a future sum.

More generally:

If the best subarray ending before the current element has a negative sum, adding it to the current element only makes the current/future result worse. Therefore, start a new subarray.

That is the greedy intuition behind Kadane.

21. Why Do We Compare Exactly Two Choices?

At index `i`, consider any maximum-sum subarray ending at `i`.

It either:

```
text
```

```
starts at i
```

or:

```
text
```

```
started earlier and extends the best possible subarray ending at i-1
```

There is no third possibility.

That gives:

```
text
```

```
current[i]
=
max(
    nums[i],
    current[i-1] + nums[i]
)
```

This recurrence is the mathematical foundation of Kadane's Algorithm.

22. Dynamic Programming Perspective

Kadane's Algorithm is essentially **space-optimized dynamic programming**.

Define:

```
text
```

```
dp[i]
=
maximum sum of a subarray ending exactly at i
```

Then:

```
text
```

```
dp[i]
=
max(
    nums[i],
    dp[i-1] + nums[i]
)
```

And the overall answer is:

text

```
max(dp[i])
```

The problem is that storing all `dp[i]` requires:

text

$O(n)$

But `dp[i]` only depends on:

text

`dp[i-1]`

So we can replace the entire DP array with:

text

current

That reduces space from:

text

$O(n)$

↓

$O(1)$

This is an excellent DP optimization pattern.

23. Approach Comparison

Approach	Time	Extra Space	Main Idea
Triple-loop brute force	$O(n^3)$	$O(1)$	Enumerate + calculate

Approach	Time	Extra Space	Main Idea
Running-sum brute force	$O(n^2)$	$O(1)$	Reuse current sum
Prefix sums	$O(n^2)$	$O(n)$	$O(1)$ range sums
Divide & conquer	$O(n \log n)$	$O(\log n)$	Left/right/cross
Kadane / DP	$O(n)$	$O(1)$	Start fresh or extend

The interviewer-expected best solution is:

```
text
```

```
Kadane's Algorithm
```

24. Pattern Recognition

This problem belongs to a very important pattern:

Running DP / State Compression

Think:

```
text
```

```
At every position,
can I summarize everything from the past
with a small state?
```

Here the complete history can be compressed into:

```
text
```

```
current
best
```

That is why we get:

```
text
```

```
 $O(n)$ 
```

time and:

```
text
```

```
 $O(1)$ 
```

extra space.

25. Another Pattern – "Extend or Restart"

This specific decision appears in many problems:

```
text
```

```
extend previous solution  
OR  
start fresh
```

Here:

```
text
```

```
current + nums[i]  
OR  
nums[i]
```

That is a very useful mental template.

Whenever an array optimization problem asks for the best contiguous segment, ask:

"Is carrying the previous state beneficial, or should I reset here?"

26. Why No Data Structure Is Needed

Unlike:

```
text
```

```
Q2 → HashSet  
Q3 → frequency array  
Q4 → HashMap  
Q5 → Heap
```

this problem needs no collection.

The required information is only:

```
text
```

```
best sum ending here  
+  
best sum overall
```


So scalar variables are sufficient.

This is an important DSA lesson:

The best solution does not necessarily require a sophisticated data structure.

27. Important Distinction — **current** vs **best**

This is probably the single most important thing to understand in Kadane.

current

Means:

text

maximum sum of a contiguous subarray
that MUST end at the current index

best

Means:

text

maximum sum of ANY contiguous subarray
seen so far

They are not the same.

Example:

text

nums = [4, -1, 2, -5]

At the end:

text

current = 0

might be lower than:

text

best = 5

because **best** remembers the optimal result found earlier.

28. Edge Case — All Negative Numbers

This is the most common Kadane trap.

Example:

text

```
nums = [-8, -3, -6, -2, -5, -4]
```

The answer is:

text

-2

not:

text

0

because the problem requires a **non-empty subarray**.

That's why we initialize:

java

```
int current = nums[0];  
int best = nums[0];
```

and not:

java

```
int best = 0;
```

29. Why `best = 0` Is Dangerous

Consider:

text

```
[-5, -2, -8]
```

If:

```
java
```

```
int best = 0;
```

the algorithm could return:

```
text
```

```
0
```

but there is no subarray with sum 0 if empty subarrays aren't allowed.

Correct answer:

```
text
```

```
-2
```

This is a classic interview test.

30. Edge Case – One Element

```
text
```

```
nums = [7]
```

Answer:

```
text
```

```
7
```

Because the only possible subarray is:

```
text
```

```
[7]
```

31. Edge Case – One Negative Element

```
text
```

```
[-10]
```

Answer:

```
text
```

```
-10
```

Again, do not return zero unless the problem explicitly allows an empty subarray.

32. Edge Case — All Positive

```
text
```

```
[1,2,3,4]
```

The entire array is optimal:

```
text
```

```
10
```

Kadane naturally keeps extending:

```
text
```

```
1  
3  
6  
10
```

33. Edge Case — Mix of Positive and Negative

```
text
```

```
[-2,1,-3,4,-1,2,1,-5,4]
```

Best:

```
text
```

```
6
```

34. Edge Case — Zero

```
text
```

```
[0, -1, 0]
```

Maximum non-empty sum:

```
text
```

```
0
```

A subarray `[0]` is valid.

35. Follow-Up – How Do You Return the Actual Subarray?

This is a very common interviewer follow-up.

Instead of only tracking:

```
text
```

```
current  
best
```

we additionally track:

```
text
```

```
start  
bestStart  
bestEnd
```

The idea:

When we decide to start fresh at `i` :

```
text
```

```
start = i
```

When we discover a new global best:

```
text
```

```
bestStart = start  
bestEnd = i
```

36. Java – Return Actual Subarray

java

```
public int[] maxSubArray(int[] nums) {  
  
    int current = nums[0];  
    int best = nums[0];  
  
    int start = 0;  
    int bestStart = 0;  
    int bestEnd = 0;  
  
    for (int i = 1; i < nums.length; i++) {  
  
        if (nums[i] > current + nums[i]) {  
            current = nums[i];  
            start = i;  
        } else {  
            current = current + nums[i];  
        }  
  
        if (current > best) {  
            best = current;  
            bestStart = start;  
            bestEnd = i;  
        }  
    }  
  
    return Arrays.copyOfRange(  
        nums,  
        bestStart,  
        bestEnd + 1  
    );  
}
```

For:

text

[-2,1,-3,4,-1,2,1,-5,4]

we get:

text

[4,-1,2,1]

37. How the Index Tracking Works

Take:

```
text
```

```
[-2,1,-3,4,-1,2,1,-5,4]
```

At 4 :

```
text
```

```
starting fresh is better
```

So:

```
text
```

```
start = 3
```

Then:

```
text
```

```
4  
4 + (-1) = 3  
3 + 2 = 5  
5 + 1 = 6
```

When 6 becomes the best:

```
text
```

```
bestStart = 3  
bestEnd = 6
```

Therefore:

```
text
```

```
nums[3...6]
```

is:

```
text
```

```
[4,-1,2,1]
```

38. Why `start = i` When We Restart?

Suppose:

```
text
current + nums[i] < nums[i]
```

The old subarray is hurting us.

So the best subarray ending at `i` is simply:

```
text
[nums[i]]
```

Therefore its starting index is:

```
text
i
```

This keeps the indices consistent.

39. Important Tie-Breaking Issue

Suppose there are multiple subarrays with the same maximum sum.

Example:

```
text
[1,-1,1]
```

Maximum sum is:

```
text
1
```

There can be several valid subarrays.

If the interviewer says:

```
"Return the earliest maximum-sum subarray."
```

then the update condition matters.

Using:


```
java
```

```
if (current > best)
```

keeps the first one found.

Using:

```
java
```

```
if (current >= best)
```

could replace it with a later equal solution.

This is a subtle but useful interview point.

40. Follow-Up – Return Start and End Indices

Often the interviewer may ask for:

```
text
```

```
[start, end]
```

Instead of copying the subarray.

That is more efficient because:

```
java
```

doesn't allocate a new array for the output subarray.

Java

```
java
```

```
public int[] maxSubArrayIndices(int[] nums) {  
  
    int current = nums[0];  
    int best = nums[0];  
  
    int start = 0;  
    int bestStart = 0;  
    int bestEnd = 0;  
  
    for (int i = 1; i < nums.length; i++) {
```

```

        if (nums[i] > current + nums[i]) {
            current = nums[i];
            start = i;
        } else {
            current += nums[i];
        }

        if (current > best) {
            best = current;
            bestStart = start;
            bestEnd = i;
        }
    }

    return new int[]{bestStart, bestEnd};
}

```

For the standard input:

text

[3, 6]

41. Java-Specific Follow-Up — `Arrays.copyOfRange`

This:

java

```
Arrays.copyOfRange(nums, bestStart, bestEnd + 1)
```

uses:

text

end exclusive

so if the desired range is:

text

3 ... 6

we pass:

text

```
3, 7
```

This creates a new array.

Time:

```
text
```

```
O(length of returned subarray)
```

That's fine unless the interviewer specifically asks for constant extra space.

42. Java-Specific Follow-Up – Null/Empty Input

The simplest algorithm assumes a non-empty array, which is usually guaranteed by the problem.

If not:

```
java
```

```
if (nums == null || nums.length == 0) {  
    throw new IllegalArgumentException("Array must not be empty");  
}
```

or return a specified fallback based on the API contract.

Don't silently invent semantics for an unspecified empty input.

43. Java-Specific Follow-Up – Integer Overflow

This is important in real applications.

For:

```
java
```

```
current + nums[i]
```

`int` can overflow if the sum exceeds the integer range.

If constraints permit large values, use:

```
java
```

```
long current  
long best
```

and potentially:

```
java  
  
long[]
```

The algorithm doesn't change.

44. Can We Use `Math.max()` ?

Yes:

```
java  
  
current = Math.max(  
    nums[i],  
    current + nums[i]  
);  
  
best = Math.max(best, current);
```

This is concise and matches the recurrence directly.

For index tracking, however, explicit `if` statements are often clearer because we need to know **which branch won**.

45. Why Explicit `if` Is Better for Returning the Subarray

With:

```
java  
  
if (nums[i] > current + nums[i]) {  
    current = nums[i];  
    start = i;  
} else {  
    current += nums[i];  
}
```

we know:

```
text  
  
if restart wins → update start  
otherwise → extend
```

You can't easily maintain the start index with only `Math.max()` unless you add extra logic.

So:

text

value-only problem
→ `Math.max` is clean

actual-subarray problem
→ explicit branch is clearer

46. Advanced Follow-Up – Maximum Subarray Sum With Circular Array

A very common extension is:

"What if the array is circular?"

For example:

text

`[5, -3, 5]`

The normal maximum is:

text

7

But in a circular array, we can take:

text

$5 + 5 = 10$

The standard solution becomes:

text

```
max(  
    normalMaximumSubarray,  
    totalSum - minimumSubarray  
)
```

with a special case for all-negative arrays.

This is a related but different problem.

Don't modify ordinary Kadane without first identifying the circular condition.

47. Why Does Circular Use Minimum Subarray?

Because the wraparound maximum consists of:

```
text
entire array
-
middle minimum-sum subarray
```

Example:

```
text
[5,-3,5]
```

Total:

```
text
7
```

Minimum subarray:

```
text
[-3]
```

So:

```
text
7 - (-3)
=
10
```

This is a beautiful extension of the same Kadane concept.

48. Advanced Follow-Up – Maximum Product Subarray

Another interviewer may change:

"Find maximum product instead of maximum sum."

You cannot directly use standard Kadane because multiplication behaves differently.

For example:

```
text
```

```
[-2,3,-4]
```

A negative value can turn the smallest negative product into the largest positive product.

So you need to track:

```
text
```

```
maximum product ending here
```

and:

```
text
```

```
minimum product ending here
```

This is an excellent example of why understanding the state transition is more important than memorizing "Kadane."

49. Advanced Follow-Up — Why Track Minimum Product There?

Because:

```
text
```

```
negative × negative = positive
```

So both states matter.

For maximum sum:

```
text
```

```
only current maximum is needed
```

For maximum product:

```
text
```

```
maximum + minimum
```

are needed.

This is a useful DP-state-design lesson.

50. Pattern Recognition – DP State Compression

The deeper lesson from Kadane is:

text

Define DP state

↓

Find recurrence

↓

Notice only previous state is needed

↓

Compress $O(n)$ DP to $O(1)$

This pattern appears across many dynamic programming problems.

For this problem:

text

$dp[i]$

=

best subarray sum ending at i

Then:

text

$dp[i]$

depends only on $dp[i-1]$

Therefore:

text

current

is sufficient.

51. Why Greedy and DP Both Describe Kadane

You may hear:

"Kadane is greedy."

You may also hear:

"Kadane is dynamic programming."

Both descriptions make sense.

DP perspective

We define:

```
text
```

```
dp[i] = best sum ending at i
```

with a recurrence.

Greedy perspective

Whenever the accumulated sum becomes harmful, we discard it and start over.

```
text
```

```
negative contribution
```

↓

```
throw it away
```

For interviews, I would say:

"Kadane can be viewed as a one-dimensional DP optimized to $O(1)$ space, with a greedy reset interpretation."

That's a strong answer.

52. Why Does a Negative `current` Get Discarded?

Suppose:

```
text
```

```
current < 0
```

and current element is:

```
text
```

```
x
```

Compare:

```
text
```

```
x
```

versus:

```
text
```

```
current + x
```

Because:

```
text
```

```
current < 0
```

we have:

```
text
```

```
current + x < x
```

Therefore:

```
text
```

```
starting fresh is always better
```

That is the formal reason we can discard a negative running sum.

53. Why Can't We Discard Every Negative Element?

This is a subtle question.

Consider:

```
text
```

```
[4,-1,5]
```

The `-1` is negative, but keeping it gives:

```
text
```

```
4 + (-1) + 5 = 8
```

which is better than:

```
text
```

```
4 + 5
```

But dropping the `-1` is not allowed because the subarray must be contiguous.

So we don't discard individual negative elements.

We discard an entire **prefix whose cumulative sum is harmful**.

That's a critical distinction.

54. Very Important: Contiguous Constraint

Suppose:

```
text
```

```
[4, -100, 5]
```

You cannot choose:

```
text
```

```
[4, 5]
```

because that skips `-100` .

You must choose a contiguous range:

```
text
```

```
[4]  
[5]  
[4, -100]  
[-100, 5]  
[4, -100, 5]
```

Therefore maximum is:

```
text
```

```
5
```

This is why this isn't a simple "pick positive numbers" problem.

55. Common Mistake – Summing Positive Numbers

A common wrong approach is:

```
text
```

add every positive number

Example:

text

[4, -100, 5]

would give:

text

9

but 9 is impossible for a contiguous subarray.

Correct answer:

text

5

This is one of the most common traps.

56. Common Mistake – Return Zero for All Negative

Wrong:

java

```
int best = 0;
```

unless empty subarrays are explicitly allowed.

For the usual problem:

text

[-5, -2, -10]

answer is:

text

-2

57. Common Mistake — Reset Whenever Current Is Negative Without Thinking

A common implementation is:

```
java

current += nums[i];

if (current < 0) {
    current = 0;
}
```

This is a common form of Kadane, but it requires careful handling of all-negative arrays.

For example:

```
text

[-5,-2,-8]
```

it can return 0, which may be wrong.

The safer formulation for the standard **non-empty subarray** problem is:

```
java

current = Math.max(nums[i], current + nums[i]);
best = Math.max(best, current);
```

This naturally handles all-negative input.

58. Can the Reset-to-Zero Version Be Correct?

Yes, with additional handling.

For example, one can separately track the maximum element.

But that adds complexity.

So for interviews, use:

```
java

current = Math.max(nums[i], current + nums[i]);
```

It directly models the actual recurrence and avoids the all-negative trap.

59. Common Mistake – Confusing Maximum Subarray With Maximum Subsequence

Maximum subarray:

text

must be contiguous

Maximum subsequence/subset:

text

can skip elements

These are different problems.

Example:

text

[5, -100, 6]

Maximum subarray:

text

6

Maximum non-contiguous subsequence sum might be:

text

11

depending on the exact definition.

60. Production Variation – Monitoring / Metrics

The pattern can be used for time-series analysis.

Suppose:

text

daily performance changes

and we want:

"Find the contiguous period where cumulative improvement was highest."

The same idea applies:

```
text

daily delta
  ↓
maximum-sum contiguous interval
  ↓
Kadane
```

Examples include:

```
text

performance deltas
profit/loss deltas
traffic growth
capacity utilization changes
```

The domain changes, but the mathematical structure remains the same.

61. Production Variation — Log/Event Impact

Suppose events have numerical impact:

```
text

+5
-2
-1
+8
-3
```

We want to find the continuous interval with the greatest cumulative impact.

This is again:

```
text

maximum contiguous sum
```

Kadane can solve it in one pass.

62. Production Variation — Streaming

A particularly nice property of Kadane is that it can operate on a stream.

You only need:

```
text

current
best
```

For each new value:

```
text

current = max(value, current + value)
best = max(best, current)
```

No need to retain the complete historical array.

So the algorithm has:

```
text

Per element:  $O(1)$ 
State:  $O(1)$ 
```

This is highly useful for streaming-style calculations.

63. Production Caveat — Returning the Actual Subarray

If you need the actual range in a live stream, you also need to track:

```
text

start
bestStart
bestEnd
```

So the memory remains constant.

You can report:

```
text

bestStart ... bestEnd
```

without retaining the entire input.

If you need the actual values, however, retrieving them later requires access to the underlying data source or stored records.

64. Production Concern – Integer Overflow

For financial/logging/metrics systems, the cumulative sum may exceed `int`.

Use:

```
java
```

```
long
```

when appropriate.

For extremely large values:

```
java
```

```
BigInteger
```

may be necessary, though that changes performance characteristics.

The algorithm remains unchanged.

65. Advanced Interview Follow-Up – "Can You Do Better Than $O(n)$?"

No.

You need to inspect the input because any individual element can affect the answer.

For example:

```
text
```

```
[1,1,1,1,...]
```

and:

```
text
```

```
[1,1,1,1,...,100000]
```

have different answers.

Therefore:

```
text
```

$\Omega(n)$

information is needed in general.

Kadane achieves:

text

$O(n)$

so it is asymptotically optimal.

66. Advanced Interview Follow-Up – Can We Parallelize It?

Interesting system/design question.

Yes, but it becomes more complicated.

A segment can be summarized by information such as:

text

```
total sum
maximum prefix sum
maximum suffix sum
maximum subarray sum
```

Then two adjacent segments can be merged.

For left segment `A` and right segment `B` :

text

```
total = A.total + B.total

prefix = max(
    A.prefix,
    A.total + B.prefix
)

suffix = max(
    B.suffix,
    B.total + A.suffix
)

best = max(
    A.best,
    B.best,
```

```
A.suffix + B.prefix  
)
```

This allows a divide-and-conquer or parallel reduction approach.

This is much more advanced than the normal interview question, but it reveals a deeper connection between:

```
text  
  
Kadane  
+  
divide-and-conquer  
+  
segment tree summaries
```

67. Advanced Connection – Segment Tree

A maximum-subarray segment tree node often stores:

```
text  
  
sum  
prefixMax  
suffixMax  
maxSubarray
```

Why?

Because two adjacent segments can be combined using exactly the formulas above.

This means the basic Kadane concept generalizes into a data structure that can answer:

```
"What is the maximum subarray sum in an arbitrary range?"
```

under updates/query requirements.

That's an excellent advanced interview connection.

68. Java-Specific Follow-Up – Why `Integer.MIN_VALUE` ?

If you're doing a brute-force or index-tracking version where you don't initialize from `nums[0]` , you may use:

```
java  
  
int best = Integer.MIN_VALUE;
```

This ensures even an all-negative array is handled correctly.

But the standard Kadane implementation can simply use:

```
java

int current = nums[0];
int best = nums[0];
```

because the array is assumed non-empty.

69. Java-Specific Follow-Up — Why Not `Arrays.stream().max()` ?

That only gives the largest individual element:

```
text

max element
```

but the answer may be a multi-element subarray.

Example:

```
text

[4,-1,2,1]
```

Largest single element:

```
text

4
```

Maximum subarray:

```
text

6
```

So this problem requires cumulative state, not simple element selection.

70. Java-Specific Follow-Up — Arrays vs Lists

For a returned actual subarray:

```
java
```

```
int[]
```

is natural and efficient because the input is an array and the output is contiguous numeric data.

A:

```
java
```

```
List<Integer>
```

would introduce boxing and collection overhead.

For DSA interviews, `int[]` is the cleaner choice.

71. Final Interview Progression

If I were solving this in a real interview, I'd make the progression explicit:

Step 1 — Brute force

"Enumerate every subarray and calculate each sum."

```
text
```

```
 $O(n^3)$ 
```

Step 2 — Running sum

"Avoid recomputing each subarray's sum."

```
text
```

```
 $O(n^2)$ 
```

Step 3 — Prefix sums

"Precompute prefix sums so each range sum is $O(1)$, but we still enumerate $O(n^2)$ ranges."

```
text
```

```
 $O(n^2)$ 
```

Step 4 — Divide and conquer

"Split into left, right, and crossing cases."

```
text
```

$O(n \log n)$

Step 5 — Kadane

"For each position, the best subarray ending here either starts here or extends the best subarray ending at the previous position."

text

$O(n)$

That is a very strong interview progression.

72. Final Optimal Code — Sum Only

java

```
public int maxSubArray(int[] nums) {  
  
    int current = nums[0];  
    int best = nums[0];  
  
    for (int i = 1; i < nums.length; i++) {  
  
        current = Math.max(  
            nums[i],  
            current + nums[i]  
        );  
  
        best = Math.max(best, current);  
    }  
  
    return best;  
}
```

Complexity

text

Time: $O(n)$
Auxiliary Space: $O(1)$

This is asymptotically optimal.

73. Final Optimal Code – Actual Subarray

When the interviewer asks for the actual range:

java

```
public int[] maxSubArrayRange(int[] nums) {  
  
    int current = nums[0];  
    int best = nums[0];  
  
    int start = 0;  
    int bestStart = 0;  
    int bestEnd = 0;  
  
    for (int i = 1; i < nums.length; i++) {  
  
        if (nums[i] > current + nums[i]) {  
            current = nums[i];  
            start = i;  
        } else {  
            current += nums[i];  
        }  
  
        if (current > best) {  
            best = current;  
            bestStart = start;  
            bestEnd = i;  
        }  
    }  
  
    return new int[]{bestStart, bestEnd};  
}
```

For:

text

[-2,1,-3,4,-1,2,1,-5,4]

the range is:

text

[3,6]

corresponding to:

```
text
[4,-1,2,1]
```

and sum:

```
text

6
```

74. Final Interviewer-Ready Answer

If the interviewer asks:

"Explain the optimal solution."

A strong 2–2.5 YOE answer would be:

"The brute-force approach considers every contiguous subarray, which is $O(n^3)$ if we calculate each sum independently. Using a running sum reduces that to $O(n^2)$, and prefix sums also keep it at $O(n^2)$. A divide-and-conquer solution can achieve $O(n \log n)$.

We can do better with Kadane's Algorithm. I define `current` as the maximum sum of a contiguous subarray that must end at the current index. For each element `nums[i]`, there are only two possibilities: either I start a new subarray at `i`, giving `nums[i]`, or I extend the previous best subarray, giving `current + nums[i]`.

Therefore:

```
current = max(nums[i], current + nums[i])
```

Then I maintain `best`, which is the maximum `current` seen anywhere in the array.

This works because if the previous running sum is negative, carrying it forward can only reduce the sum, so starting a new subarray is better.

The algorithm runs in $O(n)$ time and $O(1)$ auxiliary space, which is asymptotically optimal."

75. If Asked "Why Is It Optimal?"

Say:

"We need to inspect every element because any element can affect the maximum subarray. Therefore $O(n)$ is the lower bound in the general case. Kadane processes each element exactly once and maintains only constant state, so it is asymptotically optimal."

76. If Asked "How Do You Return the Actual Subarray?"

Say:

"I keep a `start` index for the current candidate subarray. Whenever starting fresh at index `i` is better, I set `start = i`. Whenever `current` exceeds the global `best`, I save `bestStart = start` and `bestEnd = i`. This still keeps $O(1)$ auxiliary space."

77. Final Cheat Sheet

Approach 1 — Brute Force

text

Enumerate all subarrays
calculate each sum independently

Time: $O(n^3)$
Space: $O(1)$

Approach 2 — Running Sum

text

Enumerate all subarrays
reuse running sum

Time: $O(n^2)$
Space: $O(1)$

Approach 3 — Prefix Sum

text

`prefix[r+1] - prefix[l]`

Time: $O(n^2)$
Space: $O(n)$

Approach 4 — Divide & Conquer

text

left
right
crossing

Time: $O(n \log n)$
Space: $O(\log n)$

Approach 5 — Kadane

text

```
current = max(nums[i], current + nums[i])
best = max(best, current)
```

Time: $O(n)$
Auxiliary Space: $O(1)$

78. Core Pattern

text

Best contiguous segment
↓
State for segment ending here
↓
START FRESH
OR
EXTEND
↓
Kadane / 1D DP

Core recurrence

text

```
current[i]
=
max(
    nums[i],
    current[i-1] + nums[i]
)
```

State meaning

text

current
→ best sum ending here

best
→ best sum anywhere so far

Space optimization

text

dp[i] depends only on dp[i-1]
↓
store only current
↓
O(1) auxiliary space

Key trap

text

ALL NEGATIVE
[-5,-2,-8]

Answer = -2
NOT 0

Actual-subarray follow-up

text

start
bestStart
bestEnd

Most important mental model

text

At every element:

Does my previous subarray help me?

current + nums[i]

OR

is it hurting me enough that I should restart?

```
nums[i]
```

Take the better one.

That is the real idea behind **Kadane's Algorithm**, and once you understand that "**extend or restart**" decision, you can recognize a whole family of maximum-subarray and one-dimensional DP problems rather than simply memorizing the code.

2026-09-02 21:25:52 You 

ok as aboved continue for next question,i want the solutions as interviewer expcted in the interview of 2-2.5 year experience candidate and the solutions first is basic approach then other approach then next another better approach as like that possible approaches which patterns used and why that patterns used and at last best optimize approach as per the interviewer. keep this same interview structure: brute force → better → optimal → pattern recognition → why the data structure works → dry run → edge cases → Java-specific follow-ups → production variation → final interviewer-ready answer.

Q9. Majority Element

Question

Find the element appearing more than $n/2$ times.

Best approach

Boyer-Moore Voting Algorithm

java

```
public int majorityElement(int[] nums) {  
    int candidate = 0;  
    int count = 0;  
  
    for (int n : nums) {  
        if (count == 0) {  
            candidate = n;  
        }  
  
        count += (n == candidate) ? 1 : -1;  
    }  
  
    return candidate;  
}
```

Why it works

Every different number cancels one vote of the candidate.
Because the majority is greater than $n/2$, it survives.
Complexity
text
 $O(n)$ time
 $O(1)$ space
Tricky question
What if a majority is not guaranteed?
Verify the candidate with a second pass.

 ChatGPT gpt-5-6-t-mini

Q9. Majority Element

This is another **very important product-company interview problem**, because it looks like a simple frequency-counting question, but the interviewer can use it to test whether you can move from:

text

```
Brute Force
  ↓
Sorting
  ↓
HashMap / Frequency Count
  ↓
Boyer-Moore Voting
  ↓
 $O(n)$  time +  $O(1)$  extra space
```

The most important transition is recognizing that the condition:

text

```
frequency > n / 2
```

is much stronger than just "find the most frequent element."

That strict majority guarantees that the candidate cannot be completely cancelled by all other elements.

1. Problem

Given an integer array `nums`, find the element that appears **more than $n/2$ times**, where:

```
text
```

```
n = nums.length
```

The problem usually guarantees that such an element exists.

Example

```
text
```

```
nums = [3, 2, 3]
```

Length:

```
text
```

```
n = 3
```

Majority threshold:

```
text
```

```
n / 2 = 1
```

3 appears twice:

```
text
```

```
3 -> 2
```

```
2 -> 1
```

Since:

```
text
```

```
2 > 3 / 2
```

the answer is:

```
text
```

```
3
```

2. Another Example

```
text
```

```
nums = [2,2,1,1,1,2,2]
```

Frequencies:

```
text
```

```
2 -> 4
```

```
1 -> 3
```

Length:

```
text
```

```
7
```

Majority threshold:

```
text
```

```
7 / 2 = 3
```

2 appears 4 times.

Therefore:

```
text
```

```
answer = 2
```

3. How I Would Start in a Real Interview

A strong opening would be:

"The straightforward approach is to count the frequency of each element and return the one whose count exceeds $n/2$. That takes $O(n)$ time and $O(n)$ space. We can also sort the array, in which case the majority element must occupy the middle position, giving $O(n \log n)$ time. But because the majority occurs more than half the time, we can do better with Boyer-Moore Voting. We maintain a candidate and a count, and different values cancel one vote from each other. Since the true majority has more votes than all other values combined, it survives the cancellation. This gives $O(n)$ time and $O(1)$ extra space."

That is an excellent 2–2.5 YOE interview answer.

4. Approach 1 — Brute Force

The simplest possible idea:

For every element:

1. Count how many times it appears.
 2. If its count is greater than $n / 2$, return it.
-

5. Java — Brute Force

java

```
public int majorityElement(int[] nums) {  
  
    int n = nums.length;  
  
    for (int i = 0; i < n; i++) {  
  
        int count = 0;  
  
        for (int j = 0; j < n; j++) {  
  
            if (nums[i] == nums[j]) {  
                count++;  
            }  
        }  
  
        if (count > n / 2) {  
            return nums[i];  
        }  
    }  
  
    return -1;  
}
```

The `-1` is only a fallback. If the problem guarantees a majority exists, execution will return earlier.

6. Dry Run — Brute Force

Input:

text

[2,2,1,1,1,2,2]

Take first element:

```
text
```

```
2
```

Count all 2 s:

```
text
```

```
4
```

Check:

```
text
```

```
4 > 7 / 2
```

```
4 > 3
```

True.

Return:

```
text
```

```
2
```

7. Complexity – Brute Force

For each element we may scan the entire array.

Therefore:

```
text
```

```
n × n
```

giving:

```
text
```

```
Time:  $O(n^2)$ 
```

```
Space:  $O(1)$ 
```

This is correct but inefficient.

The first obvious improvement is to avoid recounting the same value.

8. Approach 2 – HashMap Frequency Counting

This is usually the first optimized solution candidates think of.

We maintain:

```
text

value -> frequency
```

Example:

```
text

[2,2,1,1,1,2,2]
```

becomes:

```
text

2 -> 4
1 -> 3
```

As soon as a frequency becomes greater than:

```
text

n / 2
```

we can return that value.

9. Java – HashMap Solution

```
java

public int majorityElement(int[] nums) {

    int n = nums.length;

    Map<Integer, Integer> freq = new HashMap<>();

    for (int num : nums) {

        int count = freq.getOrDefault(num, 0) + 1;

        freq.put(num, count);

        if (count > n / 2) {
            return num;
        }
    }
}
```

```
    }  
}  
  
return -1;  
}
```

A modern Java version using `merge()` :

```
java  
  
public int majorityElement(int[] nums) {  
  
    int n = nums.length;  
  
    Map<Integer, Integer> freq = new HashMap<>();  
  
    for (int num : nums) {  
  
        int count = freq.merge(num, 1, Integer::sum);  
  
        if (count > n / 2) {  
            return num;  
        }  
    }  
  
    return -1;  
}
```

10. Why Can We Return Immediately?

Suppose:

```
text
```

```
n = 7
```

and the threshold is:

```
text
```

```
3
```

If some value reaches:

```
text
```

```
4
```

then:

```
text
```

```
4 > 3
```

so it has already satisfied the majority condition.

We don't need to process the rest.

This is an example of an **early-exit optimization**.

11. Complexity — HashMap

We process every element once.

Each HashMap update is expected:

```
text
```

```
O(1)
```

Therefore:

```
text
```

```
Time: O(n) average
```

```
Space: O(m)
```

where m is the number of distinct values.

Worst-case:

```
text
```

```
m = n
```

so:

```
text
```

```
Space: O(n)
```

This is already a very good solution.

But the interviewer may ask:

"Can you do it without extra space?"

That's where things become interesting.

12. Approach 3 – Sorting

There is a very elegant observation.

If an element occurs more than half the time, after sorting, it **must occupy the middle position**.

Example:

```
text
```

```
[2,2,1,1,1,2,2]
```

Sort:

```
text
```

```
[1,1,1,2,2,2,2]
```

Middle index:

```
text
```

```
n / 2 = 7 / 2 = 3
```

Element:

```
text
```

```
sorted[3] = 2
```

Therefore:

```
text
```

```
majority = 2
```

13. Why Must the Middle Element Be the Majority?

This is a very good interviewer question.

Suppose:

```
text
```

```
majority frequency > n/2
```

That means the majority occupies more than half the positions.

After sorting, all occurrences of the same value are clustered together.

A block containing more than half the elements cannot avoid crossing the center.

Therefore:

text

sorted[n / 2]

must belong to the majority element.

This gives a surprisingly simple solution.

14. Java – Sorting Solution

java

```
public int majorityElement(int[] nums) {  
  
    Arrays.sort(nums);  
  
    return nums[nums.length / 2];  
}
```

That's all.

15. Dry Run – Sorting

Input:

text

[3,2,3]

Sort:

text

[2,3,3]

Middle:

text

```
index = 3 / 2 = 1
```

Therefore:

```
text
```

```
nums[1] = 3
```

Answer:

```
text
```

```
3
```

16. Complexity – Sorting

Sorting:

```
text
```

```
 $O(n \log n)$ 
```

Middle lookup:

```
text
```

```
 $O(1)$ 
```

Therefore:

```
text
```

```
Time:  $O(n \log n)$ 
```

Extra space depends on the sorting implementation, and the input array is modified.

This solution is much better than brute force, but slower asymptotically than HashMap/Boyer-Moore.

17. Important Interview Trade-Off

Sorting has an advantage:

```
text
```

```
very simple code
```

The whole solution is:

```
java  
  
Arrays.sort(nums);  
return nums[nums.length / 2];
```

So if the interviewer prioritizes simplicity and mutation is acceptable, this is a perfectly valid solution.

But if they ask:

"Can you do $O(n)$ time and $O(1)$ space?"

you need Boyer-Moore.

18. Approach 4 — Boyer-Moore Voting Algorithm

This is the best standard solution under the majority-guaranteed condition.

The algorithm maintains:

```
text  
  
candidate  
count
```

Initially:

```
text  
  
candidate = 0  
count = 0
```

For every number:

If count is zero

Choose the current number as the candidate.

```
text  
  
candidate = n
```

If number equals candidate

Increase count:

```
text
```



```
count++
```

Otherwise

Decrease count:

```
text
```

```
count--
```

That's the whole algorithm.

19. Java — Boyer-Moore

```
java
```

```
public int majorityElement(int[] nums) {  
  
    int candidate = 0;  
    int count = 0;  
  
    for (int num : nums) {  
  
        if (count == 0) {  
            candidate = num;  
        }  
  
        count += (num == candidate) ? 1 : -1;  
    }  
  
    return candidate;  
}
```

20. More Readable Version

For an interview, I may actually write:

```
java
```

```
public int majorityElement(int[] nums) {  
  
    int candidate = 0;  
    int count = 0;  
  
    for (int num : nums) {
```

```

    if (count == 0) {
        candidate = num;
    }

    if (num == candidate) {
        count++;
    } else {
        count--;
    }
}

return candidate;
}

```

This is slightly longer but makes the logic easier to explain.

For a 2–2.5 YOE interview, I'd favor this version while whiteboarding.

21. The Core Intuition

Think of each element as casting a vote.

Suppose:

text

[2,2,1,1,1,2,2]

There are:

text

2 -> 4 votes

1 -> 3 votes

Pair a majority vote with a different vote:

text

2 vs 1

and cancel them:

text

2 ← cancelled

1 ← cancelled

We have:

text

2 votes left

1 vote left

Cancel again:

text

2 vs 1

Now:

text

1 vote of 2 remains

The majority survives.

This is the central intuition behind Boyer-Moore.

22. Why Does Cancellation Work?

Suppose the majority element appears:

text

M

times.

All other elements together occur:

text

$N - M$

times.

Because the majority is strictly greater than half:

text

$M > N / 2$

Therefore:

text

$$M > N - M$$

So even if **every non-majority element cancels one majority occurrence**, the majority still has votes left.

That is the mathematical reason the algorithm works.

23. Formal Cancellation Argument

Let:

text

majority count = M
other count = 0

Given:

text

$$M > N/2$$

and:

text

$$N = M + 0$$

we get:

text

$$M > 0$$

So:

text

$$M - 0 > 0$$

That means after pairing one majority occurrence with one non-majority occurrence repeatedly, some majority occurrences remain.

Thus the final surviving candidate must be the majority.

This is the key proof.

24. Deep Dry Run — Boyer-Moore

Input:

```
text
```

```
[2,2,1,1,1,2,2]
```

Initial:

```
text
```

```
candidate = ?  
count = 0
```

Value = 2

Count is zero:

```
text
```

```
candidate = 2
```

Now:

```
text
```

```
2 == 2
```

so:

```
text
```

```
count = 1
```

State:

```
text
```

```
candidate = 2  
count = 1
```

Value = 2

Same as candidate:

```
text
```

```
count = 2
```

Value = 1

Different:

```
text
```

```
count = 1
```

The 1 cancels one vote for 2 .

Value = 1

Different again:

```
text
```

```
count = 0
```

Now the currently accumulated votes have cancelled.

Value = 1

Count is zero.

So we choose:

```
text
```

```
candidate = 1
```

Then:

```
text
```

```
count = 1
```

Value = 2

Different:

```
text
```

```
count = 0
```

Value = 2

Count is zero.

Choose:

```
text
```

```
candidate = 2
```

Then:

```
text
```

```
count = 1
```

Final:

```
text
```

```
candidate = 2
```

Correct.

25. "Wait — Why Did Candidate Change to 1 If 2 Is Actually the Majority?"

This is an important question.

Boyer-Moore does **not** guarantee that `candidate` is the final majority at every point.

It only guarantees that after processing the full array, the surviving candidate is the majority, **provided a majority is guaranteed to exist**.

The candidate may change several times.

That's why it's called a voting/cancellation algorithm rather than a simple running frequency counter.

26. The Important Invariant

A useful way to explain the algorithm:

"`count` represents the number of un-cancelled votes for the current `candidate`."

When:

```
text
```

```
num == candidate
```

we add a vote:

```
text
```

```
count++
```

When:

```
text
```

```
num != candidate
```

we cancel one candidate vote:

```
text
```

```
count--
```

When:

```
text
```

```
count == 0
```

the accumulated candidate votes have been completely cancelled by different values, so we are free to choose a new candidate.

This is the best conceptual explanation.

27. Why Only Compare Equal vs Different?

Notice we don't care **which** other number differs.

For the cancellation argument, every non-candidate vote has the same purpose:

```
text
```

```
cancel one candidate vote
```

So we don't need to track frequencies of every other value.

This is exactly why Boyer-Moore can reduce:

```
text
```


$O(n)$ HashMap memory

to:

text

$O(1)$

state.

28. Why HashMap Is Not Necessary

HashMap stores:

text

value $\rightarrow$ count

for every distinct value.

But the majority guarantee gives us stronger information.

We don't need the exact count of every value.

We only need to know:

text

Which candidate survives cancellation?

Therefore:

text

candidate + count

are enough.

This is a powerful algorithmic lesson:

Strong problem constraints can allow you to discard information that a general solution would normally need.

29. Complexity — Boyer-Moore

We scan the array once:

text

0(n)

We maintain only:

text

candidate
count

So:

text

Time: 0(n)
Space: 0(1)

This is asymptotically optimal.

30. Can We Do Better Than O(n)?

No.

We need to inspect the input because any element could potentially affect whether a majority exists or which candidate survives.

Therefore:

text

Lower bound $\approx$ 0(n)

and Boyer-Moore already achieves:

text

0(n)

So the standard guaranteed-majority problem is asymptotically optimal.

31. Approach Comparison

Approach	Time	Extra Space	Main Idea
Brute force	$O(n^2)$	$O(1)$	Count each value repeatedly
HashMap	$O(n)$ avg	$O(n)$	Frequency map

Approach	Time	Extra Space	Main Idea
Sorting	$O(n \log n)$	Depends	Middle element
Boyer-Moore	$O(n)$	$O(1)$	Vote cancellation

Therefore:

text

Best:
Boyer-Moore Voting Algorithm

when the problem guarantees a majority.

32. Pattern Recognition

The important pattern is:

Cancellation / Voting Pattern

Whenever you have:

text

one element appears more than half

think:

text

Boyer-Moore

The condition:

text

frequency > n/2

is the critical clue.

33. Another Way to Remember It

Think:

text

Majority > everyone else combined

Therefore:

```
text

majority votes
-
all opposing votes
>
0
```

So even after maximum cancellation:

```
text

majority survives
```

That is the entire algorithm in one idea.

34. Why the `count == 0` Reset Is Valid

Suppose:

```
text

candidate = X
count = 0
```

means the votes accumulated for `X` have been completely cancelled.

So there is no reason to preserve `X` as the candidate.

The next value can become the new candidate.

This doesn't lose the true majority because any previously cancelled group had equal numbers of candidate and non-candidate votes.

Those cancelled pairs cannot affect which value has a strict global majority.

35. Advanced Proof Using Pair Removal

Another way to prove Boyer-Moore:

Imagine repeatedly removing pairs of different elements from the array.

Example:

text

[A, A, B, B, A, C, A]

Remove:

text

A + B

A + B

A + C

Remaining:

text

A

Because the majority count is greater than all other counts combined, repeatedly removing unequal pairs can never eliminate all majority occurrences.

Therefore the final remaining value must be the majority.

Boyer-Moore simply performs this cancellation **implicitly** without physically creating/removing elements.

This is a very intuitive proof.

36. Follow-Up — What If Majority Is NOT Guaranteed?

This is a very important interviewer question.

Suppose:

text

nums = [1,2,3]

There is no value appearing more than:

text

$3 / 2 = 1$

times.

What does Boyer-Moore return?

It returns **some candidate**, but that candidate is not necessarily a majority.

For example:

```
text

candidate = 3
```

could be returned.

So if the majority is not guaranteed, Boyer-Moore requires verification.

37. Two-Pass Boyer-Moore

Pass 1

Find candidate.

Pass 2

Count candidate occurrences.

Then verify:

```
text

count > n / 2
```

If not:

```
text

no majority exists
```

38. Java — Boyer-Moore Without Guarantee

```
java

public Integer majorityElement(int[] nums) {

    int candidate = 0;
    int count = 0;

    // Pass 1: find candidate
    for (int num : nums) {

        if (count == 0) {
            candidate = num;
        }
    }
}
```

```

        if (num == candidate) {
            count++;
        } else {
            count--;
        }
    }

    // Pass 2: verify candidate
    count = 0;

    for (int num : nums) {
        if (num == candidate) {
            count++;
        }
    }

    if (count > nums.length / 2) {
        return candidate;
    }

    return null;
}

```

You could return `OptionalInt` instead of `Integer` in a production API:

```
java
```

```
public OptionalInt majorityElement(int[] nums)
```

which can more explicitly represent "no majority."

39. Why Is Verification Necessary?

Because Boyer-Moore gives:

```
text
```

```
candidate
```

not:

```
text
```

```
proof of majority
```

when no majority is guaranteed.

This distinction is extremely important.

Guaranteed majority

text

Candidate

↓

answer

No guarantee

text

Candidate

↓

verify frequency

↓

answer / no majority

40. Follow-Up – Can HashMap Return "No Majority" Easily?

Yes.

With HashMap:

text

frequency count

you naturally know the exact frequency.

You can do:

java

```
if (count > nums.length / 2) {  
    return num;  
}
```

or process the complete map afterward.

So:

text

HashMap

is more general, while:

```
text
```

```
Boyer-Moore
```

exploits the strict-majority guarantee.

This is an important trade-off.

41. Java-Specific Follow-Up – Why `candidate = 0` ?

Because:

```
text
```

```
count = 0
```

means the candidate hasn't been established yet.

The initial candidate value:

```
text
```

```
0
```

doesn't actually matter because the first element causes:

```
text
```

```
count == 0
```

and therefore:

```
java
```

```
candidate = num;
```

So you can initialize candidate to any integer value.

42. Alternative Initialization

You could also write:

```
java
```

```
int candidate = nums[0];
int count = 1;

for (int i = 1; i < nums.length; i++) {
    ...
}
```

This is equally valid.

The version:

```
java

int candidate = 0;
int count = 0;
```

is compact and handles candidate initialization uniformly.

43. Java-Specific Follow-Up — Why Not Integer ?

Because this is primitive integer data:

```
java

int candidate
int count
```

Using Integer provides no advantage here.

Primitive variables are simpler and avoid unnecessary boxing.

44. Java-Specific Follow-Up — Can We Use HashMap.merge() ?

Yes:

```
java

freq.merge(num, 1, Integer::sum);
```

This is useful for frequency counting.

But Boyer-Moore intentionally avoids the map entirely.

This is a good Java interview distinction:

"I know how to count with a HashMap, but the problem constraints let me avoid storing those counts."

45. Java-Specific Follow-Up – Can We Use Streams?

Technically yes, but I would not.

The Boyer-Moore state changes sequentially:

```
text

candidate
count
```

depend on previous values.

A stream-based solution is less readable and doesn't improve the algorithm.

For DSA interviews:

```
text

simple loop
```

is the right choice.

46. Edge Case – One Element

```
text

nums = [7]
```

7 appears once.

Since:

```
text

1 > 1/2
```

the answer is:

```
text

7
```

Boyer-Moore returns 7 .

47. Edge Case — Two Equal Elements

text

[5,5]

Candidate:

text

5

Count:

text

2

Return:

text

5

48. Edge Case — Two Different Elements

text

[5,6]

No strict majority.

If the problem guarantees one, this input violates the constraint.

If majority isn't guaranteed:

text

Boyer-Moore candidate

must be verified.

49. Edge Case — All Same

text

[4,4,4,4,4]

Candidate stays:

text

4

Count grows to:

text

5

Return:

text

4

50. Edge Case — Majority Appears Exactly Half?

Suppose:

text

[1,1,2,2]

1 appears:

text

2

and:

text

$2 > n/2$

requires:

text

$2 > 2$

which is false.

So there is **no majority**.

The problem requires:

text

more than $n/2$

not:

text

at least $n/2$

This is a common trap.

51. Strict vs Non-Strict Majority

Majority

text

$\text{count} > n / 2$

At least half

text

$\text{count} \geq n / 2$

These are different requirements.

For:

text

$n = 4$

a count of:

text

2

is exactly half, not a strict majority.

Read the wording carefully.

52. Follow-Up – What If We Need the Majority Percentage?

Suppose the problem changes to:

Find an element occurring more than 60% of the time.

Standard Boyer-Moore no longer directly solves that general threshold problem in the same simple way.

The classical algorithm specifically exploits:

```
text
```

```
> n/2
```

because there can be only one such element.

For arbitrary thresholds, the problem structure changes.

This is an important reason not to memorize Boyer-Moore as:

"frequency problem → use this."

Instead:

Strict majority > 50% → Boyer-Moore is a natural candidate.

53. Why Is There At Most One Majority Element?

Suppose two different values were both strict majorities.

Then:

```
text
```

```
A > n/2
```

```
B > n/2
```

So:

```
text
```

```
A + B > n
```

which is impossible because there are only n elements.

Therefore:

text

At most one majority element exists.

This uniqueness is essential to the cancellation approach.

54. Follow-Up – What If We Need All Elements Appearing More Than $n/3$?

Now there can be at most:

text

2

such elements.

This leads to a generalized Boyer-Moore technique, often called:

text

Misra-Gries / extended Boyer-Moore

where we maintain multiple candidates.

For $> n/3$, maintain two candidates and two counts, followed by verification if the guarantee isn't provided.

This is an excellent advanced follow-up.

But don't overcomplicate the normal $> n/2$ problem.

55. Why $n/3$ Allows Two Candidates

Suppose there were three elements each appearing more than:

text

$n/3$

Then together they'd appear more than:

text

n

which is impossible.

Therefore at most:


```
text
```

```
2
```

elements can exceed $n/3$.

This generalizes the majority reasoning.

56. Follow-Up – What If We Need Elements Appearing More Than $n/4$?

At most:

```
text
```

```
3
```

such elements can exist.

This general pattern is:

```
text
```

```
threshold > n/k
```

allows at most:

```
text
```

```
k - 1
```

candidate values.

That leads to generalized frequency-cancellation algorithms.

This is an advanced DSA connection worth knowing.

57. Pattern Recognition – Frequency vs Majority

A useful decision tree:

```
text
```

```
"Count frequencies"
```

```
↓
```

```
Need exact frequency?
```

```
↓
```

```
HashMap / array
```

"Find strict majority $> n/2$ "



Boyer-Moore

"Find $n/3$ -style frequent elements"



Extended Boyer-Moore / Misra-Gries

This is much more useful than memorizing isolated solutions.

58. Why Doesn't Boyer-Moore Need Sorting?

Sorting gives us positional information:

text

middle element

Boyer-Moore doesn't need ordering.

It uses:

text

frequency dominance

instead.

Therefore it can achieve:

text

$O(n)$

without modifying the array.

This is a different kind of optimization.

59. Why Doesn't Boyer-Moore Need a HashMap?

Because it doesn't care about exact frequencies.

It only needs to know:

text

current candidate

+

```
how many un-cancelled votes it has
```

The strict-majority property guarantees that this is sufficient.

So the algorithm compresses a potentially huge frequency table into:

```
text
```

```
2 integers
```

That's the beautiful part.

60. A Useful Mental Model

Imagine every pair of different values fights:

```
text
```

```
candidate vs opponent
```

One vote is removed from each.

```
text
```

```
candidate vote --X  
opponent vote  --X
```

Since the majority has more votes than all opponents combined:

```
text
```

```
majority votes > opponent votes
```

there must be some majority votes left after all possible cancellation.

The last surviving candidate is therefore the majority.

61. Production Variation — Majority Status of an Event Type

Imagine a monitoring system receives event categories:

```
text
```

```
SUCCESS  
SUCCESS  
FAILURE  
SUCCESS
```

...

and needs to determine whether one category makes up more than half of the events.

The exact Boyer-Moore idea can process a batch using:

```
text

candidate
count
```

without constructing a complete frequency table.

This is useful when only the strict majority matters.

62. Production Variation — Streaming Data

This is particularly interesting.

Suppose events arrive one at a time:

```
text

event1
event2
event3
...
```

For strict-majority detection, the core state can remain:

```
text

candidate
count
```

So the algorithm is stream-friendly:

```
text

one event
  ↓
update candidate/count
  ↓
discard event
```

Memory:

```
text
```

0(1)

This is much smaller than maintaining a map of potentially millions of categories.

63. But Important Production Caveat

If you don't know whether a majority exists, the one-pass candidate isn't enough to **prove** it.

You need a second pass over the data or some way to verify the candidate.

In a streaming system, a second pass may not be available.

Then you may need:

text

external storage
sampling
windowed processing
counting infrastructure

depending on the requirements.

So:

text

Boyer-Moore candidate

and:

text

verified majority

are not automatically the same thing in a real-time stream.

64. Production Variation — Distributed Processing

Suppose data is split across servers:

text

Server A
Server B
Server C

Each server can compute:

```
text
```

```
local candidate/count
```

but simply choosing the candidate with the largest local count is not automatically enough to determine a global majority.

You need careful aggregation/verification.

This is a good example of an algorithm that is trivial in one array but requires more design in a distributed system.

For a normal coding interview, however, stay with the single-array model unless asked.

65. Production Consideration – Mutability

The Boyer-Moore solution:

```
text
```

```
does not modify nums
```

which is preferable to the sorting approach if the input must remain unchanged.

This gives another practical advantage beyond asymptotic complexity.

66. Common Mistake – Returning Candidate Without Understanding the Guarantee

A candidate may say:

"Boyer-Moore always returns the majority."

Not necessarily.

Correct statement:

"If a strict majority is guaranteed to exist, Boyer-Moore returns it. Otherwise, the candidate must be verified."

This distinction is important.

67. Common Mistake – Using $\geq n/2$

Incorrect for strict majority:

```
java
```

```
count >= nums.length / 2
```

Correct:

```
java
```

```
count > nums.length / 2
```

68. Common Mistake – Sorting and Claiming $O(n)$

Sorting is:

```
text
```

```
 $O(n \log n)$ 
```

not $O(n)$.

The middle lookup itself is $O(1)$, but the sorting dominates.

69. Common Mistake – Using `HashSet`

A `HashSet` only tells:

```
text
```

```
does this element exist?
```

It doesn't tell:

```
text
```

```
how many times does it occur?
```

For majority detection, exact frequency or a specialized cancellation algorithm is required.

70. Common Mistake – Thinking Candidate Is the Most Frequent So Far

Boyer-Moore's candidate is **not necessarily the most frequent element in the prefix**.

It is the current surviving candidate after cancellation.

That's a subtle but important distinction.

71. Example of Candidate Changing

text

```
[1,2,3,2,2]
```

The candidate can change during the scan.

The algorithm does not try to maintain:

text

```
"most frequent element seen so far"
```

It maintains:

text

```
"uncancelled candidate"
```

That is why the logic works with constant space.

72. Java-Specific Follow-Up – Can We Write It Using `switch` ?

There's no reason to.

The operation is fundamentally binary:

text

```
same as candidate?  
yes  → ++  
no   → --
```

An `if` is the clearest representation.

73. Java-Specific Follow-Up – Can We Use `AtomicInteger` ?

Not for this DSA problem.

`AtomicInteger` is for concurrency-related atomic operations.

This algorithm is sequential and does not require synchronization.

Using it would add complexity without benefit.

74. Java-Specific Follow-Up – Thread Safety

The local variables:

```
text  
  
candidate  
count
```

are thread-confined if the method is called normally.

The algorithm itself doesn't share mutable state.

Therefore the method is naturally thread-safe as an independent computation.

Of course, if multiple threads mutate the same input array concurrently, that's a separate issue.

75. Java-Specific Follow-Up – `Integer.MIN_VALUE` as Candidate?

You could initialize:

```
java  
  
int candidate = Integer.MIN_VALUE;
```

but there is no reason to.

Because:

```
java  
  
count = 0
```

causes the first element to become the candidate.

Using:

```
java  
  
candidate = 0
```

is simpler.

76. Edge Case – Negative Majority

```
text
```

```
nums = [-1,-1,2,-1,3]
```

-1 appears three times out of five.

Therefore:

```
text
```

```
3 > 5/2
```

and the majority is:

```
text
```

```
-1
```

Boyer-Moore works perfectly.

It doesn't depend on positive values.

77. Edge Case — Large Integer Values

The actual numeric magnitude doesn't matter.

The algorithm only compares:

```
java
```

```
num == candidate
```

so values like:

```
text
```

```
Integer.MIN_VALUE
```

```
Integer.MAX_VALUE
```

are fine.

No arithmetic is performed on the element values themselves.

That's another nice property of the algorithm.

78. Edge Case — `int[]` Length

The array length is an `int` in Java.

The code:

```
java
```

```
nums.length / 2
```

is integer division, which is exactly what we need for the threshold comparison.

For example:

```
text
```

```
n = 7  
7 / 2 = 3
```

and:

```
text
```

```
count > 3
```

means count must be at least 4.

That's equivalent to:

```
text
```

```
count > 7/2
```

for integer counts.

79. Could We Avoid the Verification Pass Without Guarantee?

Not for exact deterministic correctness using standard Boyer-Moore alone.

After pass one, you have:

```
text
```

```
candidate
```

but you haven't established:

```
text
```

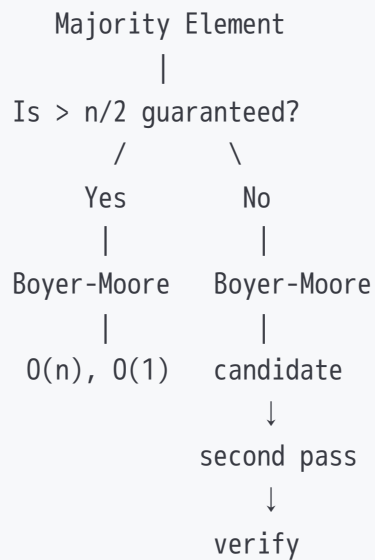
```
frequency(candidate) > n/2
```

So you need to count it.

This is an important correctness distinction.

80. Final Decision Tree

text



If you need exact frequencies of all elements:

text

HashMap

If mutation is acceptable and simplicity is preferred:

text

Sort → middle

81. Which Solution Would I Give in an Interview?

For a 2–2.5 YOE candidate, I'd demonstrate the progression:

Basic

text

Brute force
 $O(n^2)$

Then:

Better

text

HashMap
 $O(n)$ expected
 $O(n)$ space

Then:

Another approach

text

Sort
 $O(n \log n)$

Then:

Best

text

Boyer-Moore
 $O(n)$
 $O(1)$

This shows the interviewer that you understand multiple techniques rather than memorizing one code snippet.

82. Final Interview Conversation

Interviewer:

"Find the majority element."

You:

"The brute-force approach would count every element repeatedly, which is $O(n^2)$."

Then:

"I can improve that using a HashMap to maintain value -> frequency, giving $O(n)$ average time and $O(n)$ space."

Then:

"Another approach is sorting. Since the majority occurs more than half the time, after sorting it must occupy index $n/2$, so we can return the middle element. That's $O(n \log n)$."

Then:

"But we can achieve $O(n)$ time and $O(1)$ space using Boyer-Moore Voting. I treat every value as a vote. The candidate gains one vote when the current value matches it and loses one vote when it differs. When the count becomes zero, I choose the next value as the candidate. Since the majority occurs more than all other values combined, all opposing votes can cancel some majority votes, but the majority still has votes left."

That's the answer progression I'd want in the interview.

83. Final Interview-Ready Code

For the standard problem where majority is guaranteed:

java

```
public int majorityElement(int[] nums) {  
  
    int candidate = 0;  
    int count = 0;  
  
    for (int num : nums) {  
  
        if (count == 0) {  
            candidate = num;  
        }  
  
        if (num == candidate) {  
            count++;  
        } else {  
            count--;  
        }  
    }  
  
    return candidate;  
}
```

Complexity

text

Time: $O(n)$
Auxiliary Space: $O(1)$

84. If Majority Is Not Guaranteed

Use verification:

java

```
public Integer majorityElement(int[] nums) {

    int candidate = 0;
    int count = 0;

    // Find candidate
    for (int num : nums) {

        if (count == 0) {
            candidate = num;
        }

        if (num == candidate) {
            count++;
        } else {
            count--;
        }
    }

    // Verify candidate
    count = 0;

    for (int num : nums) {
        if (num == candidate) {
            count++;
        }
    }

    return count > nums.length / 2 ? candidate : null;
}
```

This remains:

text

Time: $O(n)$
Space: $O(1)$

because two passes are still linear.

85. Final Interviewer-Expected Answer

If asked:

"Why is Boyer-Moore the optimal solution?"

Say:

"Because the problem guarantees an element occurring more than $n/2$ times. That means the majority count is greater than the total count of all other elements combined. I can therefore cancel one majority vote with one non-majority vote. After all possible cancellations, the majority must still have an uncanceled vote. Boyer-Moore implements this cancellation using only a candidate and a count. So we get $O(n)$ time and $O(1)$ auxiliary space."

86. If Asked "Why Doesn't It Work Without the Guarantee?"

Say:

"Without the majority guarantee, Boyer-Moore still produces a candidate, but that candidate isn't necessarily a true majority. I therefore need a second pass to verify that its frequency is greater than $n/2$."

That is the precise answer.

87. Final Cheat Sheet

Approach 1 — Brute Force

text

For each element:
count occurrences

Time: $O(n^2)$
Space: $O(1)$

Approach 2 — HashMap

text

value $\rightarrow$ frequency

Time: $O(n)$ average
Space: $O(n)$

Approach 3 — Sorting

text


```
sort
return nums[n / 2]
```

Time: $O(n \log n)$

Extra space depends on sorting

Input modified

Approach 4 — Boyer-Moore

text

candidate

count

same $\rightarrow$ count++

different $\rightarrow$ count--

count == 0 $\rightarrow$ choose new candidate

Time: $O(n)$

Space: $O(1)$

88. Core Pattern

text

Strict majority $> n/2$

↓

Majority count $>$ all others combined

↓

Pair/cancel different values

↓

Majority survives

↓

Boyer-Moore Voting

89. Most Important Invariant

text

candidate

=

current surviving candidate

```
count
=
uncancelled votes for that candidate
```

When:

text

```
count == 0
```

the previous candidate has been completely cancelled.

90. Critical Interview Traps

text

```
count >= n/2
```



```
count > n/2
```



text

Boyer-Moore always proves a majority exists



text

Boyer-Moore finds a candidate;
verify it if no majority is guaranteed.



text

Candidate = most frequent value so far



text

Candidate = surviving candidate after cancellation



91. Connection to Earlier Problems

There is a useful progression across these questions:

text

Q2 Contains Duplicate



"Have I seen this?"



HashSet

Q3 Valid Anagram



"How many times?"



Frequency Array / HashMap

Q4 Group Anagrams



"How can equivalent objects share a key?"



Canonical key + HashMap

Q5 Top K Frequent



"How do I select highest frequencies?"



HashMap + Heap

Q6 Product Except Self



"Can I reuse prefix/suffix state?"



Prefix + Suffix

Q7 Stock



"Best previous value?"



Running Minimum

Q8 Maximum Subarray



"Extend or restart?"



Kadane / 1D DP

Q9 Majority Element

↓
"Can competing values cancel?"

↓
Boyer-Moore

The important skill you're building is not memorizing nine unrelated solutions. It is learning to identify the **structure of the problem** and select the smallest amount of state/data structure needed to solve it.

92. The One-Line Mental Model for Q9

Remember:

text

MAJORITY > HALF



MORE THAN ALL OTHERS COMBINED



CANCEL DIFFERENT PAIRS



MAJORITY SURVIVES



BOYER-MOORE

For the standard guaranteed-majority problem, **Boyer-Moore is the interviewer-expected best answer: $O(n)$ time and $O(1)$ auxiliary space.**

2026-09-02 21:29:55 You 

ok as aboved continue for next question,i want the solutions as interviewer expcted in the interview of 2-2.5 year experience candidate and the solutions first is basic approach then other approach then next another better approach as like that possible approaches which patterns used and why that patterns used and at last best optimize approach as per the interviewer. keep this same interview structure: brute force → better → optimal → pattern recognition → why the data structure works → dry run → edge cases → Java-specific follow-ups → production variation → final interviewer-ready answer.

Q10. Missing Number Question Array contains numbers from 0..n with one missing. Approach 1 Use sum. text expected = $n(n+1)/2$ Potential integer overflow. Better interview trick XOR. java public

```
int missingNumber(int[] nums) { int xor = nums.length; for (int i = 0; i < nums.length; i++) { xor ^= i; xor ^= nums[i]; } return xor; }  
Complexity 14 15 text O(n) O(1) Follow-up Why XOR? text x ^ x = 0 x ^ 0 = x All existing values cancel.
```

 ChatGPT gpt-5-6-t-mini

Q10. Missing Number

This is a classic array problem, but it is especially useful in interviews because it tests whether you can recognize a **mathematical property** and use it to reduce space.

The progression I would use in a real interview is:

text

Brute Force

↓

Sorting

↓

Mathematical Sum

↓

XOR

↓

$O(n)$ time + $O(1)$ extra space

The interesting part is that the **sum approach is already $O(n)$ and $O(1)$** , so why would we prefer XOR?

Because XOR avoids arithmetic overflow and gives us an elegant cancellation property.

1. Problem

You are given an array containing n distinct numbers taken from:

text

$0, 1, 2, \dots, n$

Exactly one number is missing.

Return the missing number.

Example

text

```
nums = [3, 0, 1]
```

There should be:

```
text
```

```
0, 1, 2, 3
```

But:

```
text
```

```
2
```

is missing.

Answer:

```
text
```

```
2
```

2. Another Example

```
text
```

```
nums = [9,6,4,2,3,5,7,0,1]
```

There are 9 elements, so the expected range is:

```
text
```

```
0..9
```

The missing number is:

```
text
```

```
8
```

3. What I Would Say First in the Interview

A strong opening:

"The basic approach is to search for every number from 0 to n, which is $O(n^2)$. We can improve this by sorting and finding the first mismatch, giving $O(n \log n)$. A better $O(n)$ and $O(1)$ -space solution uses the arithmetic sum from 0 to n and subtracts the sum of the array. However, the sum formula can overflow for large integer inputs, so an even safer $O(n)$, $O(1)$ solution is XOR. If I XOR all numbers from 0 to n with all numbers present in the array, every present number appears twice and cancels because $x \oplus x = 0$, leaving exactly the missing number."

That is a very strong interview introduction.

4. Approach 1 — Brute Force

The simplest idea is:

For every number from:

text

0

to:

text

n

check whether it exists in the array.

If one doesn't exist, return it.

5. Java — Brute Force

java

```
public int missingNumber(int[] nums) {  
  
    int n = nums.length;  
  
    for (int expected = 0; expected <= n; expected++) {  
  
        boolean found = false;  
  
        for (int num : nums) {  
  
            if (num == expected) {  
                found = true;  
                break;  
            }  
        }  
    }  
}
```

```
    }  
    }  
  
    if (!found) {  
        return expected;  
    }  
}  
  
return -1;  
}
```

6. Dry Run — Brute Force

Input:

text

nums = [3, 0, 1]

Length:

text

n = 3

Expected values:

text

0, 1, 2, 3

Check 0 :

text

found

Check 1 :

text

found

Check 2 :

text


```
not found
```

Return:

```
text
```

```
2
```

7. Complexity – Brute Force

For each of the $n + 1$ possible values, we may scan the entire array.

Therefore:

```
text
```

```
Time:  $O(n^2)$ 
```

```
Space:  $O(1)$ 
```

Correct, but inefficient.

The first obvious improvement is sorting.

8. Approach 2 – Sorting

Since the values should be:

```
text
```

```
0, 1, 2, ..., n
```

if we sort the array, the missing number creates a gap.

Example:

```
text
```

```
nums = [3, 0, 1]
```

Sort:

```
text
```

```
[0, 1, 3]
```

Expected:

text

0, 1, 2, 3

At index 2 we expect:

text

2

but find:

text

3

Therefore:

text

2

is missing.

9. Java – Sorting Approach

java

```
public int missingNumber(int[] nums) {  
  
    Arrays.sort(nums);  
  
    for (int i = 0; i < nums.length; i++) {  
  
        if (nums[i] != i) {  
            return i;  
        }  
    }  
  
    return nums.length;  
}
```

10. Why Return `nums.length` ?

Suppose:

```
text
```

```
nums = [0, 1, 2, 3]
```

Here:

```
text
```

```
n = 4
```

The expected range is:

```
text
```

```
0,1,2,3,4
```

Every array position matches:

```
text
```

```
nums[i] == i
```

So the missing number must be:

```
text
```

```
4
```

which is:

```
java
```

```
return nums.length;
```

This is an important edge case.

11. Complexity – Sorting

Sorting costs:

```
text
```

```
O(n log n)
```

Scanning costs:

```
text
```

$O(n)$

Therefore:

text

Time: $O(n \log n)$

Extra space depends on the sorting implementation.

Also:

java

```
Arrays.sort(nums);
```

mutates the input array.

So while this is better than brute force, we can do $O(n)$.

12. Approach 3 – HashSet

Another straightforward $O(n)$ approach is to store all numbers in a set.

Then iterate from:

text

$0..n$

and find which number isn't present.

Java

java

```
public int missingNumber(int[] nums) {  
  
    Set<Integer> set = new HashSet<>();  
  
    for (int num : nums) {  
        set.add(num);  
    }  
  
    for (int i = 0; i <= nums.length; i++) {  
  
        if (!set.contains(i)) {  
            return i;  
        }  
    }  
}
```

```
    }  
}  
  
return -1;  
}
```

13. Complexity – HashSet

Build set:

text

$O(n)$

Search:

text

$O(n)$

Expected HashSet lookup:

text

$O(1)$

Therefore:

text

Time: $O(n)$ average

Space: $O(n)$

This is good, but we can avoid the extra memory.

14. Approach 4 – Mathematical Sum

There is a beautiful observation.

The numbers should be:

text

$0 + 1 + 2 + \dots + n$

The sum of the first n integers is:

```
text
```

$$n(n + 1) / 2$$

So:

```
text
```

$$\text{expectedSum} = n * (n + 1) / 2$$

Then:

```
text
```

$$\text{missing} = \text{expectedSum} - \text{actualSum}$$

15. Example – Sum

```
text
```

```
nums = [3, 0, 1]  
n = 3
```

Expected sum:

```
text
```

$$3 * 4 / 2 \\ = 6$$

Actual sum:

```
text
```

$$3 + 0 + 1 \\ = 4$$

Therefore:

```
text
```

$$\text{missing} = 6 - 4 \\ = 2$$

16. Java – Sum Approach

A naive version:

```
java

public int missingNumber(int[] nums) {

    int n = nums.length;

    int expected = n * (n + 1) / 2;

    int actual = 0;

    for (int num : nums) {
        actual += num;
    }

    return expected - actual;
}
```

But there is a Java problem here.

17. Important Interview Trap – Integer Overflow

This expression:

```
java

n * (n + 1)
```

is evaluated using `int` arithmetic if `n` is an `int`.

For sufficiently large `n`, it can overflow **before** the division by `2`.

A stronger implementation is:

```
java

public int missingNumber(int[] nums) {

    int n = nums.length;

    long expected = (long) n * (n + 1) / 2;

    long actual = 0;

    for (int num : nums) {
```

```

        actual += num;
    }

    return (int) (expected - actual);
}

```

Now the multiplication happens using `long` .

This is an excellent Java-specific interview point.

18. But the Sum Approach Is Already $O(n)$, $O(1)$

Exactly.

So at this point we have:

text

```

Brute force →  $O(n^2)$ 
Sorting      →  $O(n \log n)$ 
HashSet      →  $O(n)$ ,  $O(n)$ 
Sum          →  $O(n)$ ,  $O(1)$ 

```

You could legitimately stop here.

But the interviewer may ask:

"Can you avoid the sum overflow issue?"

That's where XOR is particularly elegant.

19. Approach 5 – XOR

This is the preferred solution when we want:

text

```

 $O(n)$  time
 $O(1)$  extra space

```

without relying on arithmetic summation.

The key XOR properties are:

text

```

 $x \wedge x = 0$ 

```


and:

```
text
```

$$x \wedge 0 = x$$

XOR is also associative and commutative:

```
text
```

$$a \wedge b \wedge c$$

can be rearranged without changing the result.

20. Why XOR Finds the Missing Number

We have all numbers from:

```
text
```

$$0..n$$

except one.

Suppose:

```
text
```

$$0,1,2,3,4$$

with 2 missing.

Array contains:

```
text
```

$$0,1,3,4$$

Now XOR everything:

```
text
```

$$0 \wedge 1 \wedge 2 \wedge 3 \wedge 4$$

and XOR all array elements:

```
text
```

```
0 ^ 1 ^ 3 ^ 4
```

Combine them:

```
text
```

```
0 ^ 1 ^ 2 ^ 3 ^ 4  
^ 0 ^ 1 ^ 3 ^ 4
```

Rearrange:

```
text
```

```
(0 ^ 0)  
^ (1 ^ 1)  
^ (3 ^ 3)  
^ (4 ^ 4)  
^ 2
```

Each duplicate pair becomes zero:

```
text
```

```
0 ^ 0 = 0  
1 ^ 1 = 0  
3 ^ 3 = 0  
4 ^ 4 = 0
```

So we're left with:

```
text
```

```
2
```

That's the missing number.

21. Java – XOR Solution

```
java
```

```
public int missingNumber(int[] nums) {  
  
    int xor = nums.length;  
  
    for (int i = 0; i < nums.length; i++) {  
        xor ^= i;  
    }  
}
```

```
        xor ^= nums[i];
    }

    return xor;
}
```

This is the standard optimal solution.

22. Why Initialize xor With nums.length ?

This is a very common interviewer question.

Suppose:

text

nums.length = n

The required full range is:

text

0, 1, 2, ..., n

But the loop:

java

```
for (int i = 0; i < nums.length; i++)
```

only visits:

text

0,1,2,...,n-1

So we initialize:

java

```
int xor = nums.length;
```

to include:

text

n

Then inside the loop we XOR:

```
text
```

```
i
```

and:

```
text
```

```
nums[i]
```

Therefore, overall we XOR:

```
text
```

```
0 ^ 1 ^ ... ^ n
```

with:

```
text
```

```
nums[0] ^ nums[1] ^ ... ^ nums[n-1]
```

Exactly what we need.

23. Dry Run — XOR

Take:

```
text
```

```
nums = [3, 0, 1]
```

Length:

```
text
```

```
n = 3
```

Initialize:

```
text
```

```
xor = 3
```

i = 0

text

```
xor ^= 0  
xor ^= nums[0]    // 3
```

So:

text

$3 \wedge 0 \wedge 3$

Because:

text

$3 \wedge 3 = 0$

we get:

text

$xor = 0$

i = 1

text

```
xor ^= 1  
xor ^= nums[1]    // 0
```

So:

text

$0 \wedge 1 \wedge 0$

gives:

text

$xor = 1$

i = 2

text

```
xor ^= 2  
xor ^= nums[2] // 1
```

So:

text

$1 \wedge 2 \wedge 1$

First:

text

$1 \wedge 1 = 0$

therefore:

text

$xor = 2$

Return:

text

2

24. Another Dry Run Where Missing Number Is 0

Input:

text

$nums = [1, 2, 3]$

Length:

text

$n = 3$

Expected:

text

0,1,2,3

Missing:

text

0

Initialize:

text

xor = 3

Process:

text

i = 0:
xor = 3 ^ 0 ^ 1

i = 1:
xor = previous ^ 1 ^ 2

i = 2:
xor = previous ^ 2 ^ 3

Everything cancels:

text

1 ^ 1
2 ^ 2
3 ^ 3

leaving:

text

0

Correct.

25. Another Dry Run Where Missing Number Is n

Input:

```
text
```

```
nums = [0,1,2]
```

Here:

```
text
```

```
n = 3
```

Expected:

```
text
```

```
0,1,2,3
```

Missing:

```
text
```

```
3
```

The initial:

```
java
```

```
xor = nums.length;
```

already contains:

```
text
```

```
3
```

and the loop cancels:

```
text
```

```
0  
1  
2
```

from the full sequence with the same values from the array.

So 3 remains.

26. Why XOR Works – Formal Explanation

Let the missing number be `m`.

We calculate:

text

$$(0 \wedge 1 \wedge 2 \wedge \dots \wedge n) \\ \wedge \\ (\text{nums}[0] \wedge \text{nums}[1] \wedge \dots \wedge \text{nums}[n-1])$$

Every number except `m` appears exactly twice.

Since:

text

$$x \wedge x = 0$$

all present numbers cancel.

The only number appearing once is:

text

`m`

Therefore the final XOR is:

text

`m`

That's the correctness proof.

27. Why XOR Doesn't Overflow

This is an important advantage over the sum approach.

XOR operates on the binary representation of the integers.

There is no arithmetic expression like:

text

$$n * (n + 1)$$

that can overflow in the same way.

Java's `int` XOR operation simply performs bitwise XOR on 32-bit integers.

So for the normal problem's `int` values:

```
text

xor ^= value
```

does not have the sum-overflow issue.

28. Is XOR Always Better Than Sum?

Not necessarily.

Both are:

```
text

Time: O(n)
Space: O(1)
```

The sum solution is arguably easier for many candidates to derive:

```
text

expected - actual
```

The XOR solution has a more unusual idea but avoids the arithmetic-overflow concern.

So a good interview answer is:

"The sum approach is already $O(n)$ time and $O(1)$ space, but it needs care around integer overflow. XOR provides the same complexity and avoids that arithmetic issue, so I prefer XOR when the problem expects an $O(1)$ -space solution."

That's nuanced and strong.

29. Approach Comparison

Approach	Time	Extra Space	Main Idea
Brute force	$O(n^2)$	$O(1)$	Search every expected number
Sorting	$O(n \log n)$	Depends	Find first missing position
HashSet	$O(n)$ avg	$O(n)$	Membership lookup
Sum	$O(n)$	$O(1)$	Expected sum – actual sum

Approach	Time	Extra Space	Main Idea
XOR	$O(n)$	$O(1)$	Cancellation

For this standard problem:

text

XOR

is the strongest final answer.

30. Pattern Recognition

The key pattern is:

Cancellation / Pairing

We saw a similar conceptual idea in Q9:

text

Majority Element
↓
cancel different votes
↓
majority survives

Here:

text

Missing Number
↓
cancel identical values with XOR
↓
missing value survives

So there is a nice connection.

31. XOR Pattern

Whenever you see a problem where:

text

every value appears twice
except one

think:

text

XOR

Classic example:

text

[4,1,2,1,2]

All duplicates cancel:

text

$4 \wedge 1 \wedge 2 \wedge 1 \wedge 2$

leaving:

text

4

This is the classic **Single Number** problem.

32. Difference Between Missing Number and Single Number

Missing Number

The values belong to a known complete range:

text

$0..n$

with one missing.

Therefore we can XOR:

text

complete range
+

```
array
```

Single Number

Values may be arbitrary, but every value except one appears twice.

Therefore simply:

```
text
```

```
XOR everything
```

Same mathematical property:

```
text
```

```
 $x \wedge x = 0$ 
```

33. Why Does the Data Structure Work?

Strictly speaking, XOR is not a data structure.

The solution uses:

```
text
```

```
one accumulator variable
```

and the **XOR algebraic property**.

This is an important observation:

Sometimes the optimal solution comes from an operation's mathematical properties rather than a collection/data structure.

The "state" is just:

```
text
```

```
xor
```

which is enough to represent everything we need.

34. XOR Properties You Should Know for Interviews

The three most useful are:

Identity

text

$$x \wedge 0 = x$$

Self-cancellation

text

$$x \wedge x = 0$$

Associativity

text

$$\begin{aligned} (a \wedge b) \wedge c \\ = \\ a \wedge (b \wedge c) \end{aligned}$$

Commutativity

text

$$\begin{aligned} a \wedge b \\ = \\ b \wedge a \end{aligned}$$

Associativity + commutativity allow us to rearrange all terms so duplicate values can cancel.

35. Why Order Doesn't Matter in XOR

Suppose:

text

$$a \wedge b \wedge c$$

Because XOR is commutative and associative:

text

$$\begin{aligned} a \wedge b \wedge c \\ = \\ c \wedge a \wedge b \end{aligned}$$

```
=  
a ^ c ^ b
```

This lets us conceptually rearrange the expression into:

```
text  
  
a ^ a  
^  
b ^ b  
^  
c
```

and cancellation becomes obvious.

This is the mathematical reason the algorithm works.

36. Edge Case — Empty Array

If:

```
text  
  
nums = []
```

then:

```
text  
  
n = 0
```

The range is:

```
text  
  
0
```

so the missing number is:

```
text  
  
0
```

The XOR implementation:

```
java  
  
int xor = nums.length; // 0
```

and no loop iterations occur.

Return:

```
text
```

```
0
```

Beautifully, no special case is required.

37. Edge Case — Single Element [0]

```
text
```

```
nums = [0]
```

Expected range:

```
text
```

```
0,1
```

Missing:

```
text
```

```
1
```

XOR:

```
text
```

```
1 ^ 0 = 1
```

Correct.

38. Edge Case — Single Element [1]

```
text
```

```
nums = [1]
```

Expected:

```
text
0,1
```

Missing:

```
text

0
```

XOR:

```
text

1 ^ 0 ^ 1
= 0
```

Correct.

39. Edge Case – Missing `n`

```
text

nums = [0,1,2,3]
```

Expected:

```
text

0,1,2,3,4
```

Missing:

```
text

4
```

The initial:

```
java

xor = 4;
```

ensures `4` has only one occurrence.

40. Edge Case – Missing `0`

```
text
```

```
nums = [1,2,3,4]
```

Expected:

```
text
```

```
0,1,2,3,4
```

Everything from 1 through 4 cancels.

Remaining:

```
text
```

```
0
```

41. Edge Case — Missing Middle Value

```
text
```

```
nums = [0,1,2,4,5]
```

Here:

```
text
```

```
n = 5
```

Missing:

```
text
```

```
3
```

All other values cancel and 3 remains.

42. What If Input Contains Duplicates?

The standard problem usually guarantees:

```
text
```

```
all numbers are distinct
```

and exactly one is missing.

If duplicates are allowed, the XOR solution does **not** generally solve the modified problem.

For example:

```
text
```

```
[0,1,1,3]
```

XOR cancellation tells you something about parity, not necessarily the unique missing value from a range.

So always verify the constraints.

43. What If Two Numbers Are Missing?

Again, this is a different problem.

The simple full-range XOR would leave:

```
text
```

```
missing1 ^ missing2
```

rather than identifying both individually.

A more advanced XOR partition technique can find two missing values, but that's a modified problem.

This is a good interview lesson:

Understand exactly what assumptions make your optimization valid.

44. What If More Than One Duplicate Exists?

The standard problem doesn't allow it.

XOR detects parity, not arbitrary frequency counts.

For example:

```
text
```

```
x ^ x ^ x = x
```

So XOR doesn't tell you:

```
text
```

```
"this number appeared three times"
```

It only cancels pairs.

That is why XOR is appropriate for problems with a specific pairing/cancellation property.

45. Java-Specific Follow-Up – Why ^= ?

This:

```
java
xor ^= nums[i];
```

is shorthand for:

```
java
xor = xor ^ nums[i];
```

Similarly:

```
java
xor ^= i;
```

means:

```
java
xor = xor ^ i;
```

This is common bitwise Java syntax.

46. Java-Specific Follow-Up – Is ^ Exponentiation?

No.

This is a common mistake, especially for developers coming from mathematical notation.

In Java:

```
java
^
```

means:

```
text
```

```
bitwise XOR
```

It does **not** mean power.

For exponentiation, Java uses:

```
java
```

```
Math.pow()
```

though it returns a floating-point value.

47. Java-Specific Follow-Up — ^ With boolean

Java also supports logical XOR for booleans:

```
java
```

```
true ^ false
```

but in this problem:

```
java
```

```
int ^ int
```

is bitwise XOR.

The same operator behaves according to operand types.

48. Java-Specific Follow-Up — Why Is XOR Safe With Negative Numbers?

The standard input domain here is:

```
text
```

```
0..n
```

so values aren't negative.

But XOR itself works on Java's two's-complement integer bit representations, so the operation itself is well-defined for negative integers too.

The problem's range constraint is what matters for correctness.

49. Java-Specific Follow-Up – Why Not Use `Math.pow()` for the Sum?

Don't.

The sum formula is:

```
text

n(n+1)/2
```

and this is integer arithmetic.

`Math.pow()` works with floating-point numbers and introduces unnecessary conversion/precision considerations.

Use:

```
java

long expected = (long) n * (n + 1) / 2;
```

if you choose the sum approach.

50. Java-Specific Follow-Up – Why Cast Before Multiplication?

This:

```
java

long expected = (long) n * (n + 1) / 2;
```

is correct because:

```
text

(long) n
```

forces the multiplication into `long`.

Compare with:

```
java

long expected = n * (n + 1) / 2;
```

The multiplication can overflow as `int` **before** the result is assigned to `long`.

This is an excellent Java interview detail.

51. Sum vs XOR – What Would I Say?

If interviewer asks:

"Why XOR when sum is simpler?"

A strong answer:

"The sum approach is also $O(n)$ time and $O(1)$ space, but I need to use `long` carefully to avoid intermediate integer overflow. XOR gives the same asymptotic complexity and relies on exact cancellation, so it avoids that arithmetic-overflow concern and is especially elegant for this problem."

That's a very good answer.

52. Can We Use a HashSet Instead?

Yes:

text

$O(n)$

time and:

text

$O(n)$

space.

But we don't need to store all values.

XOR reduces the state to:

text

one integer

So:

text

HashSet
 $O(n)$ memory

versus:

```
text
XOR
O(1) memory
```

The latter is preferable under the problem's constraints.

53. Why Not Use Sorting?

Because sorting does more work than necessary.

We don't need ordering.

We need to identify one missing member of a known range.

XOR exploits exactly that property.

This is another example of:

```
text

Don't sort when a stronger mathematical property exists.
```

54. Production Variation — Detect Missing Sequence Number

Suppose a backend receives a batch of sequence IDs expected to contain:

```
text

0..n
```

with exactly one missing ID.

For a bounded batch, the XOR method can detect the missing sequence number without allocating a HashSet.

Conceptually:

```
text

expected range
  XOR
received IDs
  ↓
missing ID
```

This is useful for some:


```
text
sequence validation
batch integrity checks
packet/message sequence analysis
```

55. Production Caveat — Distributed Systems

Suppose sequence numbers arrive from multiple servers.

A local XOR can detect a missing value only relative to the set being examined.

If events are:

```
text

Server A
Server B
Server C
```

you need clearly defined:

```
text

range
partition
batch/window
```

before interpreting the result.

Also, XOR only works if the assumptions hold:

```
text

exactly one missing
no unexpected duplicates
known complete range
```

If those assumptions don't hold, use a different validation strategy.

56. Production Variation — Packet/Message Sequence Validation

Imagine receiving:

```
text

sequence numbers:
0,1,2,4,5
```

Expected:

```
text
```

```
0..5
```

XOR can identify:

```
text
```

```
3
```

as the missing sequence number.

But in a real distributed messaging system, missing messages are often better tracked with:

```
text
```

```
offsets
```

```
acknowledgements
```

```
consumer lag
```

```
persistent sequence tracking
```

rather than relying purely on XOR.

The DSA technique is useful as a lightweight mathematical operation under controlled assumptions, not as a universal distributed-consistency mechanism.

57. Pattern Recognition — Known Range + One Missing

Whenever you see:

```
text
```

```
numbers from 0..n
```

```
exactly one missing
```

```
distinct values
```

consider:

```
text
```

```
1. Sum
```

```
2. XOR
```

Then ask:

```
text
```

Need $O(1)$ space?



Sum or XOR

Concerned about integer overflow?



XOR is attractive

58. Pattern Recognition — Cancellation

The deeper connection is:

text

Q9 Majority

different values cancel each other

Q10 Missing Number

identical values cancel using XOR

Both are examples of:

text

Remove irrelevant paired information



desired information survives

This is a very useful way to remember the logic.

59. A Different XOR Formulation

You may also write:

java

```
public int missingNumber(int[] nums) {  
  
    int xor = 0;  
  
    for (int i = 0; i < nums.length; i++) {  
        xor ^= i;  
        xor ^= nums[i];  
    }  
  
    xor ^= nums.length;  
}
```

```
    return xor;
}
```

This works because the XOR includes:

```
text

0..n-1
+
n
+
all array values
```

The version:

```
java

int xor = nums.length;
```

is simply more concise.

Both are correct.

60. Which XOR Code Would I Use?

I would use:

```
java

public int missingNumber(int[] nums) {

    int xor = nums.length;

    for (int i = 0; i < nums.length; i++) {
        xor ^= i;
        xor ^= nums[i];
    }

    return xor;
}
```

because the initialization naturally handles the extra value `n`.

61. Can We Combine XOR in One Expression?

You might see:

```
java
```

```
xor ^= i ^ nums[i];
```

For example:

```
java
```

```
public int missingNumber(int[] nums) {  
  
    int xor = nums.length;  
  
    for (int i = 0; i < nums.length; i++) {  
        xor ^= i ^ nums[i];  
    }  
  
    return xor;  
}
```

This is correct.

But I'd prefer:

```
java
```

```
xor ^= i;  
xor ^= nums[i];
```

in an interview because it's easier to explain and debug.

Again:

```
text
```

```
clarity > tiny syntax compression
```

62. Important Interview Question – What Is the Time Complexity?

Answer:

```
text
```

```
O(n)
```

because the array is traversed exactly once.

63. What Is the Space Complexity?

Answer:

text

$O(1)$ auxiliary space

because we only maintain:

text

xor

The input array itself is not modified and no extra collection is created.

64. Can We Do Better Than $O(n)$?

No.

The missing value could be determined by the final element you inspect.

For example:

text

$[0, 1, 2, 3, \dots, n-1]$

has missing number n .

Changing the last relevant input can change the answer.

Therefore the algorithm must inspect the input in the general case.

So:

text

Lower bound: $\Omega(n)$

and XOR achieves:

text

$O(n)$

Thus it is asymptotically optimal.

65. Common Mistake – Sum Overflow

Wrong:

```
java  
  
int expected = n * (n + 1) / 2;
```

for sufficiently large `n`.

Better:

```
java  
  
long expected = (long) n * (n + 1) / 2;
```

Or use XOR.

66. Common Mistake – HashSet and Calling It $O(1)$ Space

HashSet requires:

```
text
```

```
O(n)
```

space in the worst case.

So:

```
text
```

```
HashSet
```

is not the optimal space solution.

67. Common Mistake – Sorting and Calling It $O(n)$

Sorting is:

```
text
```

```
 $O(n \log n)$ 
```

not $O(n)$.

68. Common Mistake – XOR Without Explaining Cancellation

Don't simply say:

```
"Use XOR because it's O(1)."
```

The interviewer wants to know **why it produces the missing number**.

The key explanation is:

```
text
```

```
Every existing number appears twice:  
once in 0..n  
once in nums
```

```
x ^ x = 0
```

```
Only the missing number appears once.  
Therefore it survives.
```

That's the important reasoning.

69. Common Mistake – Forgetting `n`

Suppose:

```
text
```

```
nums.length = n
```

The array indices only cover:

```
text
```

```
0..n-1
```

but the expected values cover:

```
text
```

```
0..n
```

So the value:

```
text
```

```
n
```


must be included separately.

That's why:

```
java  
  
int xor = nums.length;
```

is such a useful pattern.

70. Common Mistake — Assuming XOR Can Handle Arbitrary Duplicates

It can't solve arbitrary missing/duplicate problems.

Its correctness here depends on:

```
text  
  
distinct input  
+  
known complete range  
+  
exactly one missing
```

Change those assumptions and the algorithm may no longer work.

71. Advanced Follow-Up — Find the Missing and Duplicate Number

Suppose the array should contain:

```
text  
  
1..n
```

but:

```
text  
  
one number is duplicated  
one number is missing
```

Then XOR can still be part of the solution, but additional logic is required to separate the two numbers.

The basic single-XOR solution isn't enough.

A more general technique uses:

```
text
```

```
XOR all expected values  
XOR all actual values
```

giving:

```
text
```

```
missing ^ duplicate
```

Then use a distinguishing set bit to partition the values into two groups.

This is a very common advanced XOR problem.

72. Advanced Follow-Up — Why a Set Bit Helps

Suppose:

```
text
```

```
xor = missing ^ duplicate
```

Since the two numbers differ:

```
text
```

```
xor != 0
```

Choose one set bit:

```
text
```

```
rightmostSetBit = xor & -xor
```

That bit differs between the two numbers.

Partition all numbers based on that bit:

```
text
```

```
bit = 0  
bit = 1
```

The missing and duplicate values end up in different groups.

Then XOR within each group to recover each value.

This is the same cancellation idea, but with an additional partitioning trick.

73. Java-Specific Advanced Follow-Up – `xor & -xor`

This expression:

```
java  
  
int bit = xor & -xor;
```

extracts the rightmost set bit.

Why?

Because of two's-complement representation:

```
text  
  
-x = ~x + 1
```

so:

```
text  
  
x & -x
```

isolates the lowest set bit.

This is a very common bit-manipulation trick.

It is useful to know, but it is beyond what you need for the standard Missing Number question.

74. Java-Specific Follow-Up – Why XOR Is an `int`

Java's:

```
java  
  
int
```

is a signed 32-bit two's-complement integer.

The XOR operator works bit by bit over those 32 bits.

The sign doesn't create a special issue for the standard problem because all values are within the non-negative `0..n` range.

75. Production Variation – Batch Sequence Validation

Suppose a batch contains sequence IDs:

text

[100, 101, 102, 104, 105]

and expected range is:

text

100..105

We could adapt XOR by XORing:

text

100..105

with the received IDs.

Everything that exists cancels, leaving:

text

103

This demonstrates that the technique isn't restricted to starting at zero.

However, the implementation and range handling need to be adapted carefully.

76. Production Caveat — Duplicate Sequence IDs

If the batch contains:

text

100,101,101,102,104,105

XOR parity changes because the duplicate 101 appears an extra time.

The result no longer simply means:

text

missing ID

So in production sequence validation, first establish whether duplicates are possible.

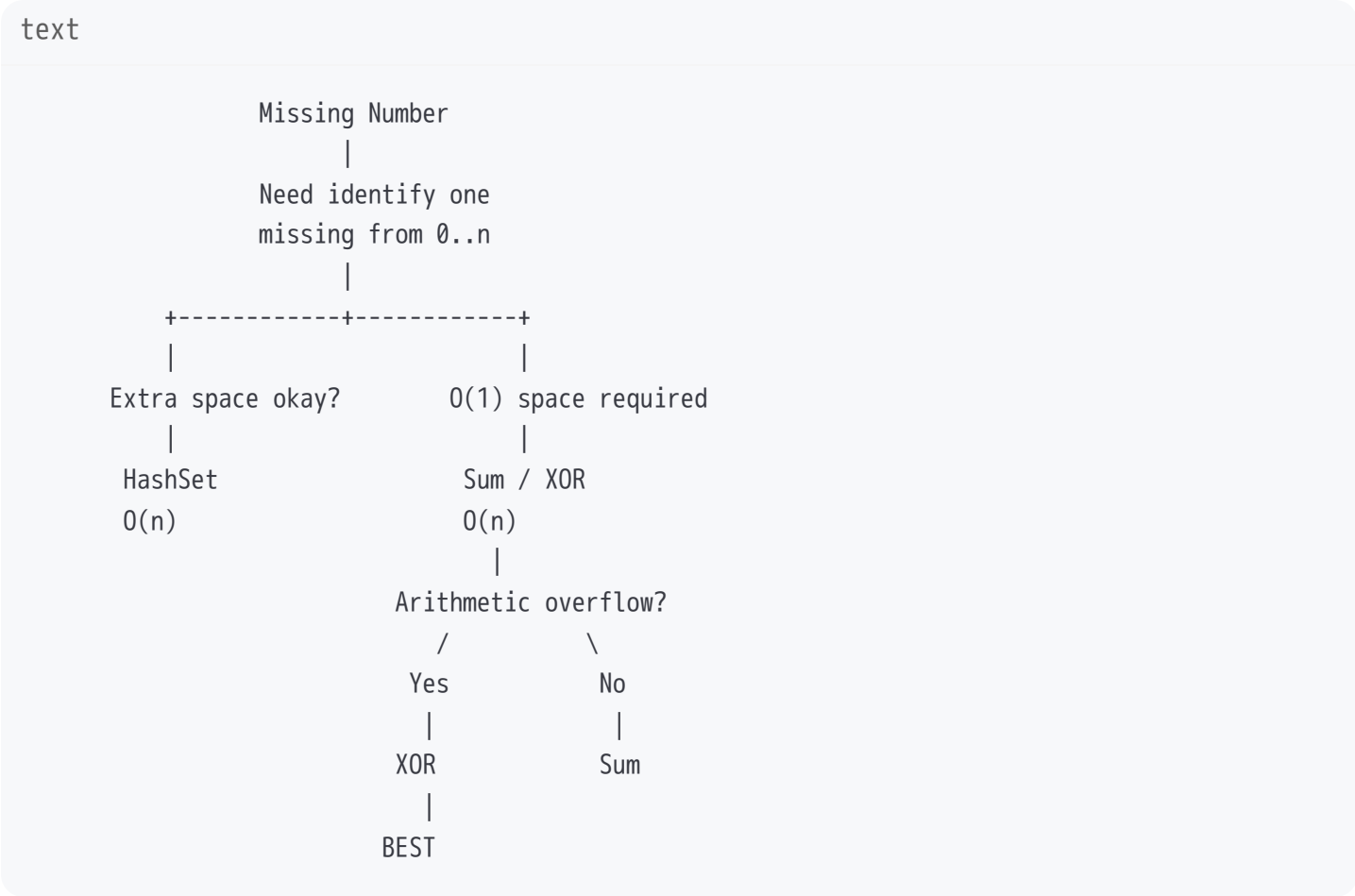
If duplicates are possible, use a structure that captures the required semantics, such as:

```
text

Set
frequency map
database constraint
```

depending on scale and requirements.

77. Final Decision Tree



For the standard interview problem, I'd select:

```
text

XOR
```

78. Final Interview Progression

This is how I'd actually solve it in front of an interviewer.

Step 1 — Brute force

"For every value from 0 to n , search the array."

text

$O(n^2)$

Step 2 – Sorting

"Sort the array and find the first index where `nums[i] != i` ."

text

$O(n \log n)$

Step 3 – HashSet

"Store all values and look for the missing number."

text

$O(n)$ average

$O(n)$ space

Step 4 – Sum

"Expected sum minus actual sum gives the missing value."

text

$O(n)$

$O(1)$

Then interviewer asks:

"What about overflow?"

Step 5 – XOR

"XOR all expected values and all actual values. Every present number appears twice and cancels, leaving the missing one."

text

$O(n)$

$O(1)$

That's the complete optimization path.

79. Final Interviewer-Ready Code

```
java

public int missingNumber(int[] nums) {

    int xor = nums.length;

    for (int i = 0; i < nums.length; i++) {

        xor ^= i;
        xor ^= nums[i];
    }

    return xor;
}
```

80. Final Interviewer-Ready Explanation

If the interviewer asks:

"Explain your optimal solution."

Say:

"The array contains all numbers from 0 to n except one. A brute-force search is $O(n^2)$, while sorting gives $O(n \log n)$. A HashSet gives $O(n)$ expected time but requires $O(n)$ space.

We can get $O(n)$ time and $O(1)$ extra space using XOR. I XOR every number from 0 to n with every number in the array. Every number that exists in the array appears exactly twice in the combined XOR, and because $x \oplus x = 0$, those values cancel. Also, XOR with zero leaves the value unchanged. Therefore, only the missing number remains.

I initialize xor with nums.length because the loop only covers 0 through n-1, while the complete range includes n.

The final complexity is $O(n)$ time and $O(1)$ auxiliary space."

That is a very strong answer for a 2–2.5 YOE candidate.

81. If Interviewer Asks "Why XOR Instead of Sum?"

Say:

"The sum solution is also $O(n)$ time and $O(1)$ space, but the arithmetic expression $n * (n + 1)$ can overflow an int, so I'd have to use long carefully. XOR avoids that arithmetic-overflow

issue and works through cancellation, so it is a clean constant-space solution."

82. If Interviewer Asks "Can You Do Better Than $O(n)$?"

Say:

"No, not asymptotically. The missing value can depend on any position, so I need to inspect the input. $O(n)$ is therefore optimal."

83. If Interviewer Asks "What If Majority/Duplicates Are Not Guaranteed?"

Say:

"The XOR solution relies on the exact constraints: values are distinct, they belong to `0..n`, and exactly one value is missing. If duplicates or multiple missing values are allowed, XOR alone is not sufficient and the problem needs a different approach."

84. Final Cheat Sheet

Approach 1 — Brute Force

text

For each expected number:
search array

Time: $O(n^2)$
Space: $O(1)$

Approach 2 — Sorting

text

Sort
Find first `nums[i] != i`

Time: $O(n \log n)$
Space: depends on sort
Input modified

Approach 3 — HashSet

text


```
Store all values
Find missing expected value
```

```
Time:  $O(n)$  average
Space:  $O(n)$ 
```

Approach 4 – Sum

```
text
```

```
expected = n(n + 1) / 2
actual = sum(nums)

missing = expected - actual
```

```
Time:  $O(n)$ 
Space:  $O(1)$ 
```

Potential issue:

```
text
```

```
integer overflow
```

Approach 5 – XOR

```
text
```

```
xor = n

for i:
    xor ^= i
    xor ^= nums[i]

return xor
```

Complexity:

```
text
```

```
Time:  $O(n)$ 
Auxiliary Space:  $O(1)$ 
```

85. Core Pattern

text

Known complete set
+
Exactly one missing
↓
Need cancellation
↓
XOR

XOR properties

text

$x \wedge x = 0$
 $x \wedge 0 = x$
 $a \wedge b = b \wedge a$
 $(a \wedge b) \wedge c = a \wedge (b \wedge c)$

Therefore:

text

all expected
XOR
all actual
↓
duplicates cancel
↓
missing survives

86. Most Important Mental Model

Don't memorize only:

java

```
int xor = nums.length;
```

Remember **why**:

text

Expected:
0 1 2 3 ... n

Actual:

0 1 3 ... n

XOR both:

$0 \wedge 0 \rightarrow \text{cancel}$

$1 \wedge 1 \rightarrow \text{cancel}$

$3 \wedge 3 \rightarrow \text{cancel}$

...

$n \wedge n \rightarrow \text{cancel}$

Only missing remains.

That's the real insight.

Connection to the previous questions

You now have another useful pattern family:

text

Q9 Majority Element



Cancellation of different values



Boyer-Moore

Q10 Missing Number



Cancellation of equal values



XOR

So don't think of Q10 as simply "**the XOR question.**" The deeper pattern is **eliminate paired/unnecessary information so only the required value remains.**

For the standard 0..n / one-missing / distinct-values problem, the interviewer-expected final answer is **XOR: O(n) time and O(1) auxiliary space**, with the sum approach as an important alternative that you should be able to explain and discuss for overflow.